



UNIVERSITY OF
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Inference from Scientific Data

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The aim of this course is to provide you with a good understanding of how quantitative conclusions are drawn from scientific data. Most of the examples in this course will come from the fields of astronomy and astrophysics, reflecting the research interests of those who have contributed in the development of the course. But the ideas are very general and can be applied much more widely. In many case, the topics covered will prove useful in Y4 research projects.

These notes have been prepared to accompany to lectures in the Y4 course Inference from Scientific Data. They are not a substitute for the lectures; in places they are incomplete while in others they contain more detail than is necessary. These notes draw heavily form those prepared for earlier versions of this course by given by Ilya Mandel, Trevor Ponman, John Veitch and Alberto Vecchio.

There are many useful textbooks, although no single one covers all the material in the way it will be done here. The following may be useful and were used in preparing these notes:

- R. J. Barlow, “Statistics”
- J. V. Wall & C. R. Jenkins, “Practical Statistics for Astronomers”
- P. Gregory, “Bayesian Logical Data Analysis”
- D. S. Sivia, “Data Analysis: A Bayesian Tutorial”
- Sections from Riley, Hobson & Bence, “Mathematical Method For Physics and Engineering”

The notes will occasionally illustrate ideas using snippets of Python code (it is assumed the reader is familiar with Python). Complete pieces of example code will be made available as Jupyter Notebooks in a public [GitHub repository](#). For example, consider the notebook [Welcome.ipynb](#); you can either clone the repository and run this notebook yourself, or else run it interactively in a Binder environment (try clicking the link to the welcome notebook).

If you have questions about the course you should email G.R.Davies@bham.ac.uk.

These notes are still relatively new and I will be adding to them during the course. Therefore, they will inevitably contain mistakes. If you spot a mistake please tell me at G.R.Davies@bham.ac.uk and I will try to fix it in the next version.

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Chapter 1

Introduction

Data is precious. Any good scientist should aim to make the best possible use of it. To do so requires an understanding of statistics, probability, and inference.

Many people studying undergraduate physics (myself included) come to regard statistics as a boring necessity. This is perhaps because of the way it is often introduced in first year laboratories in the form of a set of rules which must be followed to calculate and propagate errors. In fact (as I only came to realise later) there is much more to the subject than this. Much of the scientific enterprise can be regarded as one of making comparisons between probabilistic models and the data with which they are to be statistically tested. Moreover, this area is the subject of a lively debate between two schools of thought - frequentist and Bayesian - which disagree about the role of probability in this process. Once one understands this the subject becomes much more interesting. This course will provide you with the beginnings of such an understanding and equip you with various tools which can be applied to your own research problems.

1.1 Statistics, Probability, and Inference

I will begin by trying to give working definitions of these three terms.

Statistics concerns the collection, analysis, and presentation of data. Good statistics aims to get the maximum return from the available data, so a good scientist is always interested in it. A common complaint in all areas of science is “we do not have enough data”, so

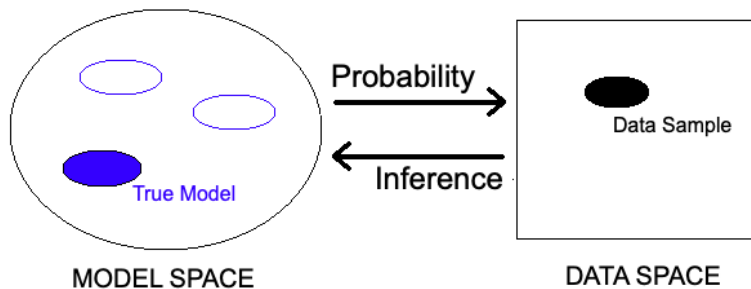


Figure 1.1: Model space $\leftrightarrow$ Data space. The roles of probability and inference.

we must endeavour to extract as much as possible from that which is available. This is especially true in astronomy where it is often not possible to make repeat observations to get additional data.

The usual situation in science is that we have some (limited) data and we want to use it to learn something about the Universe. For example, we wish to measure the value of some quantity or to test whether a particular model fits the data. This involves making predictions from our model in order to compare with the data. These predictions are almost always probabilistic in nature. Probability is the mathematical discipline concerned with quantifying the likelihood that random events will occur. We then have to reason backwards in order to draw conclusions about our model; this is the process of inference (see figure 1.1). Inference can be a purely logical argument leading to a certain conclusion; however, in the sciences, inference is almost always probabilistic with the observations providing a certain amount of evidence for or against a particular conclusion. It is our task to quantify this evidence.

There are two broad schools of thought concerning probability: frequentist and Bayesian. This has been the source of much confusion: everyone agrees that probability theory can be used to make statements about the chance of obtaining a given set of data if you know the model, what is debated is whether (and how) one can legitimately use probability theory to make a statement about the chance of that model being correct. In recent years Bayesian methods have become increasingly common, particularly in astronomy, but both are still widely used. We will encounter both frequentist and Bayesian methods in this course.

Chapter 2

Probability

2.1 Some Notation From Logic

In probability it is common to borrow notation from the field of logic.

Capital letters A , B , C ... represent events or propositions. A proposition can either be true or false. For example, we could have the proposition A = “It is raining”, or B = “I will pass this course”.

Propositions only have two possible values; they can be **true** or **false**.

The addition of an overline to a proposition denotes its negation; i.e. the event $\bar{A}$ is to be read as the event “**not** A ”. (We will not consider $\bar{\bar{B}}$.) If A is true, then $\bar{A}$ is false.

Given two propositions, e.g. A and B there are several ways of combining them to form a new proposition; these are known as logical conjunctions:

- $A \cap B$ denotes the joint event “ A **and** B ”. (This is sometimes written with the alternative notations AB , A, B or $A \wedge B$.) The following is a truth table for the **and** conjunction.

A	B	$A \cap B$
True	True	True
True	False	False
False	True	False
False	False	False

- The event $A \cup B$ denotes the event “ A **or** B ”. (This is sometimes written with the alternative notations $A \vee B$, or $A + B$.) This is the inclusive **or**; meaning either A , or B , or $A \cap B$ is true (as distinct from the exclusive **xor** which will not be used here).

Example 2.1: Or or Xor

Draw truth tables for the two logical conjunctions **or** and **xor** to clarify the difference between them.

The following simple rules come from Boolean logic. They can all be illustrated with *Venn diagrams*.

Idempotence:

$$A \cap A = A \quad (2.1)$$

$$A \cup A = A \quad (2.2)$$

Associativity:

$$A \cap (B \cap C) = (A \cap B) \cap C = A \cap B \cap C \quad (2.3)$$

$$A \cup (B \cup C) = (A \cup B) \cup C = A \cup B \cup C \quad (2.4)$$

Distributivity:

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C) \quad (2.5)$$

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C) \quad (2.6)$$

Commutativity:

$$A \cap B = B \cap A \quad (2.7)$$

$$A \cup B = B \cup A \quad (2.8)$$

Duality: or *De Morgan* after August De Morgan who formalised these.

$$\overline{A \cap B} = \bar{A} \cup \bar{B} \quad (2.9)$$

$$\overline{A \cup B} = \bar{A} \cap \bar{B} \quad (2.10)$$

Complementarity:

$$A \cap \bar{A} = \text{False} \quad (2.11)$$

$$A \cup \bar{A} = \text{True} \quad (2.12)$$

2.2 Probability

Logic only deals with certainty; propositions are either true or false. Probability extends this to cover uncertainty.

We use the notation $P(A)$ to denote the probability of event A occurring. Probabilities take real number values which, by convention, are in the range 0 to 1.

All of the above logical statements can be turned into statements about probability by writing $P(\text{LHS}) = P(\text{RHS})$ and using $P(\text{False}) = 0$ and $P(\text{True}) = 1$. E.g. the commutativity rule implies that $P(A \cap B) = P(B \cap A)$ while complementarity implies that $P(A \cap \bar{A}) = 0$.

Cox (1946) showed that certain basic desiderata for consistent reasoning lead to the conclusion that probabilities should be represented by real numbers and should obey the following laws of probability.

Range: Probabilities lie in the range 0 (impossible) to 1 (certain). (This is a convention.)

Law Of Total Probability: For exhaustive and mutually exclusive events A_i

$$\sum_i P(A_i) = 1 \quad (2.13)$$

Note: the events A and $\bar{A}$ are exhaustive and mutually exclusive; therefore we have

$$P(A) + P(\bar{A}) = 1 \quad (2.14)$$

The Sum Rule:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) \quad (2.15)$$

Negation: (this follows from duality)

$$P(\overline{A \cup B}) = P(\bar{A} \cap \bar{B}) \quad (2.16)$$

$$P(\overline{A \cap B}) = P(\bar{A} \cup \bar{B}) \quad (2.17)$$

2.3 Conditional Probability and Bayes' Theorem

The notation $P(A|B)$ means the “probability of A given that B is true”.

The final important law of probability is the **Product Rule**:

$$P(A \cap B) = P(A|B)P(B) = P(B|A)P(A). \quad (2.18)$$

Two events A and B are said to be **independent** if and only if $P(A \cap B) = P(A)P(B)$. Two events are independent if one of them occurring does not affect the probability of the other occurring; from equation 2.18 we see that

$$P(A \cap B) = P(A)P(B) \iff P(A|B) = P(A), \quad (2.19)$$

$$P(B \cap A) = P(B)P(A) \iff P(B|A) = P(B). \quad (2.20)$$

Bayes' theorem states that

$$P(A|B) = P(B|A) \frac{P(A)}{P(B)} \quad (2.21)$$

and this follows directly from the product rule. Despite its apparent simplicity, this rule is extremely important and is used in frequentist as well as Bayesian statistics.

Throughout this course we will frequently use Bayes' theorem in the form

$$P(H|D) = \frac{P(D|H)P(H)}{P(D)}, \quad (2.22)$$

where H is our hypothesis and D is our experimental data.

In practice, whenever we have some event, A , and we ask what is the probability of that event, $P(A)$, there are almost always some assumptions that we have to make. For example, what is probability of rolling a 6 on standard die? The answer is $1/6$, assuming the die is fair. Often these assumptions are so obvious we don't write them down, sometimes we even forget we making them (although we should try not to). Therefore, instead of $P(A)$ we should probably consider the conditional probability $P(A|I)$ instead, where I stands for all the relevant prior information and assumptions we make and can't be bothered to write down (e.g. if I = "the die is fair", then $P(A|I) = 1/6$). Of course, often we get lazy and slip back into writing $P(A)$. Just remember there might be some important information lurking on the right hand side of the line!

Example 2.2: Bayes' Theorem

The following is a classic example application of Bayes' theorem which yields a counter-intuitive result.

As part of a general screening program, you are given a TB test. You test positive. On a healthy person, the probability of the test giving a false positive is 0.1%, while for a person with TB the probability of a true positive is 99%. If the incidence of TB in the population is $1/100000$ what is the probability that you have TB?

What changes if you initially took the test because you showed symptoms of TB?

At first glance you might think

$$P(\text{TB}|\text{test}) \stackrel{!}{=} P(\text{test}|\text{TB}) = 0.99, \quad (\text{i})$$

so you are very likely to have TB and should be concerned.

Not so. Using Bayes' theorem

$$P(\text{TB}|\text{test}) = P(\text{test}|\text{TB}) \frac{P(\text{TB})}{P(\text{test})} \quad (\text{ii})$$

and what is sometimes called the law of total probability

$$P(\text{test}) = P(\text{test}|\text{TB})P(\text{TB}) + P(\text{test}|\overline{\text{TB}})P(\overline{\text{TB}}). \quad (\text{iii})$$

From the question, $P(\text{test}|\text{TB}) = 0.99$, and $P(\text{TB}) = 10^{-5}$, and $P(\text{test}|\overline{\text{TB}}) = 10^{-3}$, and $P(\overline{\text{TB}}) = 1 - P(\text{TB})$. Putting these into (ii) and (iii) gives $P(\text{TB}|\text{test}) = 0.0098$, i.e. less than 1%. So you should not be overly concerned.

What makes the answer so different from what we initially suspected is the fact that $P(\text{TB}) = 10^{-5}$ is small. Before taking the test, our *prior* belief was that it was unlikely that you had TB, and this influences the answer. If instead you took the test because you showed symptoms this prior belief would have been larger.

This well-known problem serves to illustrate the importance of prior belief.

2.4 Are Probabilities Objective?

Suppose that I tell everyone in a class that I am going to perform an experiment with three possible outcomes which we will label A , B , and C . I then tell just half of the class that the experiment is a coin toss and the outcomes are $A = \text{“heads”}$, $B = \text{“tails”}$, and $C = \text{“other”}$. And finally, I tell just one student that the coin is deliberately biased in some way.

I now ask the entire class, what is $P(A)$? The first half of the class will have to reason along the following lines “I know almost nothing about the experiment, so from the *principle of indifference* I will have to treat the three possible outcomes equally and answer $P(A) = 1/3$ ”. The second half of the class will answer “it’s a coin toss, so obviously $P(A) = 1/2$ ”. And finally, there will be a single student who might reach the same answer by different reasoning; “it’s a biased coin, but I don’t know to which side the coin is biased, so I still have to say $P(A) = 1/2$ ”.

Who is right?

Chapter 3

Probability Distributions

The result of an experiment or observation usually has a number of possible outcomes. If a real number x can be assigned to each possible outcome, then x is a **random variable**. The **probability distribution** quantifies the probabilities of occurrence of all the different possible values of x . We will consider separately the cases when x is discrete or continuous.

3.1 Discrete Distributions

In a discrete situation the possible observation outcomes can be counted and labelled using an integer index i ,

$$x_1, x_2, \dots, x_i, \dots \quad (3.1)$$

The number of possible outcomes can be finite (e.g. flipping a coin has two outcomes) or infinite (e.g. number of photon counts recorded in an instrument during a certain observation). The probability distribution assigns a probability $P(x_i)$ to each of the possible observation outcomes x_i . The quantity $P(x_i)$ is sometimes called the **probability mass function** (PMF) and is dimensionless.

We will now give definitions (in bold) for some common terms associated with probability distributions.

The sum of the probabilities of all possible outcomes must equal unity. This is called the

normalisation of the PMF;

$$\sum_i P(x_i) = 1. \quad (3.2)$$

Note that this implies that $0 \leq P(x_i) \leq 1$.

The **expectation**, or mean, value of the random variable x is defined to be

$$\langle x \rangle = \sum_i x_i P(x_i). \quad (3.3)$$

In addition, if we have some function of the random variable, $f(x)$, then we can define the expectation of the function as

$$\langle f(x) \rangle = \sum_i f(x_i) P(x_i). \quad (3.4)$$

The **variance** is defined as the expectation of the squared distance from the mean,

$$\text{Var}(x) = \sum_i (x_i - \langle x \rangle)^2 P(x_i). \quad (3.5)$$

Note that $\text{Var}(x) \geq 0$. The **standard deviation** is defined as the (positive) square root of the variance; $\sigma_x = \sqrt{\text{Var}(x)}$.

The **cumulative distribution** is defined as the probability that the random variable takes a value below a certain threshold;

$$C(x) = \text{Prob}(x' \leq x) = \sum_{x_i \leq x} P(x_i). \quad (3.6)$$

The expectation, or mean, defined above is the most common way of quantifying the “average” of a distribution. However, other possibilities include the **mode** (the most likely outcome(s) of the distribution; i.e. the value(s) of X which maximise $P(x_i)$) and the **median** (the value X at which the cumulative distribution $C(x)$ takes the value $1/2$).

3.2 Continuous Distributions

In a continuous situation the possible observation outcomes are described by a smoothly varying real number x . In this case probabilities are described by a **probability density**

function (PDF) denoted $P(x)$. This is defined such that $P(x)dx$ gives the probability that the random variable x lies in an interval dx centred on x ; that is

$$\text{Prob}(a \leq x \leq b) = \int_a^b dx P(x). \quad (3.7)$$

This implies that the quantity $P(x)$ must be non-negative, $P(x) \geq 0$.

The quantity $P(x)$ is a *probability density* (not a probability) and has units of $[x]^{-1}$.

We use the same symbol P for the discrete probability mass function and the continuous probability density function. It is important to remember the difference which must be understood from the context of the particular problem.

The common definitions given in section 3.1 above for discrete distributions all have analogs for continuous distributions.

Normalisation:

$$\int_{\text{all } x} dx P(x) = 1. \quad (3.8)$$

Note that this does NOT imply that $P(x) \leq 1$.

Expectation:

$$\langle x \rangle = \int_{\text{all } x} dx x P(x). \quad (3.9)$$

Or, if we have some function $f(x)$, then the expectation of f is

$$\langle f(x) \rangle = \int_{\text{all } x} dx f(x) P(x). \quad (3.10)$$

Variance:

$$\text{Var}(x) = \int_{\text{all } x} dx (x - \langle x \rangle)^2 P(x). \quad (3.11)$$

Again, we have $\text{Var}(x) \geq 0$. The **standard deviation** is defined as $\sigma_x = \sqrt{\text{Var}(x)}$.

For a constant α , it follows directly from the definition that

$$\text{Var}(\alpha x) = \alpha^2 \text{Var}(x). \quad (3.12)$$

Cumulative distribution function: the CDF is defined as

$$C(x) = \text{Prob}(x' \leq x) = \int_{x_{\min}}^x dx' P(x'). \quad (3.13)$$

The **mode** and the **median** can also be similarly defined for continuous distributions.

3.3 Moments

For a continuous random variable x with PDF $P(x)$ the **moments** provide a convenient way to quantify the shape of the distribution. The **moment generating function** provides a convenient way of computing a distribution's moments (via differentiation, instead of integrations) and is an alternative to the PDF as a way of specifying a probability distribution.

For nonnegative integer k , the k^{th} **moment** of the distribution is defined as

$$\mu_k = \langle x^k \rangle = \int dx x^k P(x). \quad (3.14)$$

We usually only consider the moments for natural numbers; i.e. $k = 1, 2, 3, \dots$. The zeroth moment is just the “total mass” and is equal to 1 for a properly normalised distribution; $\mu_0 = \int dx P(x) = 1$. The first moment is equal to the expectation value, $\langle x \rangle = \mu_1$ (it is common to drop the subscript 1 when there is no risk of confusion; i.e. $\mu = \mu_1$).

Alternatively, we can define the **central moments** of the distribution,

$$\nu_k = \langle (x - \mu)^k \rangle = \int dx (x - \mu)^k P(x). \quad (3.15)$$

Note, the variance is equal to the second central moment, $\text{Var}(x) = \nu_2$. Note also that the third and fourth central moments are related to the commonly defined concepts of **skewness** γ and **kurtosis** κ :

$$\gamma = \frac{\nu_3}{\text{Var}(x)^{3/2}}, \quad \kappa = \frac{\nu_4}{\text{Var}(x)^2}. \quad (3.16)$$

The two sets of moments contain the same information; the central moments can be computed from the non-central moments and vice versa.

Exercise 3.1: Relating the central and noncentral moments

If nC_k are the binomial coefficients, show that

$$\nu_n = \sum_{k=0}^n (-1)^k {}^nC_k \mu_1^k \mu_{n-k}. \quad (\text{i})$$

Further show that when $n = 2$ this reduces to the useful well-known result

$$\text{Var}(x) = \langle x^2 \rangle - \langle x \rangle^2. \quad (\text{ii})$$

The moments are defined as integrals and these are not guaranteed to exist for all PDFs. A distribution for which some (or all) of the moments do not exist is said to be **pathological**. A common example of a pathological distribution is the Cauchy distribution (with location and scale parameters x_0 and w respectively) which has PDF

$$P(x; x_0, w) = \frac{1}{\pi w} \left[\frac{1}{1 + \left(\frac{x-x_0}{w} \right)^2} \right]. \quad (3.17)$$

However, unless stated otherwise, we will generally assume that the moments all exist.

We have given the definitions of moments for a continuous distribution $P(x)$. Corresponding definitions for discrete distributions are obtained by replacing integrals with sums.

For a random variable x with PDF $P(x)$ and a real valued number t , the **moment generating function** (MGF) is defined by

$$M_x(t) = \langle \exp(tx) \rangle = \int dx \exp(tx) P(x). \quad (3.18)$$

As the name implies, the MGF is particularly useful for computing moments. The reason for this can be seen by using the series expansion for the exponential function;

$$M_x(t) = \left\langle 1 + tx + \frac{t^2}{2!} x^2 + \dots \right\rangle \quad (3.19)$$

$$= \langle x^0 \rangle + t \langle x^1 \rangle + \frac{t^2}{2!} \langle x^2 \rangle + \dots \quad (3.20)$$

$$= \mu_0 + \mu_1 t + \frac{\mu_2}{2!} t^2 + \dots \quad (3.21)$$

Therefore, assuming the MGF exists and is defined around the point $t = 0$, we have

$$\mu_k = \left. \frac{d^k}{dt^k} \right|_{t=0} M_x(t). \quad (3.22)$$

If the MGF is known, then the moments can be computed from equation 3.22 by differentiating, which is usually easier than computing them from equation 3.14 by integrating.

The MGF is an alternative to the PDF as a way of specifying probability distributions. If random variables x and y have the same MGF, then they also have the same PDF; i.e. they come from identical probability distributions. This **uniqueness property** is intuitively fairly obvious but technically difficult to prove.

The MGF has several useful properties that follow directly from the definition.

Shifting and scaling property: if x is a random variable and $y = ax + b$, then

$$M_y(t) = \exp(bt)M_x(at). \quad (3.23)$$

This is particularly useful for computing the central moments ν_k from the MGF.

Summation property (of independent random variables): if $x_1, x_2, \dots, x_n$ are independent random variables and $z = \sum_{i=1}^n x_i$, then

$$M_z(t) = \prod_{i=1}^n M_{x_i}(t). \quad (3.24)$$

Exercise 3.2: Useful properties of the MGF

Prove the properties of the MGF described in equation 3.23. After studying section 3.5, prove the properties of the MGF described in equation 3.24.

3.4 Some Common Distributions

Some distributions arise so frequently in the physical sciences that it is worth knowing their properties. You will probably have seen the following distributions before.

3.4.1 The Binomial Distribution

An example of a discrete distribution. The binomial distribution describes the number of successes r in a sequence of n independent experiments, each of which can have one of two outcomes: a success (which we give a value of 1) which occurs with probability p , or a failure (value 0) which occurs with probability $1 - p$.

The random variable r is the total number of successes (i.e. the sum of all the individual outcomes) and can take $n + 1$ possible values $\{0, 1, \dots, n\}$. The total number of trials n and the success probability p are the two parameters of the distribution.

$$P(r; n, p) = p^r (1 - p)^{n-r} {}^n C_r \quad (3.25)$$

where ${}^n C_r = n! / (r!(n - r)!)$ and is read as “ n choose r ”.

The binomial distribution can be derived by considering the number of combinatorial ways r successes and $n - r$ failures can occur in n trials and then multiplying the respective probabilities p and $1 - p$. Note that this assumes the events are independent.

Example 3.1: Binomial distribution

Suppose that 40% of galaxy clusters possess a central cD galaxy. A sample of 5 X-ray galaxies find 4 with central cD galaxies. What does this suggest?

The probability of obtaining this result by chance from the population described is quite small,

$$P(4; 5, 0.4) = 0.4^4 \times 0.6^1 \times \frac{5!}{4! \times 1!} = 0.077. \quad (i)$$

This tentatively suggests that the population of X-ray selected clusters might be systematically different from the general population.

Exercise 3.3: Binomial distribution

Find the (i) mean, and (ii) variance of the binomial distribution.

3.4.2 The Poisson Distribution

Another example of a discrete distribution. The Poisson distribution is used for modelling the number of independent events that occurs in a fixed interval of time (or space) if events occur at a constant mean rate. The random variable n is the number of events and has an infinite number of discrete possible outcomes, $\{0, 1, \dots\}$. The PMF for a Poisson distribution with rate parameter $\lambda > 0$ is given by

$$P(n; \lambda) = \frac{\exp(-\lambda)\lambda^n}{n!}. \quad (3.26)$$

The Poisson distribution may be derived as follows. Suppose that the probability that n events occur in time t is denoted $P_n(t)$. If the average number of events per unit time is $\tilde{\lambda}$, then in a further short interval $\delta t \ll \tilde{\lambda}^{-1}$ the probability of there being one additional event in this short interval is $\delta t \tilde{\lambda}$ (the probability of two additional events $\propto \delta t^2$ and is negligible in the limit $\delta t \rightarrow 0$). Note that $\tilde{\lambda}$ has units of inverse time while λ is dimensionless. Thus, the probability there being of exactly n events in the combined interval $t + \delta t$ is

$$P_n(t + \delta t) = P_n(t)(1 - \tilde{\lambda}\delta t) + P_{n-1}(t)\tilde{\lambda}\delta t. \quad (3.27)$$

(This follows from the law of total probability.) This holds for all $n > 0$, and rearranging and taking the limit $\delta t \rightarrow 0$ gives the differential recurrence relation for $P_n(t)$;

$$\frac{dP_n}{dt} = \tilde{\lambda} [P_{n-1}(t) - P_n(t)]. \quad (3.28)$$

We now proceed inductively. For $n = 0$, equation (3.28) simplifies to

$$\frac{dP_0}{dt} = -\tilde{\lambda}P_0(t), \quad (3.29)$$

which integrates to give $P_0(t) = P_0(0) \exp(-\tilde{\lambda}t) = \exp(-\tilde{\lambda}t)$, with the constant of integration fixed by requiring that it is certain to obtain zero events in zero time, i.e. $P_0(0) = 1$. This may be substituted back into equation (3.28) to find the following differential equation for $n = 1$,

$$\frac{dP_1}{dt} = \tilde{\lambda} [\exp(-\tilde{\lambda}t) - P_1(t)], \quad (3.30)$$

which may be solved to find $P_1(t) = \tilde{\lambda}t \exp(-\tilde{\lambda}t)$ using the initial condition $P_1(0) = 0$. Proceeding by induction, and using the initial conditions $P_n(0) = 0$ for all $N \geq 1$, we find

$$P_n(t) = \frac{(\tilde{\lambda}t)^n}{n!} \exp(-\tilde{\lambda}t). \quad (3.31)$$

By setting $\tilde{\lambda}t = \lambda$ we recover the Poisson distribution in equation (3.26).

The Poisson distribution can be generated from the following recurrence formula, which allows successive probabilities to be easily calculated once one is known.

$$\begin{aligned} p(0) &= \exp(-\lambda) \\ p(1) &= p(0) \frac{\lambda}{1} \\ p(2) &= p(1) \frac{\lambda}{2} \\ p(3) &= p(2) \frac{\lambda}{3} \\ &\dots \end{aligned} \tag{3.32}$$

For large values n of and λ , evaluating the Poisson probability distribution in equation (3.26) directly can be numerically unstable. Instead, it is preferable to use

$$P(n; \lambda) = \exp(n \log \lambda - \lambda - \log \Gamma(n+1)) , \tag{3.33}$$

which is mathematically equivalent but numerically more stable. The logarithm of the Gamma function is implemented in SciPy as `scipy.special.gammaln`.

Also, for large values n of and λ , it is often convenient to approximate the Poisson distribution using a Gaussian distribution (see section 3.4.3) with mean and variance $\mu = \sigma^2 = \lambda$. See example 3.4 for more details.

Example 3.2: Poisson distribution

Photons from a distant source arrive in our telescope at a rate of 3 s^{-1} . How long must we observe for to ensure with 99% certainty that we receive at least one photon?

Let $r = 3 \text{ s}^{-1}$. If we observe for time T the mean number of photons we expect is $\lambda = rT$ and the probability of receiving n photons is given by $P(n; \lambda)$. The probability of receiving no photons is given by

$$\text{Prob}(n = 0) = P(0; \lambda) = \exp(-rT) . \tag{i}$$

Setting $\text{Prob}(n = 0) = 0.01$ and rearranging for T gives the minimum observation

time as $T = 1.54$ s.

Exercise 3.4: Poisson distribution

Find the (i) mean, and (ii) variance of the Poisson distribution.

Example 3.3: Poisson distribution as a limiting case

The poisson distribution may be obtained from the binomial distribution in the limit that the number of trials is large, $n \rightarrow \infty$, but the probability of success tends to zero, $p \rightarrow 0$, in such a way that the expected number of successes remains finite, $np = \lambda$.

The probability of x successes in n trials is given by the binomial distribution,

$$P(x; n, p) = p^x (1 - p)^{n-x} \frac{n!}{x!(n-x)!}. \quad (\text{i})$$

Taking the limit $n \rightarrow \infty$ (with x finite) of the combinatorial factor,

$$\lim_{n \rightarrow \infty} \frac{n!}{x!(n-x)!} = \lim_{n \rightarrow \infty} \frac{n(n-1) \dots (n-x+1)}{x!} = \frac{n^x}{x!}. \quad (\text{ii})$$

Taking the limits $n \rightarrow \infty$ and $p \rightarrow 0$ (again with x finite) of the remaining terms and using $\lambda = np$,

$$\lim_{n \rightarrow \infty} \lim_{p \rightarrow 0} p^x (1 - p)^{n-x} = \frac{p^x (1 - p)^{\lambda/p}}{(1 - p)^x} = p^x \exp(-\lambda), \quad (\text{iii})$$

where, in the last step, we use the following definition of the exponential function;

$$\exp(y) = \lim_{m \rightarrow \infty} \left(1 + \frac{y}{m}\right)^m. \quad (\text{iv})$$

Multiplying ii and iii, we find the limit of the probability in i is given by

$$P(x; n, p) = \frac{p^x n^x \exp(-\lambda)}{x!} = \frac{\lambda^x \exp(-\lambda)}{x!}, \quad (\text{v})$$

which we recognise as the Poisson distribution defined in equation 3.26.

3.4.3 The Gaussian Distribution

An example of a continuous distribution, it is sometimes known as the “normal distribution”, or simply described as the “bell curve”. This is perhaps the most important, and most commonly occurring probability distribution, partly because of the central limit theorem (see section 3.6). The random variable x takes values continuously in the range $-\infty < x < \infty$ and the Gaussian PDF with parameters μ and σ^2 is given by

$$P(x; \mu, \sigma^2) = \frac{\exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right)}{\sqrt{2\pi\sigma^2}}. \quad (3.34)$$

The Gaussian distribution is properly normalised; i.e. $\int_{-\infty}^{\infty} dx P(x; \mu, \sigma^2) = 1$. For a derivation of this integral, see Appendix C.

The Gaussian (or normal) distribution is sometimes denoted with $\mathcal{N}$. If a random variable x is Gaussian distributed we write

$$x \sim \mathcal{N}(\mu, \sigma^2). \quad (3.35)$$

About 68% of values drawn from a Gaussian distribution lie within $\pm 1\sigma$ from the mean. The values for 1, 2 and 3 σ are worth remembering.

$$\text{Prob}(-1\sigma < x - \mu < +1\sigma) = 0.683$$

$$\text{Prob}(-2\sigma < x - \mu < +2\sigma) = 0.954$$

$$\text{Prob}(-3\sigma < x - \mu < +3\sigma) = 0.997$$

The **error function** is defined as

$$\text{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x \exp(-x^2), \quad (3.36)$$

so that the cumulative distribution function (CDF) for the Gaussian distribution is

$$\text{CDF}(x) = \int_{-\infty}^x dx P(x; \mu, \sigma^2) = \frac{1}{2} \left[1 + \text{erf} \left(\frac{x - \mu}{\sqrt{2}\sigma} \right) \right]. \quad (3.37)$$

We can calculate the moment generating function (MGF) for the Gaussian distribution. If the random variable $x \sim \mathcal{N}(\mu, \sigma^2)$, then from the definition in equation 3.18 we have

$$\begin{aligned} M_x(t) &= \langle \exp(tx) \rangle \\ &= \int_{-\infty}^{\infty} dx \exp(tx) P(x; \mu, \sigma^2) \\ &= \int_{-\infty}^{\infty} dx \exp(tx) \frac{\exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)}{\sqrt{2\pi\sigma^2}} \\ &= \frac{1}{\sqrt{2\pi\sigma^2}} \int_{-\infty}^{\infty} dx \exp\left(tx - \frac{(x-\mu)^2}{2\sigma^2}\right) \\ &= \exp\left(\mu t + \frac{1}{2}\sigma^2 t^2\right) \frac{1}{\sqrt{2\pi\sigma^2}} \int_{-\infty}^{\infty} dx \exp\left(-\frac{[x - \mu - \sigma^2 t]^2}{2\sigma^2}\right) \\ &= \exp\left(\mu t + \frac{1}{2}\sigma^2 t^2\right). \end{aligned} \quad (3.38)$$

In the penultimate step we have completed the square on the argument of the exponential function and in the final step we have used the normalisation of the Gaussian distribution (see Appendix C).

Exercise 3.5: Gaussian distribution

Show that μ is the mean of the Gaussian distribution. Explain why the symmetry of the distribution means that the mode and median have the same value as the mean.

Show that σ^2 is the variance of the Gaussian distribution.

Hint. You can work from the integral definitions of the mean and variance in equations 3.9 and 3.11 respectively. This approach involves two integrals. Alternatively, you can use the MGF and equation 3.22. This approach involves just the one integral already evaluated in 3.38 and a couple of derivatives. As derivatives are generally easier than integrals, the second approach is probably faster. The second approach also gives the skewness, kurtosis and higher

moments with negligible extra effort.

Example 3.4: Gaussian distribution as a limiting case

The Gaussian distribution may be obtained from the Poisson distribution in the limit that the rate is large, $\lambda \rightarrow \infty$. Taking the logarithm of the Poisson probability distribution gives

$$\log P(n; \lambda) \approx n \log \lambda - \lambda - n \log n + n - \log \sqrt{2\pi n}, \quad (\text{i})$$

where we have used Stirling's formula for the factorial of large numbers, $n! \approx \sqrt{2\pi n}(n/e)^n$ (see appendix E). The probability will only be significantly non-zero around the mean, i.e. where $n \approx \lambda$. Let's substitute $\epsilon = n - \lambda$.

$$\log P(n; \lambda) \approx (\lambda + \epsilon) \log \lambda - \lambda - (\lambda + \epsilon) \log(\lambda + \epsilon) + (\lambda + \epsilon) - \log \sqrt{2\pi(\lambda + \epsilon)}.$$

Expanding this to second order in the small quantity ϵ/λ , we find

$$\log P(n; \lambda) \approx -\frac{\epsilon^2}{2\lambda} - \log \sqrt{2\pi\lambda}, \quad (\text{ii})$$

which we recognise as the Gaussian distribution with $\mu = \sigma^2 = \lambda$.

3.5 Multivariate Distributions

In many situations we need to consider two or more random variables that are not independent from one another. These are described by joint, or multivariate probability density functions. Here we will only consider the case of several continuous random variables and will mostly focus on the bivariate case (i.e. 2 random variables, the extension to higher numbers is straightforward). Many of the definitions and properties of univariate PDFs carry over to the multivariate case.

In the bivariate case with random variables x and y , the PDF is the probability that the variable x lies in a range dx centred on x , *and* y lies in a range dy centred on y . This bivariate PDF could be denoted $P(x \cap y)$, but the more common notation is $P(x, y)$. The PDF is still a density; i.e. the dimensionality of $P(x, y)$ is $[x]^{-1}[y]^{-1}$. In order to calculate probabilities, we must perform a double integral of the bivariate PDF; i.e.

$$\text{Probability}(x_0 \leq x \leq x_1 \text{ and } y_0 \leq y \leq y_1) = \int_{x_0}^{x_1} dx \int_{y_0}^{y_1} dy P(x, y). \quad (3.39)$$

As in the univariate case, the quantity $P(x, y)$ must be non-negative and normalised (i.e. $\iint dx dy P(x, y) = 1$).

If the x and y random variables are independent then the multivariate PDF is separable, i.e. $P(x, y) = f(x)g(y)$.

Expectation: If we have some function $f(x, y)$, then the expectation of f is defined as

$$\langle f(x, y) \rangle = \int_{\text{all } x} dx \int_{\text{all } y} dy f(x, y) P(x, y). \quad (3.40)$$

The expected, or mean, value of the random variable x is defined to be

$$\langle x \rangle = \int_{\text{all } x} dx \int_{\text{all } y} dy x P(x, y). \quad (3.41)$$

A similar definition exists for $\langle y \rangle$.

Note that this definition can be rewritten as

$$\langle x \rangle = \int_{\text{all } x} dx x P_X(x), \quad (3.42)$$

where we have defined $P_X(x) = \int_{\text{all } y} dy P(x, y)$ which is called the *marginal distribution of x* . This process of integrating over one or more variables in a multivariate distribution to remove the dependence on that parameter is called *marginalisation* and will come up often later in the course.

Variance: The variance is defined analogously to the single-variable case,

$$\text{Var}(x) = \sigma_x^2 = \int_{\text{all } x} dx (x - \langle x \rangle)^2 P_X(x). \quad (3.43)$$

Covariance and Correlation: The covariance between two random variables is defined by

$$\text{Cov}(x, y) = \langle (x - \langle x \rangle)(y - \langle y \rangle) \rangle, \quad (3.44)$$

$$= \int_{\text{all } x} dx \int_{\text{all } y} dy (x - \langle x \rangle)(y - \langle y \rangle) P(x, y). \quad (3.45)$$

Note that $\text{Cov}(x, x) = \text{Var}(x)$.

The correlation is defined by

$$\text{Corr}(x, y) = \frac{\text{Cov}(x, y)}{\sigma_x \sigma_y}. \quad (3.46)$$

The covariance and correlation measure the joint variability of two random variables. For example, if you measured the mass and luminosity of a large number of stars you would expect to find a large positive correlation.

For independent random variables x and y , the covariance vanishes; $\text{Cov}(x, y) = 0$.

Exercise 3.6: Variance of a sum

Show that for any two random variables x_1 and x_2 the variance of the sum $x_1 + x_2$ is given by

$$\begin{aligned} \text{Var}(x_1 + x_2) &= \text{Var}(x_1) + \text{Var}(x_2) + 2\text{Cov}(x_1, x_2) \\ &= \sum_{i=1}^2 \sum_{j=1}^2 \text{Cov}(x_i, x_j). \end{aligned} \quad (\text{i})$$

Note that this result holds even if x_1 and x_2 come from the different distributions and are dependent random variables.

Extension; show that for random variables $x_1, x_2, \dots, x_n$ the variance of the sum is given by

$$\text{Var}\left(\sum_{i=1}^n x_i\right) = \sum_{i=1}^n \sum_{j=1}^n \text{Cov}(x_i, x_j). \quad (\text{ii})$$

Exercise 3.7: Perplexing Correlations?

Random variables x and y have a joint PDF of $P(x, y) \propto (x^2 + y^2)$, where $0 < x < 1$ and $0 < y < 1$. Show that, despite the apparent symmetry, x and y are in fact negatively correlated and find $\text{Corr}(x, y)$. Explain this result.

3.6 The Central Limit Theorem

As we have seen, the Gaussian distribution appears as a limiting case of the Poisson distribution (example 3.4) which in turn appears as a limiting case of the binomial distribution (example 3.3). It seems the Gaussian distribution pops up everywhere! This is partly a consequence of the **central limit theorem** (CLT).

Theorem: If $\{x_i | i = 1, 2, \dots, n\}$ are independent random variables each of which is drawn from PDF $P_i(x)$ (not necessarily the same for each i) with means m_i and variances σ_i^2 , then the random variable $z = (\sum_{i=1}^n x_i)/n$ has the following properties:

1. $\langle z \rangle = (\sum_{i=1}^n m_i) / n$;
2. $\text{Var}(z) = (\sum_{i=1}^n \sigma_i^2) / n^2$;
3. as $n \rightarrow \infty$ the PDF of z tends to a Gaussian with the corresponding mean and variance.

Note that for properties 1. and 2. to make sense the individual distributions $P_i(x)$ must have well defined means and variances. We will further assume that the MGF and all of the moments exist for the individual distributions $P_i(x)$.

Property 1. is easily proved using the linearity of the expectation;

$$\langle z \rangle = \frac{1}{n} \langle x_1 + x_2 + \dots + x_n \rangle = \frac{1}{n} (\langle x_1 \rangle + \langle x_2 \rangle + \dots + \langle x_n \rangle) = \frac{1}{n} \sum_{i=1}^n m_i. \quad (3.47)$$

Note that the proof of 1. *does not* use the fact that the x_i are independent.

Property 2. is also easily proved;

$$\text{Var}(z) = \text{Var} \left[\frac{x_1 + x_2 + \dots + x_n}{n} \right] = \frac{1}{n^2} (\text{Var}(x_1) + \dots + \text{Var}(x_n)) = \frac{1}{n^2} \sum_{i=1}^n \sigma_i^2, \quad (3.48)$$

where we have used the properties of the variance in equations 3.12 and in exercise 3.6 and the fact that the individual x_i random variables are independent (i.e. $\text{Cov}(x_i, x_j) = 0$ for $i \neq j$). Note that the proof of 2. *does* use the fact that the x_i are independent.

Property 3. is perhaps the most important. It is also the main reason for the ubiquity of the Gaussian distribution. This can be proved using the MGF. As we saw in section 3.3, the MGF has properties that make it very convenient for working with the sums of random variables (in particular, the *summation property* of the MGF derived in equation 3.24). This will allow us to prove the final property 3. in the CLT.

The *summation property of the MGF*; from equation 3.24, the MGF of z , can be expressed in terms of the MGFs of all the individual random variables x_i/n ,

$$M_z(t) = \prod_{i=1}^n M_{x_i/n}(t) \quad (3.49)$$

From the *scaling property of the MGF*; from equation 3.23 we have $M_{x_i/n}(t) = M_{x_i}(t/n)$. Using this, the MGF of z can be written as

$$M_z(t) = \prod_{i=1}^n M_{x_i}(t/n). \quad (3.50)$$

We now use the series expansion of the MGF (see, for example, equation 3.21) about the point $t = 0$,

$$M_{x_i}(t/n) = 1 + \frac{t}{n} \langle x \rangle + \frac{t^2}{2n^2} \langle x^2 \rangle + \dots \quad (3.51)$$

$$= 1 + m_i \frac{t}{n} + \frac{\sigma_i^2 + m_i^2}{2} \frac{t^2}{n^2} + \dots \quad (3.52)$$

In order to make further progress we must recognise that in the limit $n \rightarrow \infty$ this becomes

$$M_{x_i}(t/n) = \exp \left(\frac{m_i t}{n} + \frac{\sigma_i^2 t^2}{2n^2} \right). \quad (3.53)$$

This can be verified by re-expanding the exponential in equation 3.53, keeping terms up

to and including $(t/n)^2$ and comparing the result with equation 3.52;

$$M_{x_i}(t/n) = \exp\left(\frac{m_i t}{n} + \frac{\sigma_i^2 t^2}{2n^2}\right) \quad (3.54)$$

$$= 1 + \left(\frac{m_i t}{n} + \frac{\sigma_i^2 t^2}{2n^2}\right) + \frac{1}{2} \left(\frac{m_i t}{n} + \frac{\sigma_i^2 t^2}{2n^2}\right)^2 + \dots \quad (3.55)$$

$$= 1 + m_i \left[\frac{t}{n}\right] + \frac{\sigma_i^2 + m_i^2}{2} \left[\frac{t}{n}\right]^2 + \mathcal{O}\left(\left[\frac{t}{n}\right]^3\right). \quad (3.56)$$

Finally, substituting the result from equation 3.53 back into the expression for $M_z(t)$ in equation 3.50, we find

$$M_z(t) \approx \prod_{i=1}^n \exp\left(\frac{m_i t}{n} + \frac{\sigma_i^2 t^2}{2n^2}\right) \quad (3.57)$$

$$\approx \exp\left(\frac{t \sum_{i=1}^n m_i}{n} + \frac{t^2 \sum_{i=1}^n \sigma_i^2}{2n^2}\right) \quad (3.58)$$

$$\approx \exp\left(t \langle z \rangle + \frac{1}{2} t^2 \text{Var}(z)\right), \quad (3.59)$$

where we have used the expressions for the expectation and variance of z proved in properties 1. and 2. respectively.

Comparing the expression for $M_z(t)$ in equation 3.59 to the form of the MGF for the Gaussian distribution derived in equation 3.38, we see that the MGFs match. Using the *uniqueness property of the MGF* discussed in section 3.3, it follows that in the limit $n \rightarrow \infty$, the random variable z tends to being Gaussian distributed with the means and variances required by properties 1. and 2. respectively.

Property 3. of the CLT is perhaps the main reason that the Gaussian distribution appears so often across so many scientific disciplines.

Example 3.5: Numerical demonstration of the CLT

Experiment with this using the example notebook [CLT.ipynb](#).

Let's draw $n = 100$ random x_i from Poisson distributions with different rate parameters, $\lambda_i = i/n$, where $i = 1, 2, \dots, (n-1)$, and check that $z = (\sum_i x_i)/n$ is

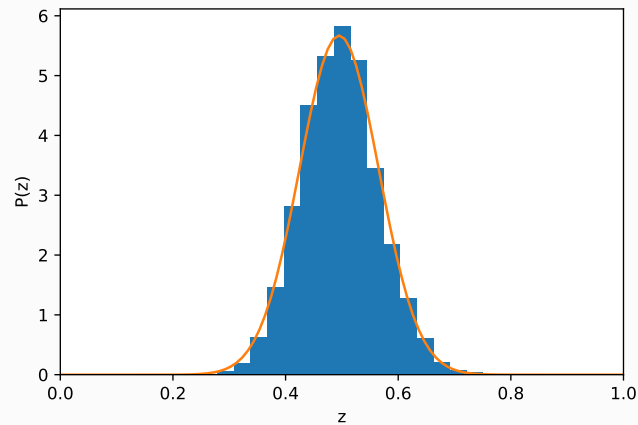
distributed as predicted by the CLT by drawing 10000 trial values for z .

```
n, trials = 100, 10000
lambdas = np.arange(n)/n
x = np.array([sp.poisson(1).rvs(size=trials) for l in lambdas])
z = np.sum(x, axis=0)/n

# plot the observed distribution
plt.hist(z, bins=20, density=True)

# plot the predicted distribution
mu, sigmasq = np.sum(lambdas)/n, np.sqrt(np.sum(lambdas)/n**2)
CLTdist = scipy.stats.norm(mu, sigmasq)
grid = np.linspace(0, 1, 100)
plt.plot(grid, CLTdist.pdf(grid))

plt.xlabel("z"); plt.ylabel("P(z)"); plt.show();
```



Chapter 4

Bayesian Parameter Estimation

The two main types of inference problems are model fitting and model selection. Model fitting, or **parameter estimation**, is the problem we face when we have a model containing several free parameters which claims to describe reality we have to determine the values of those parameters (and their associated uncertainties) from the available data. Model selection, sometimes known as **hypothesis testing**, is the problem we face when we have two or more competing models which claim to describe reality and we have to choose between them based on the data. Both problems can be tackled using either frequentist or Bayesian techniques, as we will see in this course. This chapter, as the name suggests, discusses Bayesian methods for parameter estimation.

4.1 Bayes' theorem

We encountered Bayes' theorem in chapter 2, see equation 2.21. Here our starting point will be Bayes' theorem written in the following form

$$P(M|D, I) = \frac{P(D|M, I)P(M|I)}{P(D|I)} . \quad (4.1)$$

Here M denotes our model (with particular values for all of its parameters) and D denotes the data. Every probability is conditional on I , this denotes any other prior information or assumptions that we use. The I is conceptually important and might well conceal some subtle assumption that has a profound effect on the results of the analysis; however, we often omit the I in all terms in equation 4.1.

Equation 4.1 is so important and occurs so often that we give names to each of the individual terms in this equation:

1. $P(D|M)$ is called the **likelihood**, it is the probability of obtaining the data given our model;
2. $P(M)$ is called the **prior**, it the probability we assign to our model being correct *before* performing the experiment;
3. $P(M|D)$ is called the **posterior**, it is the probability that our model is correct *after* performing the experiment;
4. $P(D)$ is called the **evidence**, it acts as a normalisation constant in Bayes' theorem. (The evidence isn't usually very important in parameter estimation problems, but is plays a central role in model selection). If the model has some free parameters, $M = \text{parameters}$, then from Bayes' theorem (4.1) we see that the evidence must be equal to the following integral;

$$P(D) = \int d(\text{parameters}) P(\text{parameters}) P(D|\text{parameters}) . \quad (4.2)$$

In a Bayesian inference problem we are generally given the likelihood and the prior as a starting point and it will be our job to compute the posterior and, in some instances, the evidence.

Before we see some examples of Bayes' theorem in action let's introduce some useful notation and a handy trick for estimation the width, or standard deviation, of a distribution.

4.1.1 Some Notation

It will be convenient to introduce the following notation to help concisely represent distributions with fixed upper (b) and lower ($a < b$) limits. We define

$$\sqcap_a^b(x) = \begin{cases} 1 & \text{if } x > a \text{ and } x < b \\ 0 & \text{else} \end{cases} . \quad (4.3)$$

Note that this is not a probability distribution, we have deliberately left off the normalisation. This is not a standard notation, but will be used throughout these notes.

4.1.2 A Handy Trick: estimating the uncertainty

We defined the concepts of the *variance* $\text{Var}(X)$ and *standard deviation* σ_X in chapter 3. These can be used to describe the “width” of a posterior distribution; i.e. if $X \sim P(X|D)$, and the posterior has a peak at $X = \hat{X}$ then we might say that we can measure the parameter $X = \hat{X} \pm \sigma_X$. However, finding the variance and/or standard deviation requires evaluating an integral. It is therefore useful to have a quick method for estimating the width of a posterior distribution.

Consider the expression for the PDF of a Gaussian distribution,

$$P(y) = \frac{1}{\sqrt{2\pi\sigma_y^2}} \exp\left(-\frac{(y - \mu_y)^2}{2\sigma_y^2}\right). \quad (4.4)$$

By taking the logarithm and differentiating twice with respect to y we can show that standard deviation σ_y given by

$$\sigma_y = \frac{1}{\sqrt{-\frac{d^2}{dy^2} \log P(y)}}. \quad (4.5)$$

Note that it does not matter at what value of y we evaluate the second derivative.

For any probability distribution $P(Y)$ that is *approximately* Gaussian we can use equation 4.5 to find a quick analytic estimate of the standard deviation. We take the logarithm of the PDF, differentiate twice, and evaluate this at the location of the peak in the PDF;

$$\sigma_Y \approx \frac{1}{\sqrt{-\frac{d^2}{dY^2} \log P(Y)|_{\text{peak}}}}. \quad (4.6)$$

This can be thought of as a quadratic Taylor series approximation to $\log P(Y)$ around the peak of the distribution. This is a special case of the *Laplace approximation*, see section 5.3.

4.2 Single Parameter Models

In this section we will see a number of examples of the application of Bayes’ theorem to parameter estimation problems where the model contains just a single parameter. The results of these calculations are fairly simple, your focus should therefore be on the methodology

which will be applied to more complicated problems in the next section. This section will also highlight the role of the prior in Bayesian analysis.

Example 4.1: Mean of a Gaussian Process

Suppose we are measuring the brightness X of a source. We have N measurements $\{x_1, x_2, \dots, x_N\}$ of the source each with a Gaussian error, or uncertainty, σ . What is our best estimate of the source brightness and the corresponding uncertainty?

We start by rewriting Bayes' theorem for this problem,

$$P(X|\{x\}, \sigma) = \frac{P(\{x\}|X, \sigma)P(X|\sigma)}{P(\{x\}|\sigma)}. \quad (\text{i})$$

Here σ plays the role of I in equation 4.1; we assume that all the measurements have the same Gaussian uncertainty with known standard deviation.

We have to choose a prior. This choice should be based on what we know (or don't know) about the source *before* we start taking measurements. In this case we know very little except that $X > 0$ (negative luminosities are impossible). We might therefore want our prior to treat all positive numbers equally, but this poses a normalisation problem (the prior is after all a probability distribution and should be properly normalised). In practice there is often some ambiguity as to what prior to use. Here we will use a uniform prior,

$$P(X|\sigma) = \frac{1}{X_{\max} - X_{\min}} \mathbb{I}_{X_{\min}}^{X_{\max}}(X), \quad (\text{ii})$$

and take care to choose the limits $X_{\min/\max}$ to be comfortably below/above the smallest/largest measurements in the list $\{x_i\}$.

From the information given in the question, the likelihood has a Gaussian form,

$$P(\{x\}|X, \sigma) = \mathcal{L}(X) = \prod_{i=1}^N \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x_i - X)^2}{2\sigma^2}\right). \quad (\text{iii})$$

Notice that the prior is normalised with respect to X , but the likelihood is not. This is as it should be; the prior is a probability distribution over X but the likelihood is a probability distribution over the data $\{x_i\}$.

Putting this all together using Bayes' theorem we find the posterior

$$P(X|\{x\}, \sigma) = \frac{1}{Z(X_{\max} - X_{\min})} \left(\frac{1}{\sqrt{2\pi\sigma^2}} \right)^N \exp \left(\sum_{i=1}^N \frac{-(x_i - X)^2}{2\sigma^2} \right) \mathbb{I}_{X_{\min}}^{X_{\max}}(X), \quad (\text{iv})$$

where the evidence is given by

$$Z \equiv P(\{x\}|\sigma) = \int_{X_{\min}}^{X_{\max}} dX \frac{(2\pi\sigma^2)^{-N/2}}{(X_{\max} - X_{\min})} \exp \left(\frac{-(x_i - X)^2}{2\sigma^2} \right). \quad (\text{v})$$

Thankfully, we do not need to actually calculate Z for this problem.

From the posterior in equation [iv](#) we can find the most likely (i.e. mode) value of the luminosity which we will denote $\hat{X}$. This is, by definition, the value that maximises the posterior,

$$\left. \frac{dP(X|\{x_i\}, \sigma)}{dX} \right|_{X=\hat{X}} = 0. \quad (\text{vi})$$

Because the prior is flat this is the same as finding the maximum of the likelihood, $\mathcal{L}(X)$. And, because the logarithm is a monotonic function, this is the same as finding the maximum of $\log \mathcal{L}(X)$,

$$\left. \frac{d \log \mathcal{L}(X)}{dX} \right|_{X=\hat{X}} = 0. \quad (\text{vii})$$

For our problem

$$\frac{d \log \mathcal{L}(X)}{dX} = \sum_{i=1}^N \frac{x_i - X}{\sigma^2}, \quad (\text{viii})$$

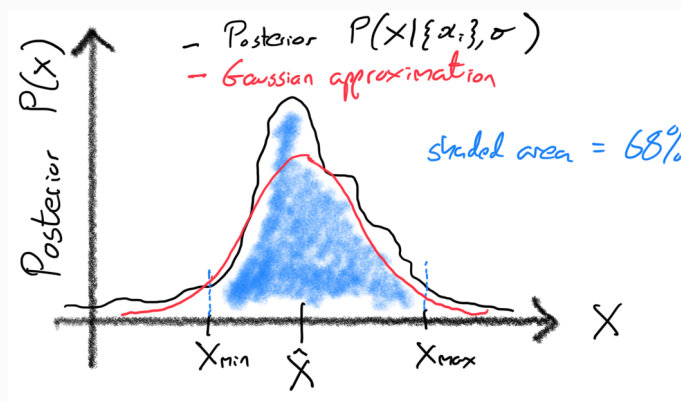
$$\Rightarrow \hat{X} = \frac{1}{N} \sum_{i=1}^N x_i. \quad (\text{ix})$$

We have just shown that the most likely value of X is the mean value of all our measurements; this result should not be a surprise.

Note that our result for $\hat{X}$ does not depend on the value of σ .

Now what about the uncertainty on this estimate? We could calculate the posterior numerically from equation [iv](#) and find the range of X that captures 60%, say, of the probability; see sketch plot. Or we could integrate the posterior numerically to find the variance (defined in equation [3.11](#)) and, after taking the square root, use

the standard deviation as a measure of the uncertainty. However, while both of these are possible, we will instead make the assumption (an excellent one it turns out!) that the posterior can be approximated as a Gaussian and we can make an analytical estimate of standard deviation.



We have (see discussion in section 4.1.2 and equation 4.6)

$$\Delta X \approx \frac{1}{\sqrt{-\frac{d^2}{dX^2} \big|_{X=\hat{X}} \log P(X|\{x_i\}, \sigma)}}. \quad (\text{x})$$

From equation iv we see that the log-posterior is quadratic in X ,

$$\log P(X|\{x_i\}, \sigma) = \text{const} - \sum_{i=1}^N \frac{(x_i - X)^2}{2\sigma^2}. \quad (\text{xi})$$

Evaluating the second derivative with respect to X and substituting into x gives

$$\Delta X = \sigma / \sqrt{N}. \quad (\text{xii})$$

Again, this result should not come as a surprise...the error decreases with the square root of the number of measurements.

Exercise 4.1: Variable errors

Consider again example 4.1, but now assume that each measurement x_i has a different Gaussian uncertainty σ_i , for $i = 1, 2, \dots, N$. Repeat the analysis

to show that the most likely value for X is the now weighted mean of the measurements,

$$\hat{X} = \frac{\sum_{i=1}^N \frac{x_i}{\sigma_i^2}}{\sum_{i=1}^N \frac{1}{\sigma_i^2}}, \quad (\text{i})$$

and the uncertainty on this value is

$$\Delta X = \left(\sum_{i=1}^N \frac{1}{\sigma_i^2} \right)^{-1/2}. \quad (\text{ii})$$

Example 4.2: Poisson process; asymmetric one parameter problem

We want to measure the density (i.e. number per square degree) of bright stars, S , on a particular patch of sky. We make a single observation and find 5 stars in an area 1 deg^2 . What is our posterior PDF on the density S ?

Let $n_1 = 5$ stars. We start by rewriting Bayes' theorem for this problem,

$$P(S|n_1) = \frac{P(n_1|S)P(S)}{P(n_1)}. \quad (\text{i})$$

We have to choose a prior on S ; here we use a uniform prior,

$$P(S) = \frac{1}{S_{\max}} \mathbb{I}_0^{S_{\max}}(S). \quad (\text{ii})$$

Assuming stars appear on the sky independently and with a constant density across the area of interest the likelihood is a Poisson distribution,

$$P(n_1|S) = \frac{\exp(-S)S^{n_1}}{n_1!}. \quad (\text{iii})$$

Putting all the ingredients together using Bayes' theorem we find the posterior

$$P(S|n_1) = \frac{1}{Z S_{\max}} \frac{\exp(-S)S^{n_1}}{n_1!} \mathbb{I}_0^{S_{\max}}(S), \quad (\text{iv})$$

where the evidence is given by

$$Z \equiv P(n_1) = \int_0^{S_{\max}} dS \frac{1}{S_{\max}} \frac{\exp(-S) S^{n_1}}{n_1!}. \quad (\text{v})$$

Again, we do not need to actually calculate the evidence for this problem.

This distribution is asymmetric; try plotting it for a sensible choice of $S_{\max}$!

The mode, $\hat{S}$, is the peak value of the distribution. This is most easily found by maximising the logarithm of the posterior probability,

$$\left. \frac{d}{dS} \right|_{S=\hat{S}} \log P(S|n_1) = 0 \quad \Rightarrow \quad \hat{S} = n_1 = 5. \quad (\text{vi})$$

Perhaps unsurprisingly, our best estimate for the density is just n_1 . However, it is worth noting that (because the posterior is asymmetric) the mean is larger; in the limit that $S_{\max} \rightarrow \infty$ you can show that

$$\langle S \rangle = \int dS S P(S|n_1) = 6. \quad (\text{vii})$$

4.2.1 Asymmetric errors

In example 4.2 we encountered a posterior distribution on the parameter S which is asymmetric about its peak. We want to quantify the uncertainty, or width, of this distribution so we can quote an error on our measurement of the density, S .

We could calculate the standard deviation σ_S of the posterior PDF on the density. Alternatively, we could use the analytic method described in section 4.1.2 to estimate σ_S , as we did in example 4.1. In either case we could then write our measurement as $S = \hat{S} \pm \sigma_S$. However, neither of these approaches is suitable here as they fail to capture the fact that our posterior PDF on S is asymmetric about the peak.

How should we proceed? There is no unique solution. If we want to quote a 68% uncertainty figure there are at least three natural possibilities we could consider:

1. draw a horizontal line that intersects the posterior at two values which enclose 68% of the probability;

2. choose two values that loose 16% of the probability in each tail, leaving 68% of the probability enclosed between them;
3. find the two values which have smallest possible range of the parameter that has 68% of the probability enclosed.

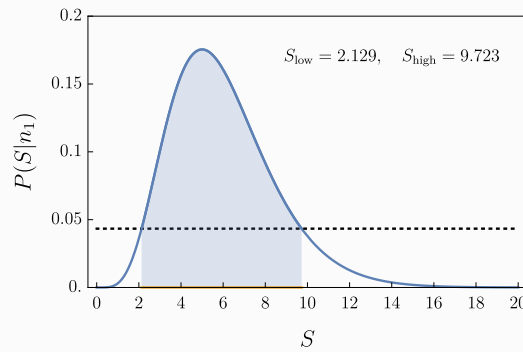
All three of these methods allow us to define an asymmetric error on the parameter S and we can write our measurement in the form $S = \hat{S}_{-\delta}^{+\Delta}$. All of these methods gives different answers. All of these methods have their pros and cons and all of these methods gives a suitable notion of the uncertainty when applied to the posterior in 4.2; see example 4.3.

The takeaway from this discussion is that the posterior PDF on a parameter can, in general, have a complicated shape and it is not always possible to adequately describe it succinctly in the form $a = b \pm c$, or even $a = b_{-d}^{+c}$. There is no substitute for plotting the full PDF.

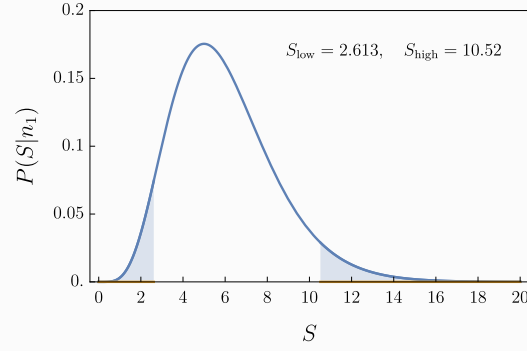
Example 4.3: Poisson process continued

Continuing from example 4.2. We now try to describe the uncertainty on S using each of the three method described in section 4.2.1. Recall that $n_1 = 5$ and we here take $S_{\max} = 100$.

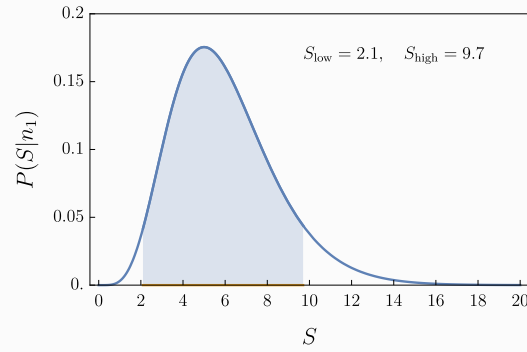
Method 1 the blue region between S_{low} and S_{high} encloses 90% of the probability.



Method 2 each blue tail beyond S_{low} and S_{high} encloses 5% of the probability.



Method 3 the shaded region is the narrowest possible region that encloses 90% of the probability.



Each of these three methods yield a satisfactory $(S_{\text{low}}, S_{\text{high}})$ that capture some notion of a 90% error.

4.2.2 Including Additional data

Suppose we have a model that contains a parameter x . We initially have a prior $P(x)$ on this parameter. We make some measurements (or perform an experiment) to get some data D_1 . The likelihood for this experiment is $P(x|D_1)$. Bayes' theorem, allows us to calculate the posterior for this parameter;

$$P(x|D_1) = \frac{P(D_1|x)P(x)}{P(D_1)}. \quad (4.7)$$

Hopefully we have learned something useful from our experiment and our posterior is different from our prior.

But we are not satisfied. We want a better measurement of x . So we make more measurements and get some new data, D_2 . We could re-perform the Bayesian analysis from scratch. As before, we say our prior $P(x)$, but now our likelihood is $P(x|D_1, D_2)$ (remembering that we are using the notation $A, B = A \text{ and } B$). Using Bayes' theorem, our new posterior for the parameter is

$$P(x|D_1, D_2) = \frac{P(D_1, D_2|x)P(x)}{P(D_1, D_2)}. \quad (4.8)$$

Hopefully we have learned something useful from our second experiment too and our second posterior is different from our first.

If we keep making more observations D_3, D_4, D_5 and so on, it will become tiresome to keep performing the entire Bayesian analysis from scratch each time. Instead, we want a more efficient way of continually updating our state of belief as new data becomes available.

To achieve this, we might say that the likelihood for the second experiment is just $P(D_2|x)$. We will incorporate the information from our first experiment by using $P(x|D_1)$ as a prior for the second experiment. Using Bayes' theorem, we arrive at the following updated posterior after the second experiment,

$$P(x|D_1, D_2) = \frac{P(D_2|x)P(x|D_1)}{P(D_2)}. \quad (4.9)$$

Substituting for $P(x|D_1)$ from equation 4.8 gives

$$P(x|D_1, D_2) = \frac{P(D_2|x)P(D_1|x)P(x)}{P(D_2)P(D_1)}. \quad (4.10)$$

If our two experiments D_1 and D_2 are independent then the probabilities multiply to give

$$P(x|D_1, D_2) = \frac{P(D_1, D_2|x)P(x)}{P(D_1, D_2)}, \quad (4.11)$$

which is exactly the result that in equation 4.8. We have shown that instead of re-performing the entire Bayesian analysis from scratch we can instead update our posterior when obtain more data by using the previous posterior as the new prior. Of course, we don't have to stop at one iteration. We can keep making more measurements, each time using the previous posterior as a prior and finding an updated posterior. If our experiments are good and yield useful information then the posterior should keep getting narrower as

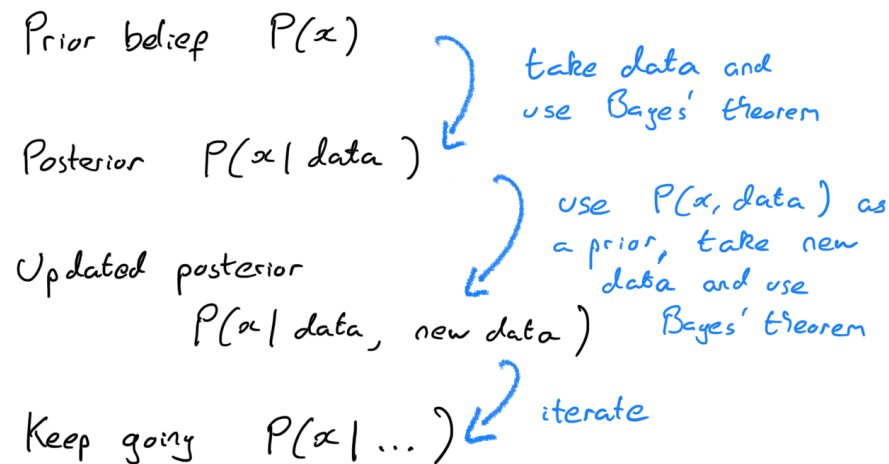


Figure 4.1: A metaphor for the scientific method?

we get an ever-better measurement. There is no limit to how many times we can use this updating procedure. See figure 4.1.

If you were philosophically inclined, and prone to grandiose generalisations, you might see this mathematical procedure using Bayes' theorem as an analogy for the whole the scientific method where we are continually learning by performing more experiments and observations.

The discussion in this section is quite formal. Example 4.4 provides a concrete example of this updating procedure and demonstrates that it is equivalent to re-performing the analysis from scratch with the original prior and the combined data D_1 and D_2 .

Example 4.4: Poisson process continued further

Continuing from example 4.2. Suppose that we observe a second 1 deg^2 patch of sky and find n_2 stars. What is our new posterior PDF on the density S ?

We can use our previous result as a prior and update our posterior with the new

data.

$$P(S|n_1, n_2) = \frac{P(n_2|S)P(n_1|S)}{Z}, \quad (\text{i})$$

where Z is the normalising evidence that we don't need to calculate here. We have

$$P(S|n_1, n_2) = \frac{\exp(-S)S^{n_2} \times \exp(-S)S^{n_1} \prod_0^{S_{\max}}(S)}{\text{normalisation}}, \quad (\text{ii})$$

$$\propto \exp(-2S)S^{n_1+n_2} \prod_0^{S_{\max}}(S), \quad (\text{iii})$$

where we have discarded normalisation constants that do not depend on S .

Note that this is the same result we would have found if we had used our original prior and done both observations simultaneously finding $n_1 + n_2$ in a patch of sky with area 2 deg^2 .

Exercise 4.2: A Large Number of Observations

Consider the calculation in example 4.4. What happens to the posterior PDF on S in the limit that we make many additional observations? I.e. $n_3, n_4, \dots$

4.2.3 Ignorance Priors

So far we have used flat, or uniform, priors in all of our examples. The motivation for this was that it was a good representation of our ignorance about the problem before starting observations. However, when dealing with positive scale parameters (such as the density S in example 4.2) we might want to instead regard our ignorance as giving equal weight to each order of magnitude. I.e. we say S has the same probability of being in the range $1 < S < 10$ as it does of being in the range $10 < S < 100$.

Question. What prior achieves this?

Answer. One that is flat, or uniform, in $\log S$ rather than S .

Changing variables in probability distributions must be done with care. Always remember that it is a probability *density* and $P(x)dx$ is the probability. Suppose we want to change to some new variable $f(x)$ which is a function of our original x ; we must equate

$$P(x) dx = P(f) df . \quad (4.12)$$

Applying this to the logarithm function, $f(x) = \log x$, gives

$$P(\log x) d \log x = P(x) dx \quad (4.13)$$

$$\Rightarrow P(x) = P(\log x) \frac{d \log x}{dx} \quad (4.14)$$

$$\Rightarrow P(x) = \frac{P(\log x)}{x} . \quad (4.15)$$

So if we want a prior that is flat in $\log x$, i.e. $P(\log x) = \text{const}$, then we must have $P(x) \propto \frac{1}{x}$. If we apply upper and lower limits, and correctly normalise this distribution by multiplying with a constant A , then we end up with the following distribution known as *Jeffreys Prior*;

$$P(x) = \frac{A}{x} \square_{x_{\min}}^{x_{\max}}(x) , \quad (4.16)$$

where $A_{\min} = B_{\min} = 1$ and $A_{\max} = B_{\max} = 100$.

Exercise 4.3: Normalising Jeffreys Prior

Show that $A = 1/\log(x_{\max}/x_{\min})$ in equation 4.16.

Jeffreys prior is uniform in $\log x$. We can compare this with a prior that is uniform in x ; see figure 4.2.

Exercise 4.4: The Impact of the Choice of Prior

Repeat the calculation from example 4.2 using Jeffreys prior between $S_{\min} = 0.1$ and $S_{\max} = 100$. Compare the posterior with that found previously. Has the mode value $\hat{S}$ shifted up or down?

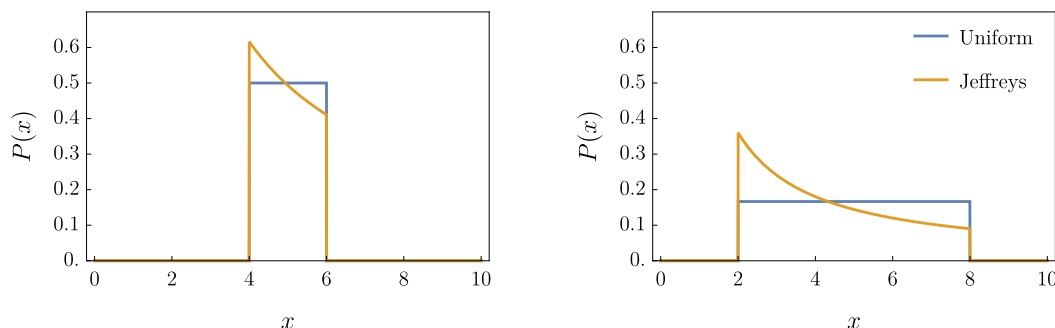


Figure 4.2: A comparison of the uniform and Jeffreys priors over the ranges $4 < x < 6$ (left) and $2 < x < 8$ (right). Note that over narrower ranges there is less of a difference between the two distributions. We often choose to use Jeffreys prior when the parameter can take values spanning a very wide range (perhaps several orders of magnitude).

4.3 Multi Parameter Models

In the previous section we saw Bayes' theorem applied to parameter estimation problems where the model contained a single parameter (i.e. X or S). In this section we encounter models with multiple parameters. The approach will be largely the same but the posterior distribution that we find can now have a much wider range of more complicated structures.

Our previous examples were also “simple” in the sense that our data consisted of a small number of discrete measurements; e.g. a list of measurements $\{x_1, x_2, \dots, x_n\}$ or simply counting n_1 stars in a particular area of sky. In this section we will encounter examples with more complicated dataset such as *time-series* or *spectral* data.

4.3.1 Time Series, or Spectral Data

Suppose we have a quantity $d(\lambda)$ where λ is a smoothly varying parameter such as time, t , or frequency, f . We measure the quantity d at several discrete values of λ_i , so our data is

$$D = \{d_i | i = 1, 2, \dots, n\} = \{d(\lambda_i) | i = 1, 2, \dots, n\}. \quad (4.17)$$

We have a model, M , for this process. The model is a continuous function of the parameter λ , and it can also depend on some other parameters which we will call θ . The model will

be written $M(\lambda; \theta)$. The model will be evaluated at the discrete set of values λ_i so that it can be compared with the data;

$$M_i(\theta) = M(\lambda_i; \theta). \quad (4.18)$$

Bayes' theorem for this particular problem is then written as

$$P(M|D) = \frac{1}{Z} P(D|M) P(M), \quad (4.19)$$

or

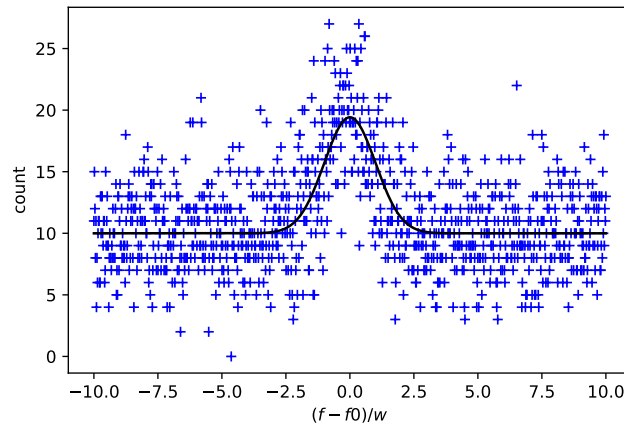
$$P(\theta|D) = \frac{1}{Z} P(D|\theta) P(\theta), \quad (4.20)$$

where the evidence is given by the integral

$$Z = P(D) = \int d\theta P(D|\theta) P(\theta), \quad (4.21)$$

Example 4.5: Spectrum Containing Gaussian Pulse

We measure a spectrum at 1000 regularly spaced frequencies, f_i . At each frequency we record a integer count (e.g. number of photons). The data are plotted below.



Our model for this data will be a constant background level plus a Gaussian pulse;

i.e. the mean number of counts expected in the bin f_i is

$$M(f_i; A, B, f_0, \sigma) = B + A \exp\left(-\frac{(f_i - f_0)^2}{2\sigma^2}\right). \quad (\text{i})$$

This model has 4 parameters: the background level B , the pulse amplitude A , the pulse spectral width σ , and the pulse central frequency f_0 . However, to simplify things a bit here we assume that f_0 and σ are already known (the data in the above figure are plotted assuming these known values).

Bayes' theorem for this problem is

$$P(A, B|D, \sigma, f_0) = \frac{1}{Z} P(D|A, B, \sigma, f_0) P(A, B|\sigma, f_0). \quad (\text{ii})$$

Assuming that the counts in each bin are independent and Poisson distributed with means M_i , the likelihood is given by

$$P(D|A, B, \sigma, f_0) = \prod_{i=1}^n \frac{\exp(-M_i) M_i^{d_i}}{d_i!}, \quad (\text{iii})$$

where $M_i = M(f_i; A, B, \sigma, f_0)$ and $n = 1000$.

We must now choose a prior $P(A, B|\sigma, f_0)$. We could use a flat prior on both A and B separately and then multiply the two distributions to get a suitable 2-dimensional prior. However, as both parameters are positive scale parameters we will here use Jeffreys priors on both A and B ;

$$P(A, B|\sigma, f_0) = \frac{A^{-1}}{\log(A_{\max}/A_{\min})} \frac{B^{-1}}{\log(B_{\max}/B_{\min})} \prod_{A_{\min}}^{A_{\max}} (A) \prod_{B_{\min}}^{B_{\max}} (B), \quad (\text{iv})$$

where $A_{\min} = B_{\min} = 1$ and $A_{\max} = B_{\max} = 100$.

The evidence, $Z = P(D|\sigma, f_0)$, is given by the following integral;

$$Z = \int_{A_{\min}}^{A_{\max}} dA \int_{B_{\min}}^{B_{\max}} dB P(D|A, B, \sigma, f_0) P(A, B|\sigma, f_0). \quad (\text{v})$$

We will not actually need to evaluate this integral for this problem.

We can numerically find the maximum of the posterior. Looking at the data plotted in the above figure a sensible initial guess for the best fitting parameters is

$A_0 = B_0 = 10$. After optimising, we find the following *maximum a posteriori* (i.e. mode) values for the parameters:

$$\hat{A} = 9.423, \quad \hat{B} = 9.997. \quad (\text{vi})$$

The model evaluated at these parameters, i.e. the maximum a posteriori model,

$$M(f; \hat{A}, \hat{B}, \sigma, f_0), \quad (\text{vii})$$

is plotted as a function of frequency on top of the data in the figure above.

4.3.2 Separable Priors

In the Gaussian pulse example 4.5 we used a multiparameter prior distribution on the two parameters A and B . In that example we used the product of two Jeffreys priors on each of the parameters. Another obvious choice would be the product of two uniform priors on each of the parameters;

$$P(A, B | \sigma, f_0) = \frac{1}{A_{\max} B_{\max}} \prod_0^{A_{\max}} (A) \prod_0^{B_{\max}} (B). \quad (4.22)$$

Both of these are examples of *separable* priors. A distribution on several parameters $\{x_1, x_2, \dots, x_n\}$ is said to be separable if it can be written in the form

$$P(x_1, x_2, \dots, x_n) = P(x_1) \times P(x_2) \times \dots \times P(x_n). \quad (4.23)$$

Separable priors make sense in the situation where our prior belief on one parameter is independent from our prior belief on another - so it makes sense to multiply the probabilities. In practice, separable priors may also be used because they are simple to write down and to work with. It also makes it particularly simple to modify the prior on one parameter without interfering with the others, should we wish to do so.

However, priors do not have to be separable. There are situations where other choices might be appropriate. For example, consider fitting temperature T and location data across the Earth. Use latitude and longitude coordinates (λ, ϕ) on the Earth. The temperature at any location might vary during the day by a certain amount σ , but the mean temperature μ depends on the latitude. Therefore, we might choose a uniform prior across the surface area of the Earth (not uniform in the angles λ, ϕ) multiplied by a Gaussian prior on the

temperature where the mean is a function of latitude;

$$P(T, \lambda, \phi) = \frac{\exp\left(\frac{-(T - \mu[\lambda])^2}{2\sigma^2}\right)}{\sqrt{2\pi\sigma^2}} \times \frac{\cos \lambda}{\pi} \varpi_{-\pi/2}^{+\pi/2}(\lambda) \times \frac{1}{2\pi} \varpi_0^{2\pi}(\phi). \quad (4.24)$$

This distribution is not separable.

4.3.3 Marginalisation

It is often the case that some of the model parameters are of little interest to us. These are known as “nuisance” parameters. (For example, in our Gaussian pulse example, the background B might be an example of a nuisance parameter.) We would like to somehow factor these out of the problem. This can be achieved by integrating out, or *marginalising*, these parameters from the posterior distribution. From the example 4.5, we can construct the *1-dimensional marginal posterior* on the amplitude parameter A ,

$$P(A|D, \sigma, f_0) = \int dB P(A, B|D, \sigma, f_0). \quad (4.25)$$

Similarly, if we wish to, we can find the 1-dimensional marginal posterior on the background parameter B ,

$$P(B|D, \sigma, f_0) = \int dA P(A, B|D, \sigma, f_0). \quad (4.26)$$

In problems with more parameters, say $m + n$ parameters $\{\theta_1, \theta_2, \dots, \theta_n, \phi_1, \phi_2, \dots, \phi_m\}$, you can marginalise over any number m of the parameters in the posterior as you want to leave a n -dimensional marginal posterior distribution;

$$P(\theta_1, \theta_2, \dots, \theta_n | \text{data}) = \int d^m \phi P(\theta_1, \theta_2, \dots, \theta_n, \phi_1, \phi_2, \dots, \phi_m | \text{data}). \quad (4.27)$$

Note, if we marginalise (integrate) over *all* of the parameters in the posterior distribution we are left with a constant which we identify as the Bayesian evidence;

$$Z \equiv P(\text{data}) = \int d^n \theta \int d^m \phi P(\theta_1, \theta_2, \dots, \theta_n, \phi_1, \phi_2, \dots, \phi_m | \text{data}). \quad (4.28)$$

For this reason you will sometimes find the *evidence* referred to by the alternative name *marginal likelihood* (see equation 4.2).

4.3.4 Plotting Multivariate distributions

In multivariate parameter estimation problems we naturally end up with a multivariate posterior probability distribution. It can be difficult to interpret such distributions. A good first step is to find the peak of the distribution (i.e. find the mode parameters), this is usually relatively easy to do numerically (usually by maximising the log-probability, see example 4.5). However this gives us no indication of the possible range of values the parameters can take, i.e. the uncertainty.

One way of investigating the uncertainty is to focus on one parameter at a time. For each parameter we can calculate the 1-dimensional marginal posterior distribution (see section 4.3.3) and plot these. However this can hide important information. Consider the distribution

$$P(x, y) = \exp \left(\frac{-1}{2} (x \ y) \cdot \Sigma^{-1} \cdot \begin{pmatrix} x \\ y \end{pmatrix} \right), \quad (4.29)$$

where $\Sigma = \begin{pmatrix} 1 & 9/10 \\ 9/10 & 1 \end{pmatrix}.$

This distribution, and its marginal distributions, are plotted in figure 4.3. The parameters x and y are correlated; we say they are *partially degenerate*¹. One combination of the parameters can be measured much more accurately than the orthogonal combination (the ellipse has one large and one small eigenvalue). But this fact is hidden from us if we only look at the marginal distributions which make both errors look the same.

Example 4.6: Contour plots

Consider again fitting the Gaussian pulse in example 4.5.

We want to plot contours of the posterior $P(A, B|D, \sigma, f_0)$. If the posterior PDF is correctly normalised, then we can choose contours as *confidence intervals*; i.e. draw contours that enclose a particular fraction of the total probability (e.g. 68.3%, 95.4%, and 99.7%).

However, we do not always have the luxury of a correctly normalised posterior. This is the case for our posterior $P(A, B|D, \sigma, f_0)$. In this case can instead find the

¹The parameters would be described as *totally degenerate* if one of the principal axes of the ellipse tended to an infinite radius.

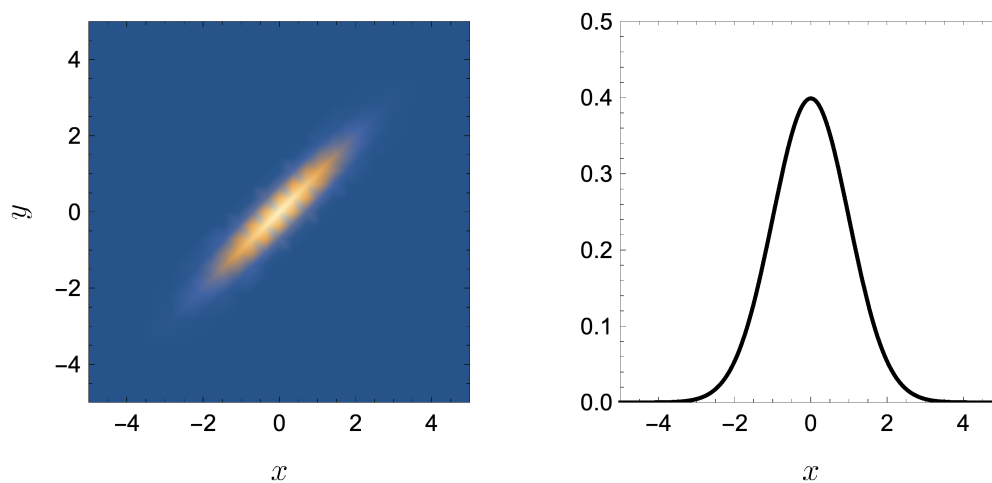
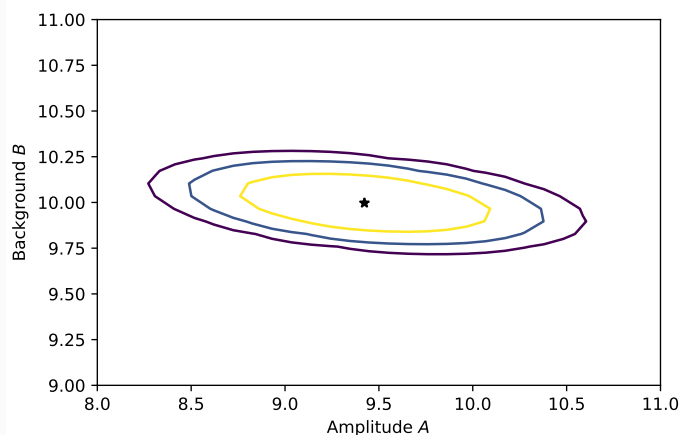


Figure 4.3: Left: density plot of the 2-dimensional distribution $P(x, y)$ from equation 4.29. Right: the 1-dimensional marginal distribution on the parameter x (the y marginal distribution looks identical). The 1-dimensional marginal distributions make it look like the errors are large ($\sim \pm 2$) on both parameters, but this hides the fact that there exists a combination of x and y that are measured with a much smaller error; this is visible in the narrow ellipse in the density plot.

peak value of the PDF (which was already done in example 4.5) and then draw contours which are below the peak value by a certain factor (e.g. factors of e^{-1} , e^{-2} and e^{-3}). This is done in the figure where $\star$ indicates the peak location $(\hat{A}, \hat{B})$.



In this problem we see that the parameters A and B are only weakly (negatively) correlated; the ellipse is only slightly misaligned from the coordinate axes.

If we have two parameters it is best to look at the full distribution in all its 2-dimensional glory by plotting a *contour* or *density* plot; this reveals extra structure not visible in the 1-dimensional marginal posteriors. But what if we have more than 2 or 3 parameters?

In higher dimensional problems it is necessary to resort to a *corner plot* to visualise the distribution. Consider the distribution

$$P(x, y, z) = \exp \left(\frac{-1}{2} (x \ y \ z) \cdot \Sigma^{-1} \cdot \begin{pmatrix} x \\ y \\ z \end{pmatrix} \right), \quad (4.30)$$

where $\Sigma = \begin{pmatrix} 1 & 9/10 & 0 \\ 9/10 & 1 & 0 \\ 0 & 0 & 10 \end{pmatrix}.$

The essential idea is that (because paper is 2-dimensional) we will plot every possible 2-dimensional and every possible 1-dimensional marginal distribution. Because this leads to a large number of plots we arrange them in a square in such a way that they share as many axes as possible, this helps to make the whole ensemble of plots (reasonably) easy to read. (There are libraries for producing such plots easily, see section 4.3.5.) The corner plot for the distribution in equation 4.30 is shown in figure 4.4. If you have not seen a corner plot before it worth spending some time staring at this figure to make sure you understand what is being plotted in each panel.

Exercise 4.5:

Explain why a corner plot is *not* the same as plotting “slices” through the distribution. It is of course possible to plot slices through the distribution, why would this be less useful than a corner plot?

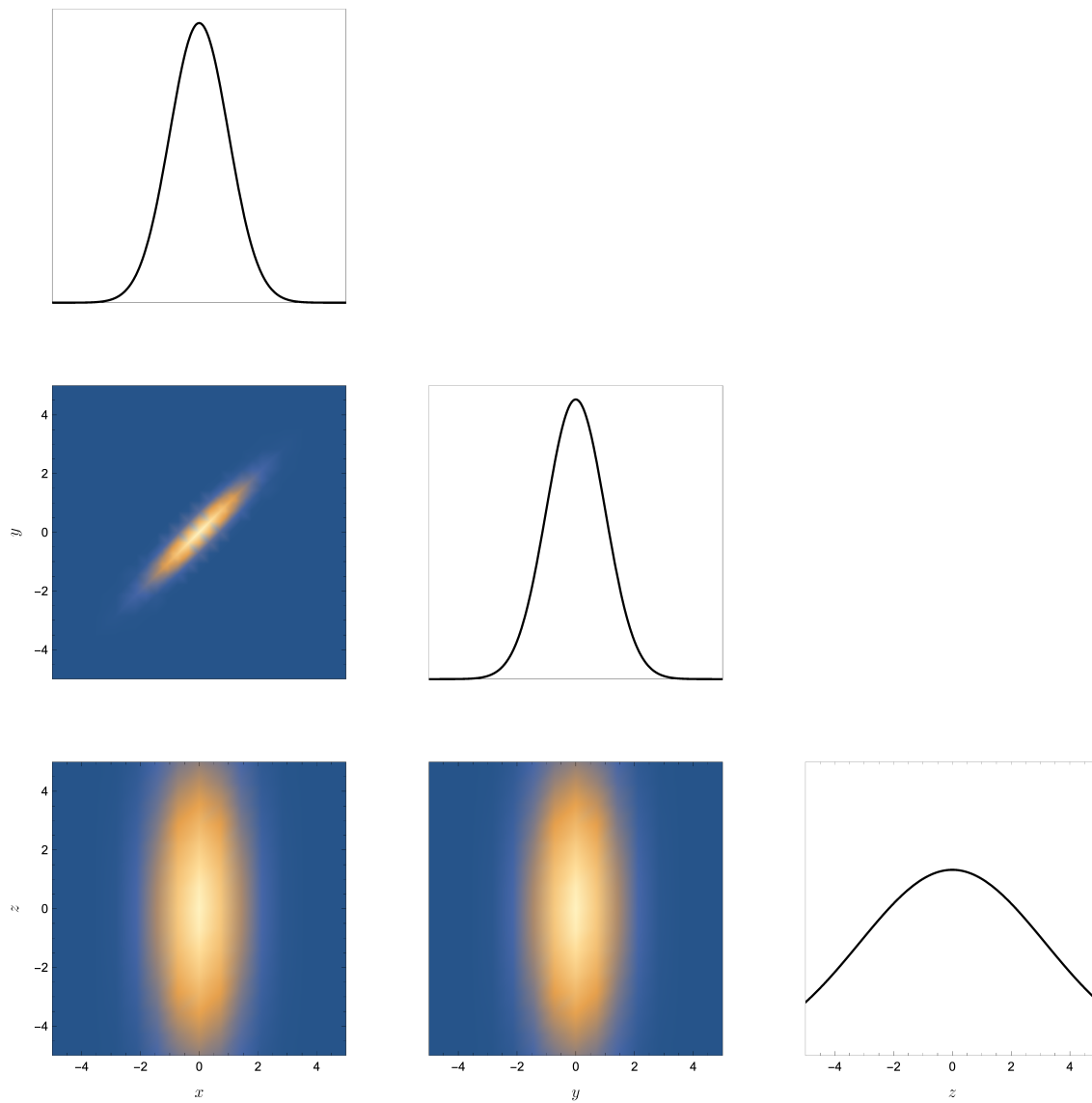


Figure 4.4: The corner plot for the distribution in equation 4.30.

4.3.5 Computational Concerns

We have now seen examples of multivariate parameter estimation problems. In order to study the shape of the posterior probability distribution we evaluated this function on a square grid of points and used the resulting values to plot a contour or density plot. The choice of the range and spacing of points in the grid was fairly arbitrary, and was mostly guided by our intuition of where we thought the posterior was going to be peaked.

This procedure quickly becomes impractical as the number of dimensions increases. How many points do you need to describe the shape of a 1-dimensional bell curve to a sensible accuracy? Maybe 10; the answer of course depends of your definition of “sensible”. So for a standard 2-dimensional Gaussian you might think a grid of points $10 \times 10 = 100$ would suffice. And you might think that in n -dimensions a grid of 10^n would suffice. Is clear that the size of our grid is quickly getting out of hand! But worse, a regular grid constructed in this way actually gets worse at describing the shape of a standard n -dimensional Gaussian as n increases (despite the huge number of points). The reason is that most of the points always end up in regions where the probability is extremely small and our grid fails to adequately cover the region of high probability density.

Example 4.7: Volume of the n -ball

The following geometric calculation, while not strictly relevant, may help illustrate why square grids are inefficient for high-dimensional probability distributions.

In a n -dimensional Euclidean space, the unit n -ball is the set of points with $x_1^2 + x_2^2 + \dots + x_n^2 < 1$. The volume (technically n -volume) of the n -ball is

$$\mathcal{B}_n = \frac{\pi^{n/2}}{\Gamma\left(\frac{n}{2} + 1\right)}. \quad (\text{i})$$

Surprisingly, this is largest when $n = 5$ and tends quickly to zero as $n \rightarrow \infty$.

Suppose we wish to completely cover the unit n -ball with an n -cube. The side lengths of the cube will have to be 2, so the volume of the n -cube is

$$\mathcal{C}_n = 2^n. \quad (\text{ii})$$

The n -cube completely covers the n -ball, but how much is wasted? I.e. how much of the n -cube's volume is in the corners that poke out beyond the n -ball? The useful fraction of the n -cube's volume is

$$\mathcal{F}_n = \frac{\mathcal{B}_n}{\mathcal{C}_n} = \frac{\pi^{n/2}}{\Gamma(\frac{n}{2} + 1) 2^n}. \quad (\text{iii})$$

It is easy to check that $f \rightarrow 0$ as $n \rightarrow \infty$. In fact, it tends to zero extremely fast. (Super-exponentially!) In high-dimensions almost all of the cube is wasted and only a tiny fraction actually covers the ball.

Exercise 4.6: Volume of the n -ball

Try plotting $\mathcal{B}_n$, $\mathcal{C}_n$, and $\mathcal{F}_n$ from exercise 4.7 versus n . Explain, in general terms, why this suggests that evaluating the posterior on a square grid of points may not be a good way of mapping the distribution in high dimensions.

It is common in modern science to encounter models with several tens or hundreds of parameters and even larger models occur in specific disciplines. We need a new tool for exploring these high-dimensional distributions...

Stochastic Sampling

Instead of pre-choosing a grid of points x where we are going to evaluate our posterior $P(\vec{x})$, we instead design a Monte Carlo (i.e. random, or stochastic) algorithm that draws random samples from the posterior distribution. There are a wide variety of algorithms available for stochastic sampling. The one feature that they all share is that the algorithm aims to output a set of N *random posterior samples*

$$\{\vec{x}_i \mid \text{for } i = 1, 2, \dots, N\}. \quad (4.31)$$

Each of these posterior samples should be an independent random variable drawn from the posterior distribution; i.e.

$$\vec{x}_i \sim P(\vec{x}) \quad \forall i. \quad (4.32)$$

Ideally we want N to be a large number (between 1000 and 10000 is typically enough for most applications).

For high dimensional problems stochastic sampling is generally the only practical way to explore the posterior distribution.

If we are able to obtain a set of random samples, they are incredibly useful; in fact, they can tell us (almost) everything we want to know about the posterior distribution. We can plot histograms of the points to map out the shape of the distribution; plotting a 1- or 2-dimensional marginal histogram is as simple as selecting the necessary columns from an array. There are packages for doing this, such as the the CORNER Python package². The following code demonstrates using CORNER to plot a histogram corner plot for the distribution defined in equation 4.30; the resulting figure is shown in figure 4.5.

```
import numpy as np
from scipy.stats import multivariate_normal

# setup the distribution using scipy
mu = np.zeros(3)
cov = np.array([[1, 9/10, 0], [9/10, 1, 0], [0, 0, 10]])
dist = multivariate_normal(mu, cov)

# draw random samples from the distribution
samples = dist.rvs(size=1000)

# plot histogram
import corner
corner.corner(samples, labels=['x', 'y', 'z'])
```

We can also use the random samples to approximate integrals of the following form

$$\int d\vec{x} f(\vec{x})P(\vec{x}) \approx \frac{1}{N} \sum_{i=1}^N f(\vec{x}_i), \quad (4.33)$$

for any function $f(\vec{x})$. In particular, if we want to compute the mean (expectation) of a single parameter which is in, say, the first entry of the vector, $x_0 \equiv \vec{x}[0]$ (using pseudo-Python notation), then we simply have to evaluate the following sum,

$$\langle x_0 \rangle \approx \frac{1}{N} \sum_{i=1}^N \vec{x}_i[0]. \quad (4.34)$$

²<https://corner.readthedocs.io/en/latest/>

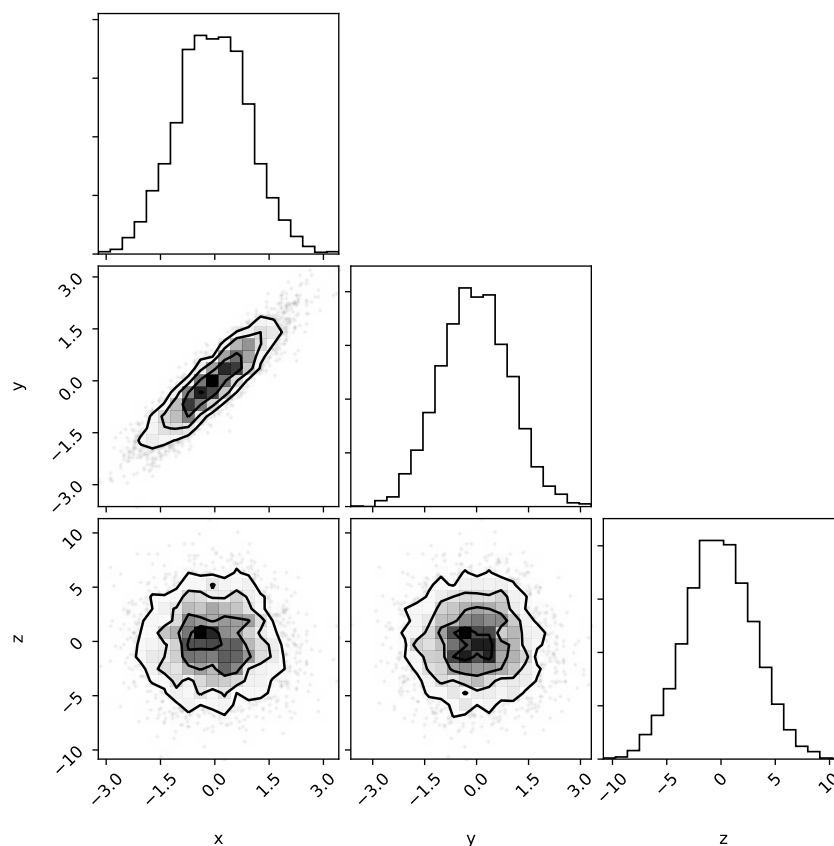


Figure 4.5: An example corner plot made using histograms of random samples. Compare this with figure 4.4.

The one key piece of information about our posterior distribution that we can't get from posterior samples alone is the Bayesian evidence.

There are many algorithms (and different implementations of these algorithms) designed to perform stochastic sampling. Perhaps the simplest and historically most widely used is the *Metropolis-Hasting Markov chain Monte Carlo* (MCMC) method which is described in appendix A. Another, which has seen rapid growth in use recently, is *nested sampling* which is described in appendix B.

We will conclude this chapter by demonstrating the use of an implementation of the *nested*

sampling algorithm called CPNEST on a higher dimensional parameter estimation problem.

Example 4.8: Gaussian Pulse Revisited

Consider again the spectral data plotted in exercise 4.5. This data is available for download from the course canvas page.

```
import numpy as np

# load spectral data from file: 2 cols are frequencies, and counts
X=np.loadtxt("GaussianPulseSpectralData.dat")
freqs, data = X[:,0], X[:,1]
```

Our model is the same Gaussian pulse as that used exercise 4.5, but now we will allow all 4 parameters to vary: the amplitude A , the background B , the pulse width σ , and the central frequency f_0 . This model is implemented in Python.

```
def model_mean(f, A, B, f0=0, w=1):
    #A Gaussian pulse with a background
    return A*np.exp(-0.5*((f-f0)/w)**2) + B
```

We now implement the likelihood and the prior for our problem. The likelihood is the same Poisson likelihood as used exercise 4.5. For our prior we use a flat prior on all parameters in the following ranges: $A \in [5, 50]$, $B \in [5, 50]$, $\sigma \in [0.3, 3]$, and $f_0 \in [-1, 1]$. CPNEST requires these functions to be implemented in a model class.

```
from scipy.stats import poisson

import cpnest
from cpnest.model import Model

class ModelClass(Model):
    def __init__(self, frequencies=freqs, counts=data):
        self.frequencies = frequencies
        self.counts = counts

        self.names = ['A', 'B', 'sigma', 'f0']
        self.bounds = [[5,50], [5,50], [0.3,3], [-1,1]]

    def log_likelihood(self, params):
```

```

        model = model_mean(self.frequencies, params['A'], params['B'],
                           f0=params['f0'], w=params['sigma'])

        # likelihood is product of Poisson probabilities with means=model
        return np.sum(poisson(model).logpmf(self.counts))

    #def log_prior(self, params): BY DEFAULT CPNEST USES A FLAT PRIOR

```

We now run the sampler, gather posterior samples and plot a histogram.

```

# create an instance of the model class
my_model = ModelClass()

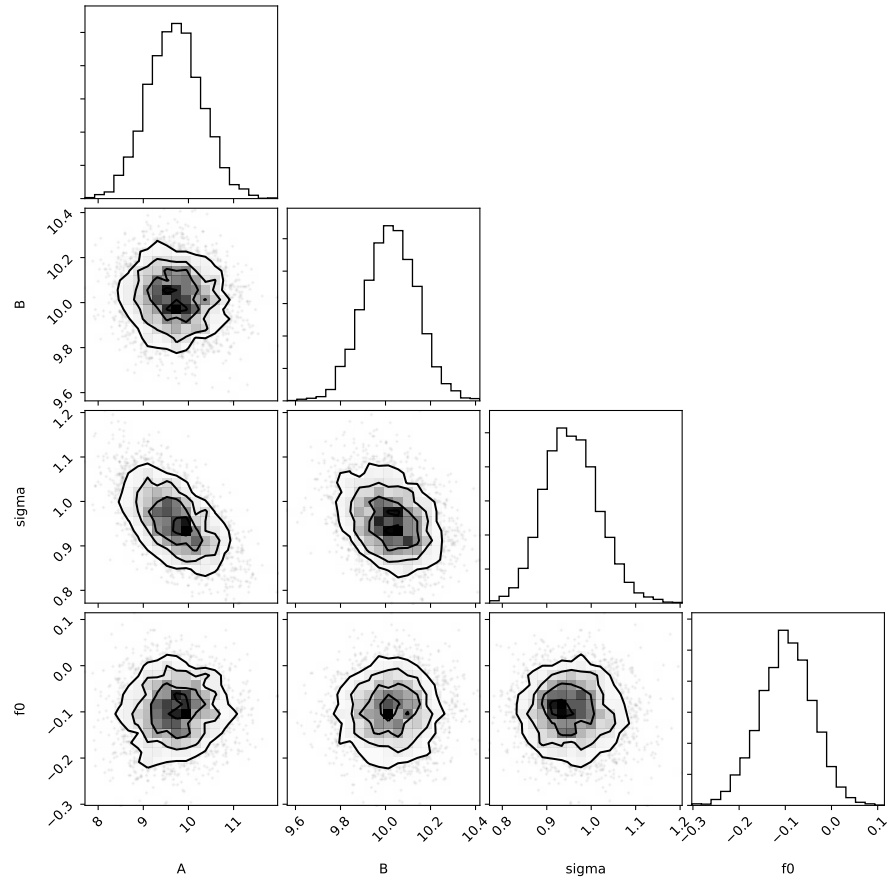
# setup the sampler
nest = cpnest.CPNest(my_model, nlive=1024, nthreads=4,
                     verbose=3, output='GaussPulse_output')

# run the sampler (this will take a few minutes)
nest.run()

# get posterior samples from nest
samples = nest.get_posterior_samples()
samples = np.loadtxt('GaussPulse_output/posterior.dat')[:, 0:4]

# produce a corner plot
import corner
corner.corner(samples, labels=my_model.names)

```



As well as producing the posterior samples which are plotted in the above histogram, the nested sampling algorithm also calculates the Bayesian evidence,

$$Z = P(D) = \exp(-2627.33) . \quad (i)$$

Here we will not use this number. Although, we will return to it in chapter 5.

Exercise 4.7: Gaussian Pulse Revisted

Repeat the analysis in example 4.8 using an MCMC algorithm instead of nested sampling.

Chapter 5

Bayesian Model Selection

In the previous chapter we encountered encountered *parameter estimation* problems where we had available to us a suitable model that described the data. There, our challenge to use Bayes' theorem to find the best possible values of the parameters and their uncertainties.

But what if we don't know what model to use? Or, what if we have several competing models which all claim to describe the data, how do we pick the best? These are questions of *model selection*. As we will see in this chapter, these questions can also be answered using Bayes' theorem, and the Bayesian evidence will play a key role.

5.1 Why Not Use the Maximum Likelihood?

A reasonable first thought would be, "why not use the model that fits the data best".

Let's formalise this idea a bit. Suppose we have two models: model A with some parameters λ_A , and model B with parameters λ_B . Given some data D we can write down the *likelihood* for both models: $P(D|\lambda_A, A)$ and $P(D|\lambda_B, B)$. We can numerically find the peak likelihood values of both of these distributions and consider the ratio

$$\text{Maximum Likelihood Ratio} = \frac{\max_{\lambda_A} P(D|\lambda_A, A)}{\max_{\lambda_B} P(D|\lambda_B, B)} = \frac{P(D|\hat{\lambda}_A, A)}{P(D|\hat{\lambda}_B, B)}. \quad (5.1)$$

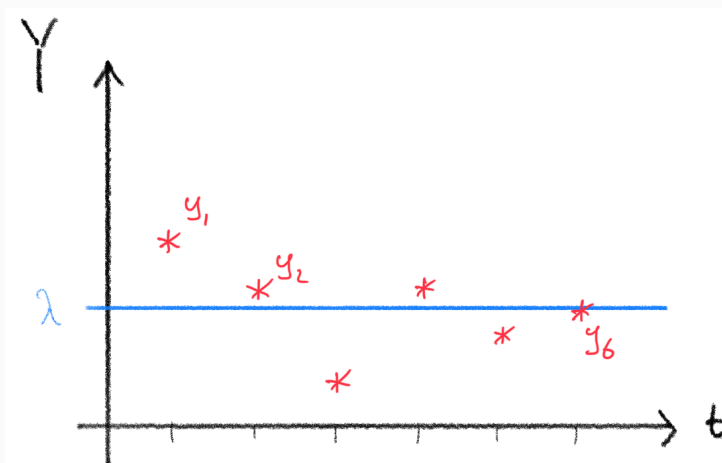
Why is it not enough to select model A if this ratio is larger than 1 and model B otherwise?

The problem with this approach is that it fails to account for the different complexity between models. Some models may have more or fewer parameters than others. Or parameters that can extend over different ranges. A model with a great many parameters allowed to vary over wide ranges will generally fit the data well (i.e. it will have a high value of the likelihood). However, this doesn't necessarily mean that this model should be chosen over a simpler model that does almost as good a job at fitting the data.

Example 5.1: Fitting the data well isn't everything

Suppose we have a sequence of measurements $\mathcal{D} = \{y(t_1), y(t_2), \dots, y(t_N)\}$ of a quantity $Y(t)$ made over an interval of time; where y . For simplicity, assume that all measurements have identical, independent Gaussian errors σ . For example,

$$\mathcal{D} = \{1.5, 1.2, 0.4, 1.1, 0.8, 1.0\} \quad \text{with} \quad \sigma = 1. \quad (\text{i})$$



We have two models:

- A - the quantity $Y = 0$.
- B - the quantity $Y = \lambda$, where λ is a constant, unknown parameter.

For model B, choosing $\lambda = \text{average}(y_i)$ will fit the data best.

There could be more models: e.g. model C, with two parameters ($Y = \lambda_1 + \lambda_2 x$), model D with three ($Y = \lambda_1 + \lambda_2 x + \lambda_3 x^2$), and so on.

It is clear that in (almost^a) any circumstances B will fit the data better than A (i.e. model B will have a higher peak likelihood value). Model B cannot possibly fit the data any worse than model A because A is nested inside B and can be recovered by setting $\lambda = 0$.

Similarly, model C will almost surely fit the data better than B , and D will fit better than C , and so on indefinitely.

Does this necessarily mean the data favours model Z ? No, we have failed to account properly for the different complexity of the models.

^aThe only exception being when the average of the measurements is exactly zero, in which case the two models will fit equally well

A model that fits the data well using a small number of free parameters is sometimes described as being *parsimonious*.

5.2 The Odds Ratio

Considering the material covered in previous chapters, it is perhaps no surprise that instead of the likelihood we are going to consider the *posterior odds ratio*,

$$\begin{aligned} \text{Posterior Ratio} \equiv \mathcal{O}_{A,B} &= \frac{P(A|D)}{P(B|D)}, \\ &= \frac{P(D|A)}{P(D|B)} \times \frac{P(A)}{P(B)}. \end{aligned} \tag{5.2}$$

In going to the second line we have used Bayes' theorem and the factor $P(D)$ has cancelled from the top and bottom of the fraction.

The second term in equation 5.2 is called the *prior odds ratio*. As is usual in a Bayesian analysis, our final answer depends partly on what we believed before performing the experiment. In some (but not all) situations we might want to try and be fair to both models and we might take the prior odds ratio to be unity.

Notice that in our first attempt in equation 5.1 we evaluated the ratio of the likelihoods at the maximum points. Here, in equation 5.2, we do *not* wish to do this. Instead of maximising over the free parameters we will marginalise over them, thereby allowing them

to take on all possible values.

$$\mathcal{O}_{A,B} = \frac{\int d\lambda_A P(D, \lambda_A|A)}{\int d\lambda_B P(D, \lambda_B|B)} \times \frac{P(A)}{P(B)}. \quad (5.3)$$

We will now use the *product rule* (see section 2.3) to rewrite the integrands in this ratio in terms of more familiar probability distributions;

$$\mathcal{O}_{A,B} = \frac{\int d\lambda_A P(D|\lambda_A, A)P(\lambda_A|A)}{\int d\lambda_B P(D|\lambda_B, B)P(\lambda_B|B)} \times \frac{P(A)}{P(B)}. \quad (5.4)$$

Here we see again the likelihoods for the two models reappearing. We have also been forced to introduce *priors* for the free parameters in both models: $P(\lambda_A|A)$ and $P(\lambda_B|B)$.

But now we recognise the terms inside the integrals in both the numerator and denominator on equation 5.4, they are the likelihood times the prior for each model. But the integral of the likelihood times the prior over the full parameter space of the model is, by definition, just the *Bayesian evidence* for that model. Using the notation introduced in chapter 4 we can write the posterior odds as

$$\mathcal{O}_{A,B} = \frac{Z_A}{Z_B} \times \frac{P(A)}{P(B)}. \quad (5.5)$$

The take home message is that when comparing two models for the same data, *compute the ratio of the model evidences*. This ratio then acts to update our relative state of belief in the two models. Before performing the experiment we have the prior odds ratio. After performing the experiment we multiply this by the ratio of evidences to get the posterior odds ratio.

You might worry about the presence of the prior odds ratio in the final answer for $\mathcal{O}_{A,B}$. It might feel unscientific in some way to allow our prior beliefs to affect the answer in this way; “would it not be better to let the data decide?” Unfortunately, this is misguided; it is important to realise that our prior beliefs are always present (and rightly so) when we interpret any data. (Although, in some situations it might be appropriate to set the prior odds ratio to unity.) In any case, in most situations the evidence ratio ends up dominating over the prior odds ratio. If one model fits the data significantly better than the other then it will have a much higher evidence and this will lead naturally lead us to favour that model regardless of the prior odds. The prior odds ratio is generally significant only in situations where both theories give similarly good fits to the data and/or the more complicated model gives a fit that is not sufficiently better to justify its extra complexity.

Of course, we could have chosen to define the odds ratio the other way around, with B in the numerator and A in the denominator. It makes no real difference, we have

$$\mathcal{O}_{A,B} = \frac{1}{\mathcal{O}_{B,A}}. \quad (5.6)$$

Example 5.2: An example using the odds ratio

Consider again the data in example 5.1 and the models A and B.

Let us choose a prior odds ratio that treats both models as being equally likely;

$$\frac{P(A)}{P(B)} = 1. \quad (\text{i})$$

First, consider the simpler model A. This model has no free parameters. We can write down the likelihood for this model as follows,

$$P(D|A) = \prod_{i=1}^6 \frac{\exp\left(-\frac{1}{2}y_i^2\right)}{\sqrt{2\pi}} = 1.41 \times 10^{-4}. \quad (\text{ii})$$

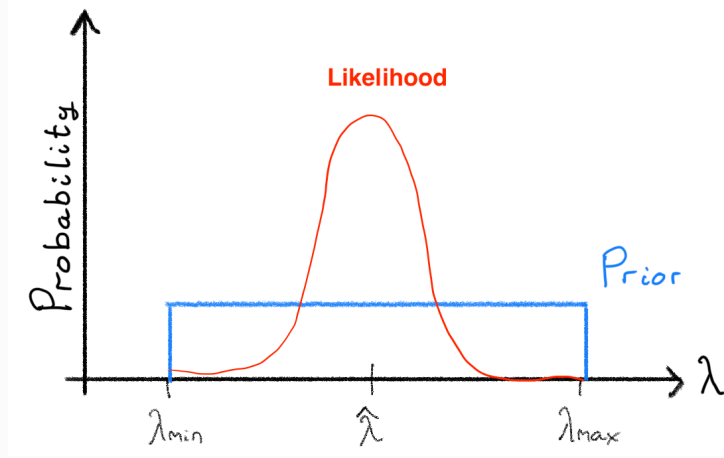
As there are no free parameters in this model there is no need to perform a marginalisation integral.

Second, consider the more complicated model B. This model has a single free parameter, λ . We can also write down the likelihood for this model as follows,

$$P(D|\lambda, B) = \prod_{i=1}^6 \frac{\exp\left(-\frac{1}{2}(y_i - \lambda)^2\right)}{\sqrt{2\pi}}. \quad (\text{iii})$$

The likelihood for model B has a maximum value of 2.84×10^{-3} at $\lambda = \hat{\lambda} = 1$. Therefore, if we were using the maximum likelihood ratio in equation 5.1 to perform model selection, we would favour the more complicated model B.

However, in order to compute the posterior odds ratio in equation 5.2 we must first evaluate the evidence for model B. This involves first specify a prior for the free parameter λ . Here, I will choose $P(\lambda|B) = (1/2\Lambda)\Pi_{-\Lambda}^{+\Lambda}(\lambda)$, where $\Lambda \gg 1$. The likelihood and prior for model B are sketched in the figure.



The evidence for model B is now given by the integral

$$P(B|D) = Z_B = \int_{-\Lambda}^{\Lambda} d\lambda \frac{1}{2\Lambda} \prod_{i=1}^6 \left[\frac{\exp(-\frac{1}{2}(y_i - \lambda)^2)}{\sqrt{2\pi}} \right]. \quad (\text{iv})$$

Because $\Lambda \gg 1$, this integral can be well approximated by

$$P(B|D) \approx \frac{1}{2\Lambda} \int_{-\infty}^{\infty} d\lambda \prod_{i=1}^6 \left[\frac{\exp(-\frac{1}{2}(y_i - \lambda)^2)}{\sqrt{2\pi}} \right] = \frac{1.45 \times 10^{-3}}{\Lambda}. \quad (\text{v})$$

Therefore, the posterior odds ratio is given by

$$\mathcal{O}_{A,B} = \frac{P(A|D)}{P(B|D)} \times \frac{P(A)}{P(B)} = \frac{1.41 \times 10^{-4}}{1.45 \times 10^{-3}/\Lambda} \times 1 = 0.0973\Lambda. \quad (\text{vi})$$

Notice that if we choose Λ to be larger than about 10, then the odds ratio becomes greater than unity and we favour the simpler model A. However, if we choose $\Lambda \lesssim 10$ then we favour model B.

At first, it is surprising (and a little worrying) that the answer seems to depend so much on the apparently arbitrary choice of the prior range Λ . In particular, if we were to allow the free parameter in model B to take any value (i.e. $\Lambda \rightarrow \infty$) it looks like the posterior odds would always be large and we would be forced to favour model A, regardless of what the data looks like. While this is formally true, in practice models never have parameters that can vary over infinite ranges

and after more careful consideration it is only right models with parameters that can vary over wide ranges (or multiple parameters) are properly penalised for this extra freedom.

In this example calculation, if we choose a reasonable value for Λ , we find a moderate posterior odds ratio ($\log \mathcal{O}_{A,B} \sim 0$) indicating there is no clear preference for either model. This raises an interesting question; how large (or small) does the odds ratio have to be for us to claim that there is “decisive” evidence in favour of a particular model? This is really more of a semantic question than a mathematical one; the answer depends on what you mean by “decisive”. However, for Harold Jeffreys answer to this question it is interesting to consider the table shown at https://en.wikipedia.org/wiki/Bayes_factor.

Occam’s Razor: At this point it is customary to mention the philosophical idea that can be stated something like “the simplest explanation is usually the right one”. Some people like to see an analogy between this concept and the Bayesian model selection procedure described above. Some part of this idea seems to be captured by the fact that the evidence ratio penalises a complicated model with many parameters (or parameters that can take values spanning a wider range) and that does not fit the data sufficiently better than a simpler model. This idea is usually attributed to William of Ockham (c. 1287-1347) who was a philosopher, theologian, and Franciscan friar. Ockham did not express the idea mathematically and certainly did not use the idea to perform Bayesian model selection in the way described here.

5.3 The Laplace Approximation

As we have seen, in Bayesian model selection problems it is necessary to find the Bayesian evidence. Finding the Bayesian evidence involves evaluating a integral over the parameter space of the model. In example 5.2 this was simply a 1-dimensional integral that could be easily evaluated numerically (in fact, it is also possible to evaluate it analytically). However, most models used the physical sciences are much more complicated and have more than one free parameter. Therefore finding the Bayesian evidence usually involves evaluating a high dimensional integral. In all but the simplest possible problems, these types of integrals are impossible to perform analytically. Worse, they are usually extremely difficult to evaluate numerically. In this section we will describe a useful method for approximating the evidence called the *Laplace approximation*.

To recap, if our problem involves parameters $\vec{x}$, and we have prior and likelihood functions

$\pi(\vec{x}) \equiv P(\vec{x})$ and $\mathcal{L}(\vec{x}) \equiv P(D|\vec{x})$ respectively, then the *unnormalised* posterior distribution $P(\vec{x}) \equiv P(\vec{x}|D)$ is given by

$$P(\vec{x}) = \mathcal{L}(\vec{x})\pi(\vec{x}), \quad (5.7)$$

then the normalising evidence $Z \equiv P(D)$ is given by the following integral;

$$Z = \int d\vec{x} P(\vec{x}). \quad (5.8)$$

Let us suppose we know that this distribution has a maximum value at the parameters $\hat{\vec{x}}$. (It is usually relatively easy to find the parameters $\hat{\vec{x}}$ numerically.) We now Taylor expand the quantity $\log P(\vec{x})$ about this peak location. In order to keep things simple, initially let's work in just 1-dimension; i.e. when $\vec{x} = x$. We have

$$\log P(x) = \log P(\hat{x}) - \frac{c}{2}(x - \hat{x})^2 + \dots, \quad (5.9)$$

where $c = -d^2/dx^2|_{x=\hat{x}} \log P(x)$. There is no linear term because we have chosen to expand about the peak where the first derivative vanishes. We have discarded terms $\mathcal{O}([x - \hat{x}]^2)$. Taking the exponential of this equation we find that we can approximate the unnormalised posterior as a Gaussian,

$$\log P(x) \approx \log P(\hat{x}) - \frac{c}{2}(x - \hat{x})^2. \quad (5.10)$$

A Gaussian is simple to integrate analytically (see appendix C); we have

$$Z \approx P(\hat{x}) \sqrt{\frac{2\pi}{c}}. \quad (5.11)$$

This is the Laplace approximation for the evidence in 1-dimension.

Notice that the Laplace approximation for the evidence in equation 5.11 is *not* invariant under a change of parameters. If we change variables $x \rightarrow x' = y(x)$, where $y(x)$ a suitable function of the original x , then the second derivative c transforms as $c \rightarrow c'$, where

$$c' = \frac{c}{\left(\frac{dy}{dx}\right)^2}. \quad (5.12)$$

If we instead use c' in equation 5.11 then we obtain a different approximation for Z . This is undesirable because the true evidence (defined in 5.8) is invariant under such a change of parameters; the accuracy of the Laplace approximation depends on which parameterisation we choose to use.

Exercise 5.1:

Derive the result in equation 5.12.

We can repeat this calculation for a general multivariate D -dimensional problem. In this case the parameters vector $\vec{x}$ has components x_i where $i = 1, 2, \dots, D$. We can approximate the logarithm of the unnormalised posterior distribution by the following multivariate Taylor series where we have expanded about the peak,

$$\log P(\vec{x}) \approx \log P(\hat{\vec{x}}) - \frac{1}{2} C_{ij} (x_i - \hat{x}_i)(x_j - \hat{x}_j). \quad (5.13)$$

The constants C_{ij} form an $D \times D$ matrix and are given by the negative of the matrix of second derivatives (aka the *Hessian*) of the log-posterior;

$$C_{ij} = - \left. \frac{\partial^2}{\partial x_i \partial x_j} \right|_{\vec{x}=\hat{\vec{x}}} \log P(\vec{x}). \quad (5.14)$$

The derivatives may be evaluated either analytically or numerically. Again, the integral of this multivariate Gaussian is a well known result (see appendix C) and involves the determinant of the matrix $\mathbf{C} = C_{ij}$,

$$Z \approx P(\hat{\vec{x}}) \sqrt{\frac{(2\pi)^D}{\det \mathbf{C}}}. \quad (5.15)$$

This result is the full *Laplace approximation*. This is a useful semi-analytic method for approximating the Bayesian evidence. It is only semi-analytic because the location of the maximum $\hat{\vec{x}}$ (and possibly the derivatives C_{ij}) must still be found numerically; however, this is usually much easier than numerically evaluating a high dimensional integral.

Again, as with the 1-dimensional version, the *Laplace approximation* in equation 5.15 is *not* invariant under a change of parameterisation, $\vec{x} \rightarrow x' = x'(x)$.

Exercise 5.2:

Study the derivation of the multivariate Gaussian integral in appendix C that was used to obtain the Laplace approximation in equation 5.15.

Exercise 5.3: Using the Laplace approximation

Use the 1-dimensional Laplace approximation to find the evidence in example 5.2.

5.4 Numerically Evaluating the Evidence

As stated above, finding the Bayesian evidence usually involves evaluating a complicated high dimensional integral. This is almost always impossible to do analytically. Therefore, in situations where the Laplace approximation is insufficiency accurate, our only option is to attempt to evaluate the integral numerically.

Any numerical integration algorithm can, in principle, be used to find the Bayesian evidence. However, very often the integrals involved have features that prove particularly problematic for standard numerical routines (e.g. Gaussian quadrature). For example, (i) it is common for the evidence integrals to be very high-dimensional (after all, physically interesting models usually have many free parameters). (ii) The integrands are often very strongly peaked (the parameter space might be large, but the if the data are high quality and the model parameters are well measured then the posterior may only have significant support in a very small region of this space). (iii) The integrands are often multi-modal (there may be several disjoint sets of parameters that fit the data almost equally well) (iv) The integrand may contain degeneracies long narrow and curving degeneracies (there may be a continuous range of parameters that fit the data almost equally well). These possibly problematic situations are sketched in figure 5.1.

Fortunately, numerical integration algorithms exist that have been specifically designed for evaluating these types of integrals. The most widely use family of algorithms is *nested sampling* (see appendix B).

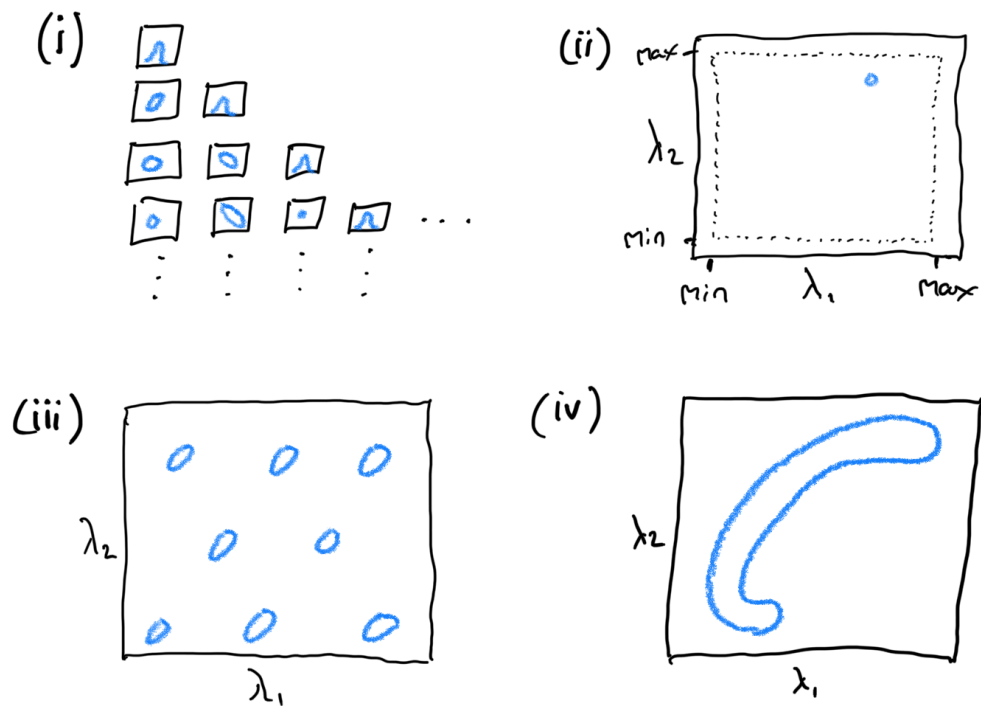


Figure 5.1: Sketches of contour plots for posterior probability distribution where evaluating the Bayesian evidence integral numerically may prove difficult. Roman numerals correspond to the numbered points in the discussion in section 5.4.

Example 5.3: A more complicated model selection example

Consider again the Gaussian pulse example studied in examples 4.5 and 4.8. This data is available for download from the course canvas page and was plotted in example 4.5.

Previously we analysed this data under the hypothesis that the pulse shape was described by a Gaussian model in some background noise; see 4.8. There, we found an evidence value of $Z_{\text{Gaussian}} = \exp(-2627.33)$.

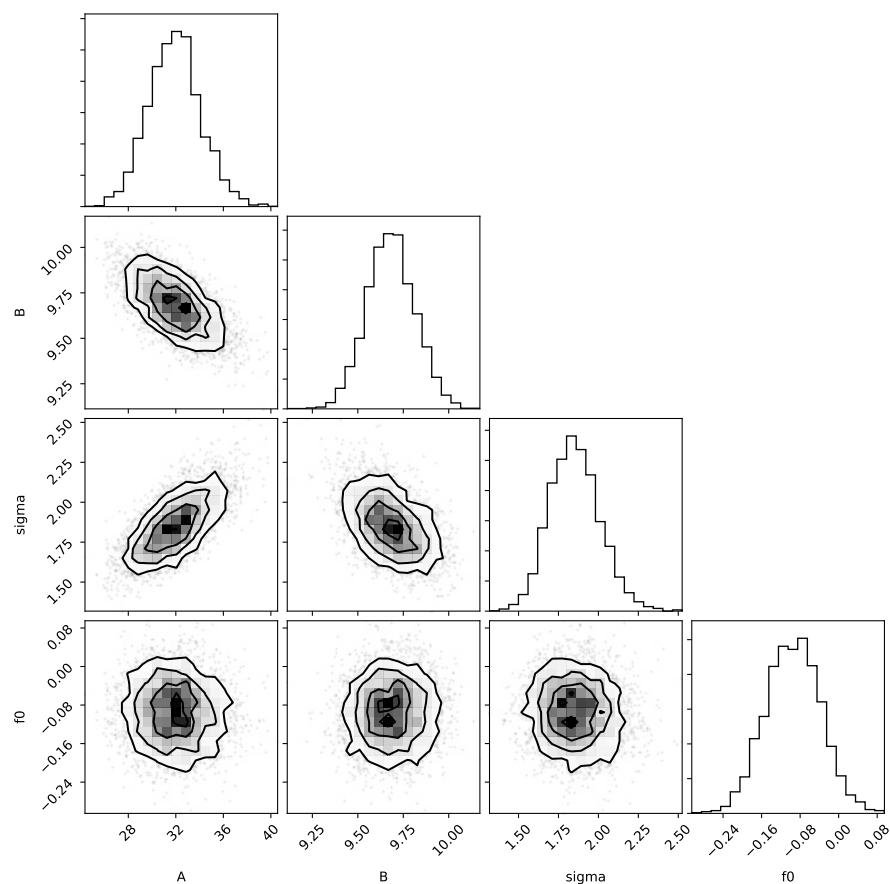
Here we will reanalyse the data under the alternative hypothesis that the pulse has a Lorentzian shape. Like the Gaussian model, this function is described by 4 parameters: the amplitude A , the pulse width w , the central pulse frequency f_0 , and the background strength B .

```
def model_mean(f, A, B, f0=0, w=1):
    #A Lorentzian pulse plus a background
    return (A/np.pi) * (w/2) / ( (f-f0)**2 + (w/2)**2 ) + B
```

We can now reuse most of the code from the previous example to repeat the analysis using the new Lorentzian model. In particular, we will use the same prior distribution as before on the 4 free parameters. Rerunning the example code with this new model we find the new posterior probability distribution shown below; this should be compared with the posterior distribution that we found before with the Gaussian model.

As well as producing the posterior samples which are plotted in the histogram below, the nested sampling algorithm also calculates the Bayesian evidence,

$$Z_{\text{Lorentzian}} = P(D|\text{Lorentzian model}) = \exp(-2630.03). \quad (\text{i})$$



If we assume equal prior probabilities for the two models (Gaussian and Lorentzian) then the posterior odds ratio is just the ratio of the evidences;

$$\mathcal{O}_{\text{Gaussian, Lorentzian}} = \frac{Z_{\text{Gaussian}}}{Z_{\text{Lorentzian}}} = \exp(2.70). \quad (\text{ii})$$

This is larger than 1 and so we favour the Gaussian model for the spectral shape. However, the evidence is not so overwhelming; using the language from Harold Jeffreys table, we would describe this as “substantial” evidence in favour of the Gaussian model. Perhaps it is not so surprising that the evidence is not more clear cut. After all, looking at the plotted spectrum shown in example 4.5, we can see that the data is reasonably noisy and it might be possible to fit it quite well with a variety of different curves.

5.5 Summary

This chapter has discussed the topic of *Bayesian model selection*. The same problem is sometimes given the alternative name *hypothesis testing*. We will discuss hypothesis testing again in chapter 6 where we revisit it from a frequentist point of view. Here, we simply point out that these two problems are not really very different. In both types of problem we have some data and two or more models (aka hypotheses; possibly with different numbers of free parameters) and it is our task to decide which model (hypothesis) is preferred.

In this section we will summarise the main points of this chapter, but this time using the alternative terminology of *hypothesis testing* in order to emphasise the similarities with *model selection*. Up until this point we have also been suppressing the other relevant prior assumptions (denoted $\mathcal{I}$, see equation 4.1) that are always present from our notation; in this summary these will be reinstated into our notation for completeness.

1. We are given some observations/measurements/data, $\mathcal{D}$ and prior assumptions $\mathcal{I}$.
2. We have several competing *hypotheses* which will be denoted $\mathcal{H}_0, \mathcal{H}_1, \mathcal{H}_2 \dots$.
Under hypothesis $\mathcal{H}_i$ the data $\mathcal{D}$ is described by model $\mathcal{M}_i$ with free parameters $\vec{\theta}_i$.
3. For each hypothesis we can write down the *likelihood*, $P(\mathcal{D}|\vec{\theta}_i, \mathcal{H}_i, \mathcal{I})$.
4. For each hypothesis we must choose a *prior*, $P(\vec{\theta}_i|\mathcal{H}_i, \mathcal{I})$.
5. For each hypothesis we can write down *Bayes' theorem*,

$$P(\vec{\theta}_i|\mathcal{D}, \mathcal{H}_i, \mathcal{I}) = \frac{1}{\mathcal{Z}_i} P(\mathcal{D}|\vec{\theta}_i, \mathcal{H}_i, \mathcal{I}) P(\vec{\theta}_i|\mathcal{H}_i, \mathcal{I}), \quad (5.16)$$

where we have defined the *evidence* for hypothesis $\mathcal{H}_i$ as

$$\mathcal{Z}_i = \int d\vec{\theta}_i P(\mathcal{D}|\vec{\theta}_i, \mathcal{H}_i, \mathcal{I}) P(\vec{\theta}_i|\mathcal{H}_i, \mathcal{I}), \quad (5.17)$$

and we recall that we are using the shorthand notation $\mathcal{Z}_i \equiv P(\mathcal{D}|\mathcal{H}_i, \mathcal{I})$.

6. For any pair of hypotheses $\mathcal{H}_i$ and $\mathcal{H}_j$, we can compute the *posterior odds ratio*,

$$\mathcal{O}_{i,j} \equiv \frac{P(\mathcal{H}_i|\mathcal{D}, \mathcal{I})}{P(\mathcal{H}_j|\mathcal{D}, \mathcal{I})} = \frac{P(\mathcal{H}_i|\mathcal{I})}{P(\mathcal{H}_j|\mathcal{I})} \times \frac{\mathcal{Z}_i}{\mathcal{Z}_j}, \quad (5.18)$$

which is the product of the *prior odds ratio* and the ratio of the *evidences*. The ratio of evidences acts to update our state of belief in light of the data $\mathcal{D}$ and quantifies the relative goodness-of-fit and parsimoniousness of the two models.

7. If the *posterior odds ratio* is greater (less) than 1 then we favour *hypothesis* $\mathcal{H}_i$ ($\mathcal{H}_j$).

Chapter 6

Frequentist Inference

Frequentist statistical inference is a different point of view on the types of problems explored in chapters 5 and 4. It was the dominant view throughout the 20th century and is still widely used today.

In the frequentist approach probabilities can only be attributed to the outcome of a random process (such as a noisy measurement) and not to models or model parameters. Naively, this makes some sense; if you were to count the number of photons per second N coming off the screen with luminosity L you are reading this on, then N is a random variable (probably following a Poisson distribution of the form $P[N; L]$) but there is nothing random about the underlying constant luminosity L . Therefore, a frequentist would object to defining the *posterior* probability $P(L|D)$ as this implies the value of L is itself uncertain.

In this picture, when we write $P(A)$, the quantity A is a random variable which can take on different values in repeated experiments, or in an *ensemble*. Notice, there is no actual value of A . The values are regarded as being *drawn* (or *sampled*) from a population. We assume that, at least in principle, an infinite number of repetitions are possible. The quantity $P(A)$ represents the sampling distribution of A , it gives the relative frequency of obtaining different outcomes from the experiment, not an uncertainty.

6.1 Statistics

In this section we are using the term *statistic* in a precise way where it has a technical meaning.

If we have some data D which is regarded as being drawn randomly from some population, then a *statistic* is any function of this random data; e.g. $S(D)$. Therefore, the statistic is also a *random variable* and we can describe the relative frequencies of its possible outcomes $P(D)$ in terms of its *sampling distribution*.

The simplest statistics are *estimators* which are used to estimate some property of the population from which the data have been drawn. The sample mean is an example, see exercise 6.1 - it estimates the population mean.

A good estimator $\bar{a}$ of the population property a should be

1. unbiased, $\langle \bar{a} \rangle = a$
2. consistent, $\lim_{N \rightarrow \infty} \bar{a} = a$
3. efficient, $\text{Var}(\bar{a})$ should be small

Example 6.1: The Sample Mean

Suppose we have a continuous random variable x with some unknown $P(x)$. Our task is to estimate the population mean $\langle x \rangle$.

We have N independent observations of this random variable $\{x_1, x_2, \dots, x_N\}$.

Claim: the best estimator of $\langle x \rangle$ is the *sample mean*, $\bar{\mu} = (1/N) \sum_{i=1}^N x_i$.

The sample mean $\bar{\mu}$ is function of the random data x_i (therefore it is a *statistic*) and the claim is we can use it to estimate the population mean (therefore it is also an *estimator*). Let's check that it is a good estimator.

To check that the sample mean is an *unbiased* estimator we evaluate its expecta-

tion;

$$\begin{aligned}
 \langle \bar{\mu} \rangle &= \left\langle \frac{1}{N} \sum_{i=1}^N x_i \right\rangle \\
 &= \frac{1}{N} \sum_{i=1}^N \langle x_i \rangle \\
 &= \frac{1}{N} (N \times \langle x \rangle) = \langle x \rangle ,
 \end{aligned} \tag{i}$$

as required for a good estimator. In going from the first to the second line we have used the linearity of the expectation (i.e. $\langle a + b \rangle = \langle a \rangle + \langle b \rangle$) and in going to the third line we have used the definition of the population mean for each of the observation (i.e. $\langle x_i \rangle = \langle \bar{x} \rangle$ for all i).

To check that the sample mean is *consistent* we evaluate its variance;

$$\begin{aligned}
 \text{Var}(\bar{\mu}) &= \text{Var} \left(\frac{1}{N} \sum_{i=1}^N x_i \right) \\
 &= \frac{1}{N^2} \text{Var}(x_1 + x_2 + \dots + x_N) \\
 &= \frac{1}{N} \text{Var}(x) \rightarrow 0 \quad \text{as } N \rightarrow \infty ,
 \end{aligned} \tag{ii}$$

as required for a good estimator. In going from the first to the second lines we have used quadratic scaling of the variance (i.e. $\text{Var}(\alpha a) = \alpha^2 \text{Var}(a)$ for constant α) and in going to the third line we have used the additive property of the variance (i.e. $\text{Var}(a + b) = \text{Var}(a) + \text{Var}(b)$ for independent random variables a and b).

Although we won't prove it, there is no other *unbiased* and *consistent* estimator of the population mean with a smaller variance; the sample mean $\bar{\mu}$ is the most *efficient* possible estimator.

Exercise 6.1: Properties of the expectation and variance

Verify the properties of the expectation and variance used in exercise 6.1 starting from the definitions given in chapter 3. For the additive property of the variance show that, in general, for random variables a and b

$$\text{Var}(a + b) = \text{Var}(a) + \text{Var}(b) + 2\text{Cov}(a, b) , \tag{i}$$

and for independent random variables $\text{Cov}(a, b) = 0$.

6.2 Error Propagation

A measurable quantity is a random variable which is drawn from some population. The width of this distribution is related to the measurement uncertainty.

If a random variable x lies in the interval $x \pm \sigma_x$ with a certain probability then the value of $f(x)$ lies in the interval $[f(x - \sigma_x), f(x + \sigma_x)]$ with the same probability (assuming the function f has no turning points within this interval). The usual error propagation formula $\sigma_f = (df/dx)\sigma_x$ is just the linear approximation to this.

Example 6.2:

Let x and y be two independent random variables with 1σ uncertainties of σ_x and σ_y respectively. Furthermore, let $f(x, y) = xy$. The 1σ uncertainty on the value of f satisfies

$$\frac{\sigma_f}{f} = \sqrt{\left(\frac{\sigma_x}{x}\right)^2 + \left(\frac{\sigma_y}{y}\right)^2}. \quad (\text{i})$$

To see this simply note that

$$df = \frac{\partial f}{\partial x}dx + \frac{\partial f}{\partial y}dy = ydx + xdy, \quad (\text{ii})$$

from which it follows that

$$\sigma_f^2 = x^2\sigma_y^2 + y^2\sigma_x^2 + 2xy\sigma_{xy}, \quad (\text{iii})$$

and $\sigma_{xy} = 0$ because x and y are independent (i.e. uncorrelated; see next section) random variables. Using $f = xy$ and rearranging gives the desired result.

Averaging N measurements reduces the scatter by a factor of $\sqrt{N}$ because

$$\text{Var}\left(\frac{1}{N}\sum_{i=1}^N x_i\right) = \frac{1}{N^2}\text{Var}\left(\sum_{i=1}^N x_i\right) = \frac{N}{N^2}\text{Var}(x) = \frac{1}{N}\text{Var}(x). \quad (6.1)$$

However, always beware of systematic errors! These do not average out and will eventually always dominate over statistical errors for large enough N .

6.3 Correlation

If we have several measurements of two or more variables, it is often of interest to test to see if they are *correlated* (or *anti-correlated*).

Recall from section 3.5, the *covariance* between random variables x and y is defined as

$$\begin{aligned} \text{Cov}(x, y) &= \langle (x - \langle x \rangle)(y - \langle y \rangle) \rangle, \\ &= \langle xy \rangle - \langle x \rangle \langle y \rangle, \end{aligned} \quad (6.2)$$

and the *correlation*, which will be denoted $\rho(x, y)$, is defined as the normalised quantity

$$\rho(x, y) \equiv \text{Corr}(x, y) = \frac{\text{Cov}(x, y)}{\sqrt{\text{Var}(x)\text{Var}(y)}}. \quad (6.3)$$

The normalisation is such that $\rho(x, y)$ is in the range -1 (perfect anti-correlation) to $+1$ (perfect correlation).

Exercise 6.2: Range of possible correlations

Show that $-1 < \text{Corr}(x, y) < +1$. (This is not easy.)

An easier exercise; find $\text{Corr}(x, y)$ when (i) $y = x$, and (ii) $y = -x$.

We will assume that our data consists of N observations of pairs of numbers (x, y) ;

$$\text{data} = \{(x_i, y_i) \mid \text{for } i = 1, 2, \dots, N\}. \quad (6.4)$$

Given data of the form in equation 6.4, several statistics are available to test for correlation. However, you should first always make a *scatter plot* of your data and beware of *rouge points* and *selection effects*.

Rouge Points: Beware situations such as those shown in the centre panel of figure 6.2. Try the “thumb test”; if covering a small fraction of the data with your thumb makes any sign of the correlation disappear then you should be very cautious.

Selection Effects: A selection bias that causes us to have some missing data points from our sample can cause a spurious correlation; see right panel of figure 6.2.

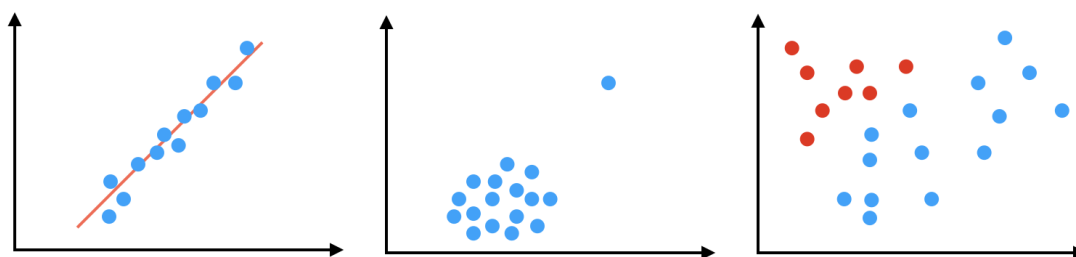


Figure 6.1: Left: an example of a correlation that looks good. Centre: fails the “thumb test”, covering up one point removes any hint of a correlation. Right: If the red points were missing due to some selection effect we would be in danger of detecting a spurious correlation in the remaining blue points.

Assume that we have made our scatter plot and are satisfied that it is safe to proceed. The following two sections will describe frequentist tests for (anti-)correlation.

6.3.1 Pearson Correlation Coefficient

Given observations or measurement data of the form in equation 6.4 we aim to test for the presence of any (anti-)correlation between x and y . To that end we calculate the following number, called the *Pearson correlation coefficient*;

$$r_p = \frac{\frac{1}{N} \sum_{i=1}^N (x_i - \bar{x})(y_i - \bar{y})}{\sigma_x \sigma_y}, \quad (6.5)$$

where

$$\sigma_x^2 = \frac{1}{N} \sum_{i=1}^N (x_i - \bar{x})^2 \quad \text{and} \quad \bar{x} = \frac{1}{N} \sum_{j=1}^N x_j, \quad (6.6)$$

and similar definitions for σ_y and $\bar{y}$.

The Pearson correlation coefficient is an example of *statistic*. Specifically, we will use the Pearson correlation coefficient as an estimator of the correlation $\rho(x, y)$.

Note that it does not matter whether we use N or $N - 1$ in the normalisation of the sums over i in equations 6.5 and 6.6 because these factors cancel in the fraction. (Usually, $N - 1$ must be use to obtain an unbiased estimator of the population variance.)

The Pearson correlation coefficient as an *unbiased* estimator of the correlation; this means that the mean satisfies $\langle r_p \rangle = \rho(x, y)$ (see section 6.1). We will also state without proof that the standard deviation of the Pearson correlation coefficient is given by $\sigma_{r_p} = (1 - r_p^2)/(N - 1)$.

In order to test for correlation we further calculate the number

$$t_p = \frac{r_p \sqrt{N - 2}}{\sqrt{1 - r_p^2}}. \quad (6.7)$$

Both r_p and t_p are statistics; that means they are functions of our random data. Therefore, r_p and t_p are themselves random variables. This means that they will both have particular means (i.e. $\langle r_p \rangle = \rho(x, y)$) and variances (i.e. $\sigma_{r_p} = (1 - r_p^2)/(N - 1)$). However, they are not necessarily distributed in any simple way; for example, it is extremely unlikely that either r_p or t_p will follow, say, a Gaussian distribution. What distributions they in fact follow will depend on the random data from which they are calculated; i.e. it depends on the distributions of the x and y and also on N .

If x and y are *uncorrelated* and are both distributed as *Gaussian* random variables (in other words, if x and y are independent Gaussian random variables) then it is possible to show that t_p follows a *Student's t-distribution* with $N - 2$ degrees of freedom. (It is worth taking a moment to appreciate what a huge assumption is involved here.)

The PDF for a random variable t that follows a *Student's t-distribution* with ν degrees of

freedom is given by

$$f_\nu(t) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\sqrt{\nu\pi}\Gamma\left(\frac{\nu}{2}\right)} \left(1 + \frac{t^2}{\nu}\right)^{-\frac{\nu+1}{2}}, \quad (6.8)$$

see figure 6.2.

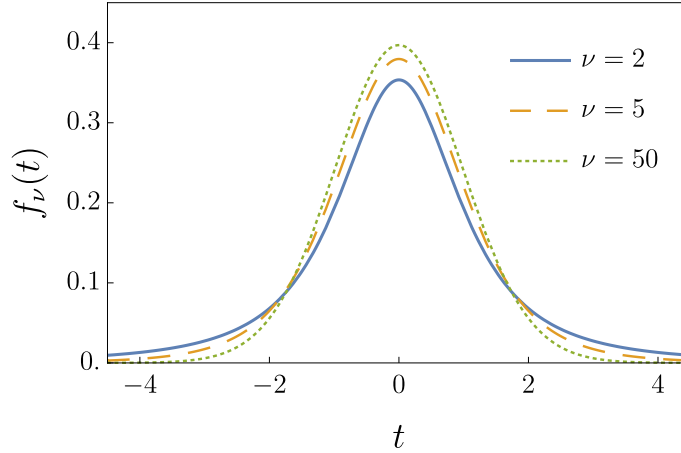


Figure 6.2: The student's t -distribution $f_\nu(t)$ (see equation 6.8) plotted for different numbers of degrees of freedom: $\nu = 2, 5$, and 50 .

Now we will describe how we can use this fact to test for a positive correlation.

We will assume that the data x and y are Gaussian distributed. We will further assume that the x and y are uncorrelated (this is called the *null hypothesis*). We will calculate t_p ; let's say that we find the value $t_p = T$. We then ask “how likely is it that we find a correlation at least this extreme under the null hypothesis?”; or, slightly more quantitatively, “how likely is it that we find a value $t_p \geq T$ if the null hypothesis is in fact true?”. This probability is given by

$$p = \int_T^\infty dt_p f_{N-2}(t_p). \quad (6.9)$$

If this probability is very small, then it is very unlikely that the null hypothesis is correct and we can conclude that there is a correlation between the two variables. The smaller the value of p the more confident we are that there is a correlation

Caution: The null hypothesis involved two key assumptions: the x and y were assumed to be uncorrelated *and* Gaussian distributed. If we find a small value for p it could be due

to a failure of the x and y to be Gaussian distributed and not due to a correlation. The next section will discuss an alternative statistical test that does not assume a particular distribution for the null hypothesis. In practice, sometimes taking the logarithm of the data prior to performing the Pearson test can help to achieve a more approximately Gaussian distribution.

It is easy to modify this procedure in order to test for negative correlation; the procedure is identical except we use

$$p = \int_{-\infty}^T dt_p f_{N-2}(t_p). \quad (6.10)$$

And to test for either positive or negative correlation we use a *two-tailed* test with

$$p = \int_{|t_p| > |T|} dt_p f_{N-2}(t_p). \quad (6.11)$$

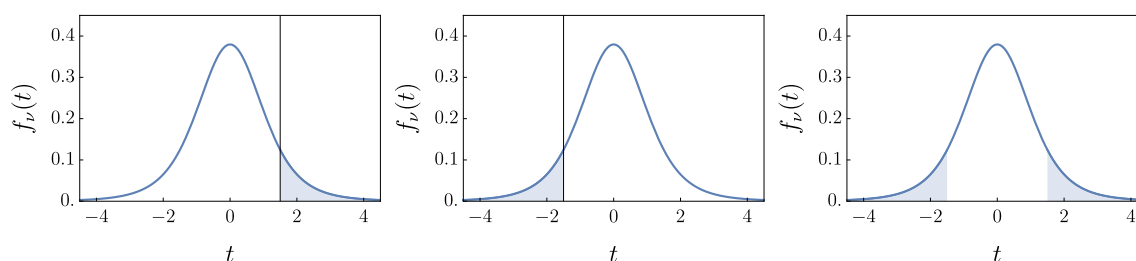


Figure 6.3: Left: to test for positive correlation, see equation 6.9. Centre: to test for negative correlation, see equation 6.10. Right: to test for either positive or negative correlation (a “two-tailed” test), see equation 6.11.

In order to actually calculate these p -values we can either evaluate the integral numerically for ourselves or else look up the result in a *statistical table* (a.k.a. *look-up table*). Statistical tables are rather old fashioned now that we all have laptops with libraries of code for the statistical distributions and numerical integration readily available. However, it is still worth going through the process of using them once or twice to help build an understanding of how the test works.

Example 6.3: Using r_p to test for correlation

A sample of 20 galaxies are measured and show signs of an anti-correlation between their radio luminosities and their optical luminosities. A calculation gives $r_p = -0.4$, is this result significant?

First we calculate

$$t_p = \frac{-0.4\sqrt{18}}{\sqrt{1-0.4^2}} = -1.85. \quad (\text{i})$$

A priori we have no information about whether to expect a positive or a negative correlation, therefore we will use a two-tailed test for any type of correlation.

Consulting a t -statistic table for a 2-tailed test with $20 - 2 = 18$ degrees of freedom we find $P < 0.1$. Therefore we can have $> 90\%$ confidence that there is an anti-correlation present.

If, instead of using a lookup table, we want to perform the test by integral numerically evaluating the integral, then we must compute (the factor of 2 is because we are performing a two-tailed test)

$$P = 2 \int_{1.85}^{\infty} dt_p f_{\nu=18}(t_p) = 0.08055 < 0.1, \quad (\text{ii})$$

and we reach the same conclusion.

Exercise 6.3:

Repeat the test in example 6.3 using both (i) a statistical table, and (ii) a numerical integration of the student's t -distribution to assess the significance of the correlation.

Exercise 6.4:

With an example page from a statistical table for the student's t -distribution

to use as a guide for the formatting, write a program that uses numerical integration to produce your own such table for any number of degrees of freedom. Check it agrees with the example.

It is worth carefully thinking through the logical arguments in this section as it serves as a prototype for all frequentist hypothesis testing. In this case, the hypothesis we wanted to test was “the variables x and y are correlated”. We began by assuming the opposite; the null hypothesis was “the variables x and y are uncorrelated”. We then computed a statistic (t_p in this case). We then asked “how likely is it for our statistic to be at least this extreme if the null hypothesis is true?” If this is a small probability then we have some confidence that the null hypothesis fails and our original hypothesis is true.

6.3.2 Spearman Rank Correlation Coefficient

One potential drawback with the Pearson correlation coefficient was that it assumed a Gaussian distribution for the data. The *Spearman rank correlation coefficient* is an alternative that does not assume any particular distribution; it is called a *distribution free* test, or sometimes (confusingly) *non-parametric*.

Distribution free tests typically reply on ranking the data (e.g. placing into ascending order) and basing the test only on the order. In this way the actual values are less important.

Given observations or measurement data of the form in equation 6.4 the Spearman test for the presence of (anti-)correlation proceeds by first ranking the data by the x -value, and then by the y -value. For each observation we record the integer difference d_i between these two rankings. We then calculate the number

$$r_s = 1 - \frac{6 \sum_{i=1}^N d_i^2}{N(N^2 - 1)}. \quad (6.12)$$

Example 6.4: Calculating r_s

Consider the following data value pairs with $N = 5$.

(x, y)	x -ranking	y -ranking	d_i
(2.0, 3.3)	2	1	1
(4.1, 8.0)	3	3	0
(1.3, 4.3)	1	2	1
(10, 8.1)	5	4	1
(9.1, 9.4)	4	5	1

From the differences d_i between the x - and y -rankings, we calculate

$$r_s = 1 - \frac{6(1^2 + 0^2 + 1^2 + 1^2 + 1^2)}{5(5^2 - 1)} = 0.8. \quad (\text{i})$$

Provided $N \gtrsim 10$ (which it is *not* in example 6.4), then under the *null hypothesis* of no correlation the quantity

$$t_s = \frac{r_s \sqrt{N-2}}{\sqrt{1-r_s^2}} \quad (6.13)$$

is distributed according to the student's t -distribution with $N - 2$ degrees of freedom.

We can use this to test for positive and/or negative correlation in the same way as we did with the Pearson correlation coefficient. The procedure is as follows:

1. Given data $\{(x_i, y_i) | i = 1, 2, \dots, N\}$ calculate r_s and t_s using equations 6.12 and 6.13.
2. Calculate the p under the null hypothesis using either a single- or two-tailed integral as appropriate (or look it up in a table).
3. The smaller the p the less likely is the null hypothesis and the more confident we can be in claiming a correlation.

At this point I feel obliged to point out that *correlation does not imply causation*. This point is emphasised in exercise 6.5 which also serves as an opportunity to practice implementing the various tests for correlation we have discussed.

Exercise 6.5: Testing for correlation

Consider the data in table 1 of R. Matthews¹ “Storks Deliver Babies ($p = 0.008$)”, Teaching Statistics 36-38 **22** (2000) doi:10.1111/1467-9639.00013.

Before doing anything else make a scatter plot, storks Vs babies.

Perform both the Pearson and Spearman tests for correlation between storks and babies. Appreciate the absurdity of what I am asking you to do.

6.3.3 Kendall’s Test

Kendall’s rank correlation coefficient, or *Kendall’s τ -test*, is another example of a distribution free test for correlation, similar in spirit to the Spearman test. It is mentioned here only to emphasise that alternative tests also exist.

Exercise 6.6: Testing for correlation, again

Look up the details of Kendall’s τ -test elsewhere and implement it for the data in exercise 6.5.

6.3.4 The Covariance Matrix

In the previous sections we saw tests for correlation between two variables. If there are $N > 2$ variables then we can quantify the covariance between each pair as described above and display the results in an $N \times N$ *covariance matrix*;

$$V_{ij} = \text{Cov}(x_i, x_j), \quad (6.14)$$

where x_i and x_j are two variables, like x and y .

¹This is a nice short pedagogical paper and well worth reading

The covariance matrix is symmetric (i.e. $V_{ij} = V_{ji}$) and its diagonal entries are equal to the variances (i.e. $V_{ii} = \text{Var}(x_i) = \sigma_i^2$).

The *correlation matrix* can also be defined,

$$\rho_{ij} = \frac{V_{ij}}{\sigma_i^2 \sigma_j^2}. \quad (6.15)$$

Any estimator for the correlation between 2 variables (e.g. ρ_p) can be used to estimate the entries in this matrix.

6.4 Frequentist Model Fitting

In chapter 4 we discussed *model fitting*, or estimating the parameters of model given some observed data, from a Bayesian perspective. In this section we discuss the same problem from a frequentist point of view.

Our task will be to fit some observed data of the form $\{d_i | i = 1, 2, \dots, N\}$, where each measurement has a known uncertainty σ_i . We will fit these data with a model M that has some adjustable parameters $a, b, c, \dots$.

In a Bayesian analysis, as we have seen, we would write down a likelihood and prior for our problem and then compute the posterior PDF on the model parameters. From this PDF we can then compute the best fitting, or mode, parameters and uncertainty ranges/regions.

In a frequentist analysis it does not make sense to assign a probability PDF to the model parameter. After all, there is only one true value for the model parameters and so it does not make sense to define a frequentist probability. However, we can use least squares, χ^2 minimisation, or maximum likelihood methods which we will now discuss in turn.

6.4.1 Least Squares

The idea of *least squares* fitting is to choose the model parameters such that the sum of the squares of the differences between the model predictions and the data is minimised. I.e.

$$\text{minimise } R^2 = \sum_{i=1}^N [d_i - M_i(a, b, c, \dots)]^2, \quad (6.16)$$

where M_i denotes the model prediction for the i^{th} data point.

Example 6.5: Linear Regression

Suppose we have real-valued measurements d_i taken at values x_i of some real-valued independent variable x .

We wish to fit a straight line to our data; our model is $M_i(a, b) = ax_i + b$ and has two free parameters. We have

$$R^2 = \sum_{i=1}^N [d_i - ax_i - b]^2, \quad (\text{i})$$

and this is minimised by solving the pair of simultaneous equations

$$\left. \frac{\partial R^2}{\partial a} \right|_{a=\hat{a}, b=\hat{b}} = 0 \quad \Rightarrow \quad -2 \sum_{i=1}^N x_i [d_i - ax_i - b] = 0, \quad (\text{ii})$$

$$\left. \frac{\partial R^2}{\partial b} \right|_{a=\hat{a}, b=\hat{b}} = 0 \quad \Rightarrow \quad -2 \sum_{i=1}^N [d_i - ax_i - b] = 0, \quad (\text{iii})$$

for the unknowns $\hat{a}$ and $\hat{b}$. It is left as an exercise to check that solution is

$$\hat{a} = \frac{\overline{xd} - \bar{x}\bar{d}}{\bar{x}^2 - \bar{x}^2}, \quad (\text{iv})$$

$$\hat{b} = \bar{d} - \hat{a}\bar{x}, \quad (\text{v})$$

where $\bar{x} = \frac{1}{N} \sum_{i=1}^N x_i$, and $\overline{xd} = \frac{1}{N} \sum_{i=1}^N x_i d_i$, and so on.

Note that least squares fitting takes no account of potentially different errors on different data points. If the errors are known to vary significantly then consider using χ^2 minimisation (see section 6.4.2).

Note also that if we have a scatter plot there is not always clear choice for the independent variable. In the notation of example 6.5 then it is not clear choice for the x -variable. The results of a least squares minimisation are *not* independent of this choice. For problems of

this sort consider using an *orthogonal regressor* which is independent of which way around we choose the axes (see figure 6.4).

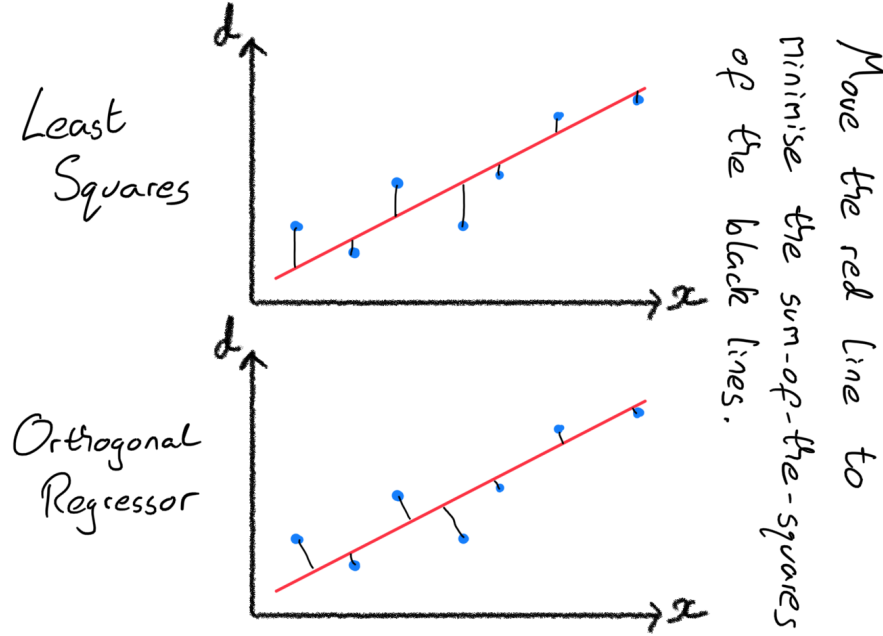


Figure 6.4: Top: least squares fitting is appropriate for linear regression when there is a clear distinction between the dependent and independent variables; least squares fitting minimises the sum-of-the-squares of the vertical distances between the data and model. Bottom: an orthogonal regressor might be appropriate when no such clear distinction exists; an orthogonal regressor minimises the sum-of-the-squares of the shortest (i.e. orthogonal) distances between the data and model.

6.4.2 χ^2 Minimisation

The idea of χ^2 *minimisation* is to choose the model parameters such that the *weighted* sum of the squares of the differences between the model predictions and the data is minimised. I.e.

$$\text{minimise } S = \sum_{i=1}^N \left[\frac{d_i - M_i(a, b, c, \dots)}{\sigma_i} \right]^2, \quad (6.17)$$

where M_i denotes the model prediction for the i^{th} data point.

In simple cases (e.g. linear regression or polynomial fits) the best fitting parameters can sometimes be found analytically (as was done in example 6.5). In more complicated cases one must perform this minimisation numerically.

Minimising χ^2 typically works well when a reliable error value σ_i can be attributed to each measurement and if the expected scatter about the model prediction is symmetric.

Sometimes we encounter counting problems; e.g. spectral fitting. Since we can't put an error bar on a single photon it is necessary to "bin" the counts and make a histogram and associate a Poisson error $\sqrt{n}$ to each bin count. However, this binning process discards some information, and the larger the bins the more information is lost. And the bins can't be made too small because if they contain $\lesssim 10$ counts the Poisson distribution will be too skewed.

For these reasons, with sparse counts it is better to use a *maximum likelihood method* (see section 6.4.3).

6.4.3 Maximum Likelihood

We previously encountered the *likelihood* $P(\{d_i\}|a, b, c \dots)$ in a Bayesian context.

Here we encounter it again as a *statistic* in a frequentist analysis. It is possible to consider the likelihood as a statistic because, for fixed values of the model parameters, the likelihood is a function of the data. We can maximise this statistic with respect to the model parameters to find the best fitting model. However, as we have discussed, in frequentist usage we may *not* use the likelihood to produce PDFs on the model parameters.

Example 6.6: A One-Parameter Maximum Likelihood Example

Suppose our data is some photon counts (i.e. integers) $\{x_1, x_2, \dots, x_N\}$ each collected over the same integration time, T . Our model of the source is that it has a constant luminosity such that we expect photons to arrive independently in our detector at an average rate of ρ photons per unit time.

The (Poisson) likelihood is given by

$$\mathcal{L}(\rho) \equiv P(\{x_i\}|\rho) = \prod_{i=1}^N \frac{\exp(-\rho T)(\rho T)^{x_i}}{x_i!} \quad (\text{i})$$

If we insert the measured values of the photon counts $\{x_i\}$, then we can view the likelihood ρ as being a function of our single model parameter, ρ . We then aim to maximise the likelihood with respect to ρ (in practice, it is easier to maximise $\log P(\{x_i\}, \rho)$ instead) to find the maximum likelihood value of the parameter, $\hat{\rho}$.

$$\log \mathcal{L}(\rho) = \sum_{i=1}^N -\rho T + x_i \log(\rho T) - \log(x_i!) \quad (\text{ii})$$

$$\left. \frac{d}{d\rho} \log \mathcal{L}(\rho) \right|_{\rho=\hat{\rho}} = 0 \quad \Rightarrow \quad \rho = \frac{1}{NT} \sum_{i=1}^N x_i. \quad (\text{iii})$$

Generally, for more complicated models with multiple parameters, the maximisation of the likelihood will have to be done numerically.

Example 6.7: A More Interesting Maximum Likelihood Example

Consider the problem of identifying a source (or more generally sources) in an image with a sparse number of photons. Our square image is pixelated and we observe photon counts for each pixel; therefore, our data is the matrix of integers

$$\{d_{ij} | \text{for } i = 1, 2, \dots, N \text{ and } j = 1, 2, \dots, N\}. \quad (\text{i})$$

See the image below with the corresponding data file available on the Canvas page. Suppose we know the form of the instrument point spread function,

$$h_{ij} = \begin{pmatrix} 0.05 & 0.1 & 0.05 \\ 0.1 & 0.4 & 0.1 \\ 0.05 & 0.1 & 0.05 \end{pmatrix}. \quad (\text{ii})$$

This describes what a single source looks like when viewed through our instrument; notice that $\sum_{\mu\nu} h_{\mu\nu} = 1$. You can interpret this as saying a photon that is aimed at the (α, β) pixel of our instrument has a 5% change of instead landing on the $(\alpha + 1, \beta - 1)$ pixel.

If we believe that the background noise level in our image has a constant level b and that there is a single source of strength s at point (α, β) in the image, then our model for the expected number of photons in each bin is

$$M_{ij} = b + sh_{(i-\alpha)(j-\beta)} \quad (\text{iii})$$

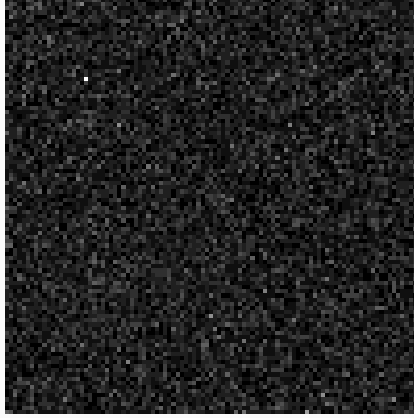
If photons to arrive independently in our detector at a constant average rate, then the (Poisson) likelihood for this problem is given by

$$\mathcal{L}(s, \alpha, \beta, b) \equiv P(\{x_i\} | s, \alpha, \beta, b) = \prod_{i,j=1}^N \frac{\exp(-M_{ij}) M_{ij}^{d_{ij}}}{d_{ij}!}, \quad (\text{iv})$$

or

$$\log \mathcal{L}(s, \alpha, \beta, b) = \sum_{i,j=1}^N -M_{ij} + d_{ij} \log M_{ij} - \log(d_{ij}!). \quad (\text{v})$$

Here I have opted to include the background noise b as a parameter. In practice, this is probably an example of a nuisance parameter. Alternatively, in some cases the value of b may already be known by other means, in which case it can be omitted from the arguments of $\mathcal{L}$.



Exercise 6.7: Sparse Photon Image

Use the maximum likelihood statistic to estimate the strength s and the integer pixel position (α, β) in the image shown in example 6.7. The data file is available on the course Canvas page. You may assume that the background noise is known to be $b = 2$.

(The parameter s is continuous and can be easily maximised over numerically by many standard numerical optimisation routines. But the parameters (α, β) are integers; you will therefore have to think a little more carefully about how best to maximise over these parameters.)

6.4.4 The χ^2 Distribution

This is the distribution following by the sum of the squares of normal (aka Gaussian) random variables.

If the random variables $u_i \sim \mathcal{N}(0, 1)$, then the random variables $X = \sum_{i=1}^{\nu} u_i^2$ follows a χ^2 distribution with ν degrees of freedom.

The PDF of the χ^2 distribution is (see figure 6.5 and the derivation in appendix D)

$$f(X; \nu) = \frac{X^{(\nu/2)-1} \exp(-X/2)}{2^{\nu/2} \Gamma(\nu/2)} \quad \text{for } X > 0. \quad (6.18)$$

The χ^2 distribution has support only for values $X > 0$.

Exercise 6.8: The χ^2 Distribution

Pick a ν . Numerically, generate $\nu \times N$ normal (aka Gaussian) random variables, $u_i \sim \mathcal{N}(0, 1)$. Group these into N sets of size ν and compute the sum of the squares of each set. Plot a histogram of the N resultant values and verify that these do indeed follow a χ^2 distribution with ν degrees of freedom.

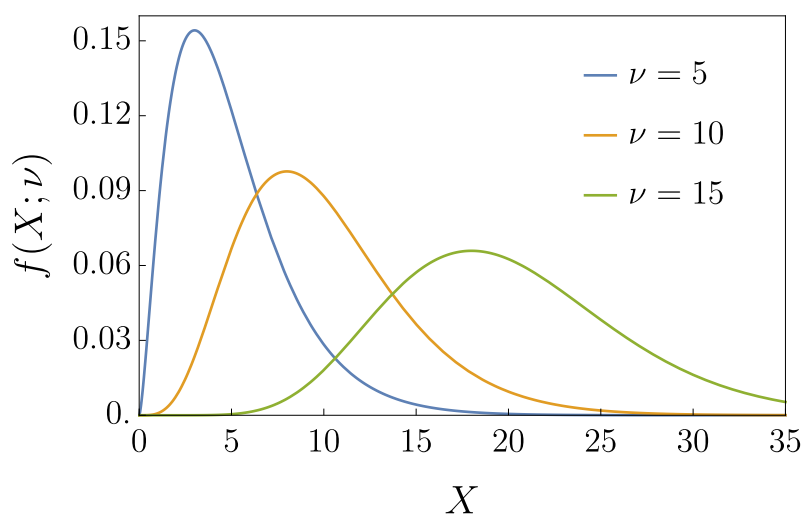


Figure 6.5: The PDF of the χ^2 distribution for different numbers of degrees of freedom.

Exercise 6.9: The Mean of the χ^2 Distribution

Show that the mean of the χ^2 distribution for ν degrees of freedom is ν ; i.e.

$$\int_0^\infty dX \, X f(X; \nu) = \nu. \quad (\text{i})$$

It can be shown that if a constraint is placed on the u_i random variables (for example, if we require that $\sum_i \alpha_i u_i = c$ for some constants α_i and c) the $X = \sum_{i=1}^\nu u_i^2$ still follows a χ^2 distribution, but now with $\nu - 1$ degrees of freedom.

6.4.5 Confidence Interval for Minimum χ^2 Fit

In a minimum χ^2 fit the model parameters are adjusted to minimise the following quantity,

$$S = \sum_{i=1}^N \left(\frac{d_i - M_i}{\sigma_i} \right)^2, \quad (6.19)$$

where M_i are the model predictions which depend on several free parameters: $\theta = (a, b, c \dots)$. Let's say that the number of free parameters is p .

The best-fitting values of the parameters the values $\hat{\theta} = (\hat{a}, \hat{b}, \hat{c} \dots)$ that minimise S ; at these best-fitting values we have $S = S_{\min}$.

Assuming the data follow a Gaussian distribution, we expect the quantity S to follow a χ^2 distribution with $N - p$ degrees of freedom.

This can be used to produce a confidence interval for a fitted parameter. To do this, we look at how the value of S increases as the parameter moves away from its optimum value (see the sketch in figure 6.6).

For example, consider the parameter a with best-fitting value $\hat{a}$. As a moves away from $\hat{a}$ the value of S will increase from $S_{\min}$ to some new value S_a . It is important that, if there are other parameters $(b, c, \dots)$, then these must be re-optimised for at each new value of a .

The increase $\Delta S = S_a - S_{\min}$ is also distributed as a χ^2 random variable with 1 degree of freedom. Looking at a table of χ^2_1 values we find the values in the table below. By seeing how far we have to move a from $\hat{a}$ in order to increase S by, say, $\Delta S = 2.71$ we can define a 90% confidence interval for the parameter a .

ΔS	Probability
1	0.68
2.71	0.90
3.84	0.95

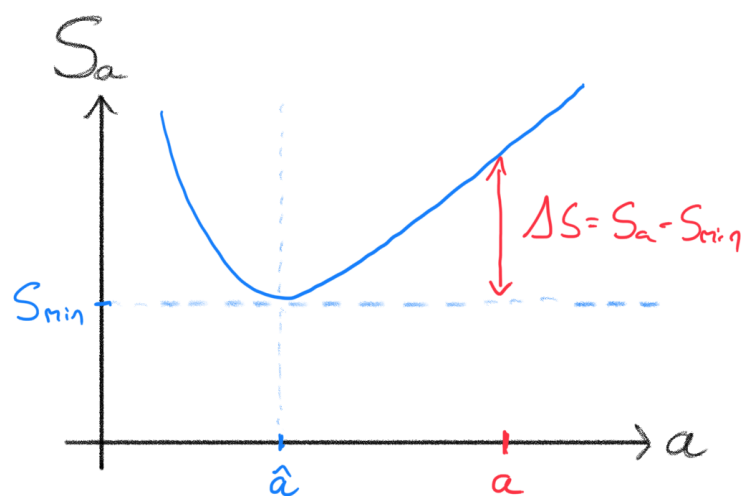


Figure 6.6: Sketch of the behaviour of S near the optimum parameters. The increase in S_a as we move away from the optimum parameters can be used to define confidence intervals.

6.4.6 Quality of Fit

The χ^2 statistic can also be used to evaluate the adequacy of the fit of a model to some data.

Again, the statistic is given by

$$S = \sum_{i=1}^N \left(\frac{d_i - M_i}{\sigma_i^2} \right)^2, \quad (6.20)$$

where M_i are the model predictions which depend on p parameters. If the model correctly describes the data, and if the errors are really Gaussian with the correct variances σ_i^2 , then S should be close to 1. Specifically, S should be distributed as a χ^2 random variable with $\nu = N - p$ degrees of freedom.

If $S \gg 1$ then the model does not fit the data well within the errors σ_i . For example, the following table for $\alpha = 0.1$ indicates the values of S that should only be exceeded with probability 0.1 if the model is correct.

ν (DOF)	$S(\alpha = 0.1)$
1	2.71
2	4.61
3	6.25
10	15.99

If $S \ll 1$ this might indicate that the errors σ_i have been overestimated.

6.5 KS test

The Kolmogorov-Smirnov (KS) test is used to test whether some data $\{x_i | i = 1, 2, \dots, N\}$ follows a predicted distribution.

The null hypothesis is that the data $\{x_i | i = 1, 2, \dots, N\}$ *does* follow the predicted distribution. We compute a statistic $\tilde{D}$ that, assuming the null hypothesis, follows a known distribution (the Kolmogorov distribution) and we compute the P -value; $P = \text{Prob}(D < \tilde{D})$. If P is smaller than, say, 0.1 then we can reject the null hypothesis at the 90% level.

The KS test is simple to perform:

1. Calculate $S_e(x)$ - the expected cumulative distribution.
2. Calculate $S_o(x)$ - the observed cumulative distribution.
3. Calculate $D = \max|S_e(x) - S_o(x)|$.
4. Calculate $P = \text{Prob}(D < \tilde{D})$. This can be done either by consulting a table for the Kolmogorov distribution, or by numerically evaluating the cumulative Kolmogorov distribution.

This procedure is illustrated in example 6.8.

Example 6.8: Example Applying KS Test

Let's generate ourselves N random numbers

```
import numpy as np

N = 20
x = np.random.normal(size=N)
```

We now perform a KS test of the hypothesis that the data $\{x_i\}$ are consistent with coming from a standard normal distribution. (Of course, we know they are because we generated the data ourselves; pretend we don't know where the $\{x_i\}$ came from.)

The expected cumulative distribution for a standard normal distribution can be expressed in terms of the error function,

$$S_e(x) = \frac{1}{2} \left(1 + \text{erf}(x/\sqrt{2}) \right) . \quad (\text{i})$$

We will use the implementation in SCIPY.STATS.

```
from scipy.stats import norm

Se_grid = np.linspace(-3, 3, 100)
Se_values = norm.cdf(Se_grid)
```

The observed cumulative distribution $S_o(x)$ of the data can be found by sorting.

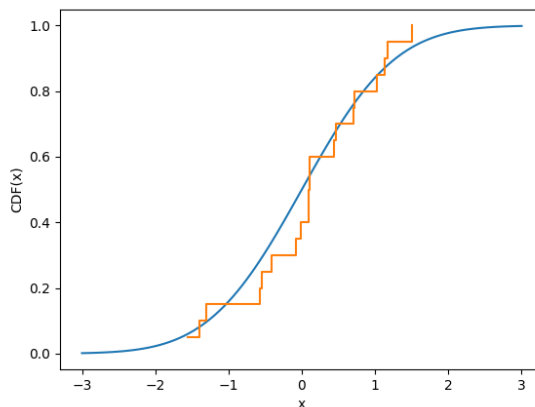
```
So_points = np.sort(x)
So_values = np.arange(1,1+N)/N
```

Let's plot the expected and the observed cumulative distributions together.

```
import matplotlib.pyplot as plt

plt.plot(Se_grid, Se_values)
plt.step(So_points, So_values, , where='post')

plt.xlabel("x"); plt.ylabel("CDF(x)")
plt.show()
```



This agreement looks reasonable, but the aim of the KS test is to test this agreement quantitatively.

Now let us compute $D = \max |S_e(x) - S_o(x)|$; this is the largest distance between the two curves on the plot.

```
# largest distance orange above blue
D_plus = ( np.arange(1, 1+N)/N - norm.cdf(np.sort(x)) ).max()

# largest distance blue above orange
D_minus = ( norm.cdf(np.sort(x)) - np.arange(0, N)/N ).max()
```

```
# largest distance overall
D = max(Dplus, Dminus)
```

For my random data, I find $D = 0.1696$.

Assuming the data really do come from a standard normal distribution (the *null hypothesis*) how likely is it that we would get at least this value? The value of D should follow a Kolmogorov distribution; let's examine this distribution graphically.

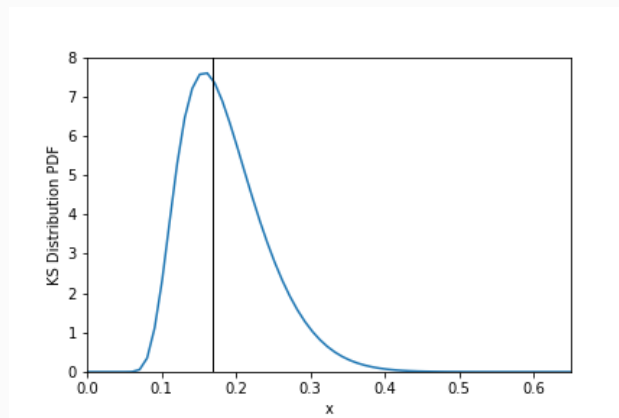
```
from scipy.stats import kstwo

grid = np.linspace(0, 1, 100)
plt.plot(grid, kstwo(N).pdf(grid))

plt.xlabel("x"); plt.ylabel("KS Distribution PDF");

plt.axvline(D, c='k', lw=1)
plt.xlim(0, 0.65); plt.ylim(0, 8);

plt.show()
```



Our value of D looks to be nicely consistent with this distribution. In fact, assuming the null hypothesis, we would expect to find larger values of D with approximately 50% probability; this is the P -value.

```
P = 1-kstwo(N).cdf(D) # Finding P value
```

For my random data, I find $P = 0.5559$.

Larger values of D are unlikely under the null hypothesis; if D was larger than some critical value (0.2647 for $N = 20$) then we could reject null hypothesis at the 90% level.

```
1-kstwo(N).cdf(0.2647) # Testing 90\% critical value found in table
>>> 0.0999990449482987
```

The KS, along with many others, are already implemented for us in `SCIPY.STATS`; the same result can be obtained in one line.

```
from scipy import stats
stats.kstest(x, stats.norm.cdf)
>>> KstestResult(statistic=0.16964940846164633,
                  pvalue=0.5558866014796376)
```

Chapter 7

Lindley's Paradox

In this course we have encounter both Bayesian and frequentist approaches to inference problems. In most problems, either style of reasoning can be applied and similar conclusions obtained.

However, this is not always the case. In this chapter we will see that an example where the application of Bayesian and frequentist approaches to a hypothesis testing problem lead to apparently contradictory conclusions. This situation is sometimes called *Lindley's paradox*¹.

7.1 A Coin Fair?

We flip a coin 150 time and record 64 heads and 86 tails. Is this coin fair?

This seems a fairly straightforward question that, although we will never be able to answer with absolute certainty, we should nevertheless be able to give a fairly straightforward answer to.

¹D. V. Lindley, "A statistical paradox", *Biometrika*, **44** 1-3 187–192 (1957).

7.1.1 A Frequentist Approach

First, we will attempt to answer the question using a frequentist style of reasoning.

We will start with the following *null hypothesis*:

$$H_0 = \text{"the coin is fair"} . \quad (7.1)$$

The alternative hypothesis, H_1 , is that the coin is loaded in some way.

Assuming H_0 , the possible outcomes of a coin toss are $P(\text{heads}) = P(\text{tails}) = 1/2$. Given a 150 trials we would expect, on average, to obtain 75 heads. The actual number of 64 seems a bit low. We can compute a P -value for this;

$$P = \text{Prob}(64 \text{ or fewer heads}) . \quad (7.2)$$

If this P -value is less than, say, 5% then we will reject the null hypothesis at 95% significance.

Using the Binomial distribution (see section 3.4.1), we can compute the P -value in equation 7.2 and we find

$$P = 0.043 < 0.05 . \quad (7.3)$$

Under the null hypothesis, it is unlikely ($< 5\%$ probability) that we would obtain such small a number of heads from 150 trials by chance alone. Therefore, we reject the null hypothesis at the 95% significance level.

7.1.2 A Bayesian Approach

Second, we will attempt to answer the question using a Bayesian style of reasoning.

We propose two competing models for the coin.

- Model A: The coin is fair; $P(\text{heads}) = P(\text{tails}) = 1/2$.
- Model B: The coin is loaded in some way; $P(\text{heads}) = p$ and $P(\text{tails}) = 1 - p$, where p is a free parameter of the model.

For both models we can use the Binomial distribution (see section 3.4.1) with probability mass function $p(r; n, p)$ (see equation 3.25) to compute the likelihood.

Model A has no free parameters. Therefore, the Bayesian evidence for model A is simply the likelihood function evaluated on the data;

$$Z_A = P(64 \text{ heads} | A) \quad (7.4)$$

$$\begin{aligned} &= p(r = 64; n = 150, p = 1/2) \\ &\approx 0.01302. \end{aligned} \quad (7.5)$$

For model B the likelihood $P(64 \text{ heads} | p, B)$ is again a Binomial probability. However, this time the likelihood depends on a single free parameter, p . We have to choose a prior $P(p)$ for the parameter p ; here, we use a flat prior between 0 and 1. The Bayesian evidence for model B is given by

$$\begin{aligned} Z_B &= \int_0^1 dp' P(64 \text{ heads} | p, B) P(p) \\ &= \int_0^1 dp' p(r = 64; n = 150, p') \\ &\approx 0.006623. \end{aligned} \quad (7.6)$$

If we adopt uniform prior odds ($P(A)/P(B) = 1$) then the posterior odds ratio between the models is simply the ratio of the evidences;

$$\mathcal{O}_{A,B} = \frac{Z_A}{Z_B} \approx 1.966 > 1. \quad (7.7)$$

This analysis favours model A; i.e. the coin is fair. Although we should note that the strength of the evidence in favour of model A is “Barely worth mentioning” according to Jeffreys’ descriptive table for interpretation.

7.1.3 A Paradox?

Is this a paradox?

Chapter 8

Final Remarks

In the broadest possible terms, the two main classes of problem in inference from scientific data are fitting models (parameter estimation) and choosing between models (model selection, or hypothesis testing).

In this course we have encountered a range of such problems and seen how they can be tackled by a mix of Bayesian and frequentist techniques.

These ideas find a wide range of applications across many fields, and certainly not limited to physics and astronomy.

I hope you have enjoyed the course, and I hope that some of you are able to successfully use the ideas covered here in your final year research projects.

Appendix A

Markov Chain Monte Carlo

Markov chain Monte Carlo (MCMC) algorithms aim to draw random samples from a target probability distribution (usually the posterior distribution, in a Bayesian analysis). This is achieved by constructing a *Markov chain* which, after many iterations, converges to the target distribution.

A *Markov chain* is a random sequence of points where the probability of each point depends only on the previous point in the chain.

Various algorithms exist for constructing such a Markov chain. The simplest and most well-known is the *Metropolis–Hastings* algorithm which is described here. The Metropolis–Hastings algorithm is very flexible and, as we shall see, is extremely simple to implement. It can be used to sample from any distribution $P(x)$; in particular, we do not care about the dimensionality (x can be vector of any length) and we do *not* require that $P(x)$ be properly normalised. (This later point is useful because, as we have seen, normalising posterior distributions in high dimensions involves evaluating the tricky evidence integral.)

The algorithm generates a sequence of points, called a *chain*, $x_1, x_2, \dots, x_n$. The construction of this chain is designed such that, in the limit $n \rightarrow \infty$, the distribution of these points matches the target distribution $P(x)$; i.e. $x_i \sim P(x)$. The points are produced iteratively, and the location of the next point x_{i+1} depends only on the current point x_i (i.e. a Markov chain). At each iteration x_i , the algorithm randomly proposes a possible next point u . This proposed point is either accepted (in which case the chain moves to the proposed point; $x_{i+1} = u$) or rejected (in which case the chain stays where it is; $x_{i+1} = x_i$). The probability of the proposed point being accepted is decided by the relative values of the

probability density at the two points: $P(u)$ and $P(x_i)$. If $P(u) > P(x_i)$ then the chain will always move to u , if $P(u) < P(x_i)$ then the chain will only move to u with probability $P(u)/P(x_i)$; i.e. the chain “wants” to move uphill, but can sometimes move downhill too.

The algorithm requires us to specify a proposal distribution to choose the point u . This distribution is denoted Q , and at each step the proposed point u is drawn randomly from this distribution, $u \sim Q(u|x)$. Here, we will assume that this proposal is symmetric, i.e. $Q(x|u) = Q(u|x)$.

The algorithm then proceeds as follows.

1. **Initialise.** Pick an arbitrary starting value, x_0 .
2. **Iterate.** For $i = 0, 1, 2, \dots$ Generate a proposed point $u \sim Q(u|x_i)$, evaluate the *acceptance ratio* $\alpha = P(u)/P(x_i)$, and accept the proposed point (i.e. $x_{i+1} = u$) with probability $\min(\alpha, 1)$ otherwise reject the proposed point (i.e. $x_{i+1} = x_i$).

The result of the algorithm is a Markov chain of points $\{x_0, x_1, x_2 \dots\}$ which should be distributed according to the target distribution $P(x)$; i.e. $x_i \sim P(x)$. Although, see the practical discussion of *thinning* and *burnin* below.

A.1 An example MCMC implementation

Let’s illustrate the Metropolis–Hastings algorithm by implementing it ourselves for a simple example distribution in 2-dimensions; $\vec{x} = (x, y)$. Let us suppose we wish to sample points from the following target distribution which looks like a ring of radius $r_0 = 5$ in the 2D plane.

```
import numpy as np

def P(x, r0=5, w=1):
    """
    Target distribution, a ring in the 2D plane
    """
    r = np.linalg.norm(x)
    return np.exp(-0.5*((r-r0)/w)**2)
```

Notice that the above probability distribution $P(x)$ is *not* properly normalised; this does not matter for the Metropolis–Hastings algorithm.

We choose to use a Gaussian proposal distribution $Q(u|x)$; i.e. $u = \mathcal{N}(x, I_2)$, where I_2 is the 2×2 identity matrix.

```
from scipy.stats import multivariate_normal

def Q(x, cov=np.identity(2)):
    """
    proposal distribution, a 2D Gaussian centred on the current point
    """
    u = multivariate_normal(x, cov).rvs()
    return u
```

The following is a simple implementation of the Metropolis–Hastings MCMC algorithm.

```
x0 = np.array([0, 0]) # initial starting point

chain = [x0] # chain stores all of the points visited

num_iterations = 10000
for i in range(num_iterations): # iterate the algorithm

    x = chain[-1] # current point is last in the chain

    u = Q(x) # proposed point

    alpha = P(u)/P(x) # compute acceptance ratio

    u = np.random.uniform() # draw random number in range 0 to 1

    if u < alpha: # accept the proposed point with probability
        x_new = u
    else: # reject the proposed point
        x_new = x

    chain.append(x_new) # append new point to the chain
```

The result of the algorithm is a Markov chain of points stored in the variable “chain”. Let’s look at the first 100 points in this chain, these are plotted in figure [A.1](#). We can see that the chain executes a random walk across the 2-dimensional parameter space. The size of the steps in the random walk is governed by our choice of the proposal distribution.

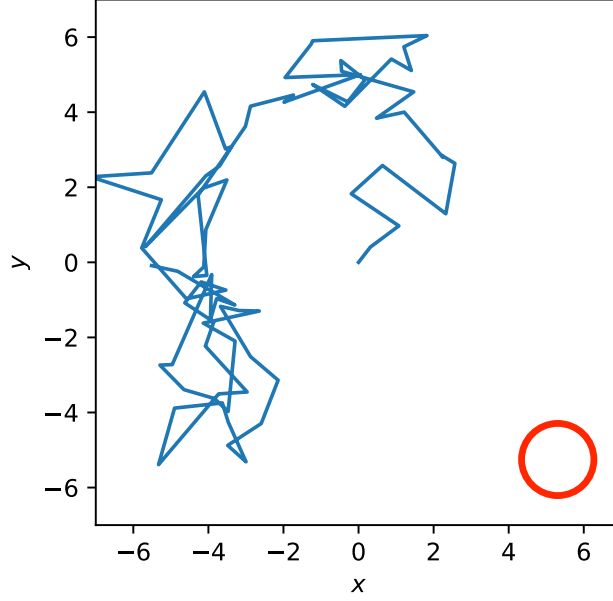


Figure A.1: The chain starts at the origin, where we initialised it. It immediately (in about 10 steps) heads outwards to a region of higher probability near a radius of $r_0 = 5$. Thereafter, it is possible for the chain to go to regions of lower probability, but there is a tendency for it to stay in regions of high probability. In fact, we can already see the chain begins to diffuse around the circle $x^2 + y^2 = r_0^2$. However, in these first 100 steps it doesn't get all the way around the circle; there is an excess of points at the 9 o'clock position. (The red circle in the lower right of the plot indicates the width of the proposal distribution.)

We can use the full chain to visualise the target distribution $P(x)$ by plotting a histogram. However, it is always a good idea to clean up the chain before plotting such histogram. The first few points in the chain may not be representative of the whole; this is because we choose the starting point arbitrarily and it might have been in a region of low probability (see figure A.1). Therefore it is common to remove some points from the start of the chain; this portion of the chain is called the *burnin* phase. After the burnin, we ideally want each point in the chain to be an independent random variable drawn from the target distribution. However this is clearly not the case for our chain; in figure A.1 we see that in the first 100 points there are none in over half of the circle. This is because the MCMC chain takes some time to move from place to place. Therefore, it is common to *thin* the chain by taking every n^{th} point. The choice of n should be guided by how long it takes the chain to move around the parameter space (what is known as the *autocorrelation length*). Here we choose to use a burnin length of 100 and a thinning of 50 (although this is probably

a bit too small).

The corner package¹ provides an easy way to plot multidimensional histograms, also known as *corner plots*, see figure A.2.

```
samples = np.array(chain)[100::50] # remove the burnin and perform the thinning

import corner
corner.corner(samples, labels=[r'$x$', r'$y$'])
```

The ring structure of the target distribution $P(x)$ can be clearly seen in figure A.2.

There are several potential problems that can arise with the Metropolis-Hastings MCMC algorithm. Some of these can be seen even in our above example

1. **Autocorrelation.** We want independent random samples from the target distribution. However, as we have seen it takes time for the MCMC chain to move around. Points in the chain separated by less than the autocorrelation length are not independent. If the autocorrelation length is L , then this means that to get N independent samples we have to run for $L \times N$ iterations which can take a long time. You might think of increasing the width of the proposal distribution to allow the chain to make bigger steps and hence move around faster decreasing the autocorrelation length. This can work, however it also runs into the acceptance problem...
2. **Acceptance.** At each iteration of the algorithm a new point u is proposed. If this point has a lower value of P than the current point then it is only accepted with probability $P(u)/P(x) < 1$ (if it has a higher value of P the is definitely accepted). If the proposal distribution keeps proposing points with low probability then it could take many iterations before our chain even moves a single step. A poor choice of proposal distribution typically leads to a low acceptance rate.
3. **Getting stuck.** Sometimes the target distribution can have multiple disconnected regions of high probability. (This was not the case is our example). For the chain to converge we need it to move backwards and forwards between these regions many times. This poses a serious problem for MCMC algorithms. When the chain is in one region it is very unlikely that our proposal distribution will propose a new point in the other. This can lead to our chain getting stuck in just one region and failing to adequately explore the full space.

¹<https://corner.readthedocs.io/en/latest/>

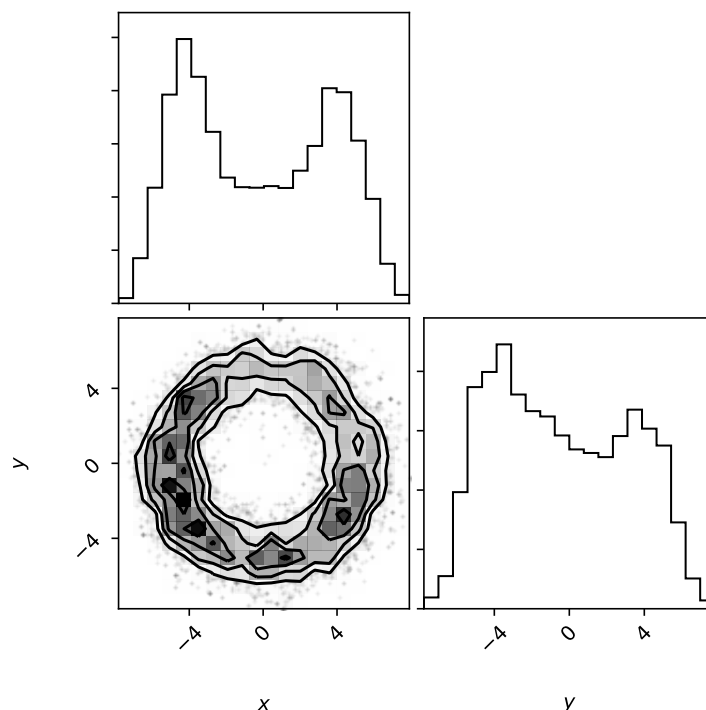


Figure A.2: An example of a *corner plot*, or *triangle plot* for the distribution $P(x)$. The lower left panel shows a 2-dimensional histogram of the point in our chain. The other two panels show the 1-dimensional marginalised histograms for the x (right) and y (top) dimensions. Our MCMC algorithm has performed fairly well. Although, it looks like we haven't run our MCMC for quite long enough. There seem to be more points near the 9 o'clock position than the rest of the ring. If we look back at figure A.1 we see that the chain initially went in this direction. We should probably run our algorithm for more iterations (say 100000) and apply more thinning to the resultant chain (say, take every 100th point). If we did this we would expect to see this slight imbalance disappear.

4. **Serial.** The basic Metropolis-Hasting algorithm described does not lend itself well to being run in parallel.
5. **Doesn't evaluate the evidence.** The basic Metropolis-Hasting algorithm described can be used to draw random points from a posterior distribution. This allows us to estimate parameter values and find their uncertainties. However, it does not give us the Bayesian evidence.

Fortunately, a lot of research has been done into MCMC algorithms and this has resulted in variations to the Metropolis-Hastings algorithm designed to overcome some (all?) of these problems. There are many variations and implementations available and free to use. Implementing Metropolis-Hastings for yourself is a good way to learn about MCMC algorithms, but for serious research problems we should use a tried and tested package.

A.2 Variations of the MCMC algorithm

One such package is EMCEE². This is a Python package that implements another type of MCMC algorithm (not Metropolis-Hasting) which is called *affine invariant Markov chain Monte Carlo ensemble sampling*. We will not describe this algorithm at all here³, we will just briefly demonstrate how to use this package to sample from our ring distribution $P(x)$.

```
import emcee

ndim = 2 # our problem is 2 dimensional
nwalkers = 10 # instead of a single chain this algorithm uses several

x0 = np.random.randn(nwalkers, ndim) # choose random starting points

def log_P(x): # the algorithm needs the log probability function as an input
    return np.log(P(x))

sampler = emcee.EnsembleSampler(nwalkers, ndim, log_P) # setup the sampler

state = sampler.run_mcmc(x0, 100) # run 100 burnin iterations

sampler.reset()

sampler.run_mcmc(state, 10000) # run algorithm for 10000 iterations

samples = sampler.get_chain(flat=True, thin=50) # get the thinned mcmc chain

corner.corner(samples, labels=[r'$x$', r'$y$'])
```

The histogram of the resulting chain, stored in “samples”, is plotted in figure A.3.

²<https://emcee.readthedocs.io/en/stable/>

³For details, see Goodman, J., Weare, J., *Ensemble samplers with affine invariance*, Commun. Appl. Math. Comput. Sci. 5 , 1, 65-80 (2010)) doi:10.2140/camcos.2010.5.65.

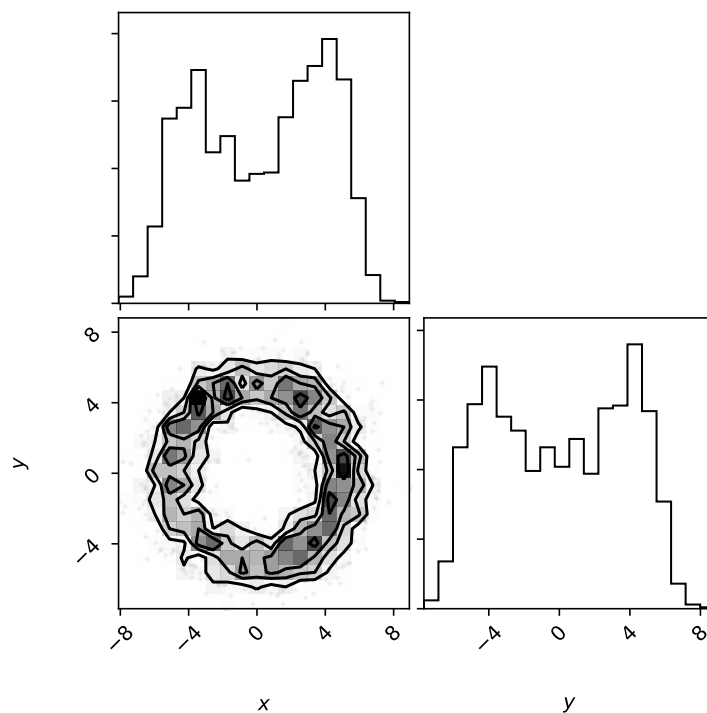


Figure A.3: The 2-dimensional histogram of the chain produced by the EMCEE algorithm. It should not be a surprise that for similar parameter to those used in figure A.2 these samples appear to be slightly better (there is no imbalance visible in the ring). The emcee algorithm includes several improvements over the “vanilla” Metropolis-Hastings MCMC algorithm implemented above.

Appendix B

Nested Sampling

The *nested sampling* algorithm is another example of a stochastic sampling algorithm that can be used to generate a sequence of independent, random samples from a target distribution.

One of the draw backs we identified with the Metropolis-Hastings MCMC algorithm was that it didn't give us the Bayesian evidence. Nested sampling does calculate the evidence. In fact, nested sampling is primarily an integration algorithm and produces random samples from the target distribution as something of a by-product.

The nested sampling algorithm is significantly more complicated than the Metropolis-Hastings algorithm and will not be described here; for more details see chapter 9 of D. S. Sivia, "Data Analysis: A Bayesian Tutorial" or the the paper by Skilling, J. *Nested Sampling*, AIP Conference Proceedings. 735: 395–405 (2004) DOI:10.1063/1.1835238.

As was the case with MCMC, there are ready to use implementations of the algorithm available. One such implementation is CPNEST¹

The following code snippet demonstrates how to use CPNEST to sample from the ring distribution defined in appendix A.

```
import cpnest
from cpnest.model import Model

class ModelClass(Model):
```

¹<https://github.com/johnveitch/cpnest>

```

r0 = 5 # the radius of the ring
names = ['x', 'y'] # we give a name to each of our parameters
bounds = [[-8,8], [-8,8]] # we have to provide ranges for each parameter

def log_likelihood(self, params):
    r = np.sqrt( params['x']**2 + params['y']**2 )
    return -0.5*(r-self.r0)**2

```

The above code defines a model class which is required by the CPNEST package. This class contains a method called “log_likelihood”. We can optionally also provide a “log_prior” method, but by default CPNEST will use a flat prior. The code below creates an instance of this class (called “my_model”) and samples from the distribution. The algorithm requires as input an integer number of live points (typically, a large number will give a more accurate estimate of the evidence integral).

```

my_model = ModelClass() # create an instance of the model class

nest = cpnest.CPNest(my_model,
                    nlive=1024,
                    output='output_folder') # setup the sampler

nest.run() # run the sampler

samples = nest.get_posterior_samples() # load the posterior samples
samples = np.loadtxt('output_folder/posterior.dat')[:, 0:2]

import corner # produce a corner plot
corner.corner(samples, labels=[r'$x$', r'$y$'])

```

The algorithm also produces the following estimate of the evidence integral,²

$$Z = \int dx P(x) \approx \exp(-1.18). \quad (\text{B.1})$$

²Warning: what CPNEST calls the “evidence” in the output files is really the “log-evidence”.

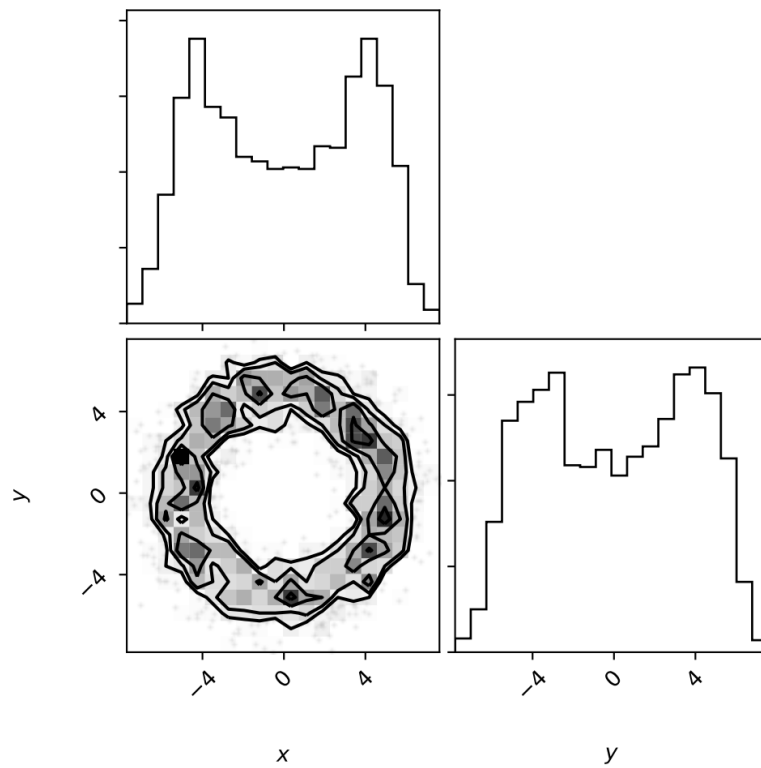


Figure B.1: The 2-dimensional histogram of the chain produced by the CPNEST algorithm.

Appendix C

Gaussian Integrals

The Gaussian integral is the integral over the real line of the function $\exp(-x^2/2\sigma^2)$, where $\sigma^2 > 0$. Surprisingly the solution involves π , and is given by

$$\mathcal{I} = \int_{-\infty}^{\infty} dx \exp\left(-\frac{x^2}{2\sigma^2}\right) = \sqrt{2\pi\sigma^2}. \quad (\text{C.1})$$

Proof:

The exponential function has the property that $\exp(-x^2)\exp(-y^2) = \exp(-r^2)$, where $r^2 = x^2 + y^2$ is the squared distance on the xy plane; this property will allow us to write $\mathcal{I}^2$ as the integral over the plane.

Define the following function and consider its square;

$$\mathcal{I}(a) = \int_{-a}^a dx \exp\left(-\frac{x^2}{2\sigma^2}\right), \quad (\text{C.2})$$

$$\mathcal{I}(a)^2 = \left\{ \int_{-a}^a dx \exp\left(-\frac{x^2}{2\sigma^2}\right) \right\}^2 \quad (\text{C.3})$$

$$= \int_{-a}^a dx \exp\left(-\frac{x^2}{2\sigma^2}\right) \int_{-a}^a dy \exp\left(-\frac{y^2}{2\sigma^2}\right) \quad (\text{C.4})$$

$$= \int_{-a}^a dx \int_{-a}^a dy \exp\left(-\frac{x^2 + y^2}{2\sigma^2}\right). \quad (\text{C.5})$$

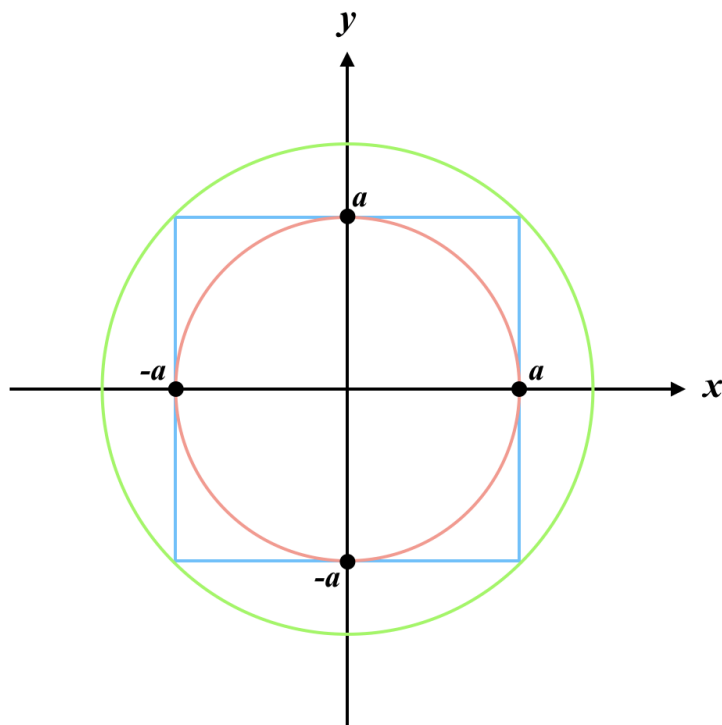


Figure C.1: The cyan square with side length $2a$ is the integration region for $\mathcal{I}(a)^2$. The large green circle has radius $\sqrt{2}a$ while the smaller red circle has radius a . Integrating over the circles yields upper and lower bounds respectively for $\mathcal{I}(a)^2$.

Therefore $\mathcal{I}(a)^2$ is the integral over a square with sides $2a$ (see figure). Changing from Cartesian to polar coordinates ($x = r \cos \phi$ and $y = r \sin \phi$) gives

$$\mathcal{I}^2(a) = \int_{\text{square}} r dr d\phi \exp\left(-\frac{r^2}{2\sigma^2}\right). \quad (\text{C.6})$$

Because the integrand is strictly positive we may bound $\mathcal{I}(a)^2$ from above and below by considering the integrals over two disks which graze the square at the corners and the midpoints of the sides respectively (see figure).

$$\int_{\text{small circle}} r dr d\phi \exp\left(-\frac{r^2}{2\sigma^2}\right) < \mathcal{I}(a)^2 < \int_{\text{large circle}} r dr d\phi \exp\left(-\frac{r^2}{2\sigma^2}\right) \quad (\text{C.7})$$

The circular integrals are easy to evaluate as the integrand is rotationally symmetric; this

pair of inequalities become

$$2\pi \int_0^a dr \, r \exp\left(-\frac{r^2}{2\sigma^2}\right) < \mathcal{I}(a)^2 < 2\pi \int_0^{\sqrt{2}a} dr \, r \exp\left(-\frac{r^2}{2\sigma^2}\right). \quad (\text{C.8})$$

Changing the integration variable to $u = r^2$ gives

$$\pi \int_0^{a^2} du \exp\left(-\frac{u}{2\sigma^2}\right) < \mathcal{I}(a)^2 < \pi \int_0^{2a^2} du \exp\left(-\frac{u}{2\sigma^2}\right) \quad (\text{C.9})$$

$$2\pi\sigma^2 \left(1 - \exp\left(-\frac{a^2}{2\sigma^2}\right)\right) < \mathcal{I}(a)^2 < 2\pi\sigma^2 \left(1 - \exp\left(-\frac{a^2}{\sigma^2}\right)\right). \quad (\text{C.10})$$

In the limit $a \rightarrow \infty$ the upper/lower bounds both tend to the same value

$$\lim_{a \rightarrow \infty} \mathcal{I}(a)^2 = 2\pi\sigma^2, \quad (\text{C.11})$$

which implies that

$$\mathcal{I} = \sqrt{2\pi\sigma^2}. \quad (\text{C.12})$$

C.1 Multi-Dimensional Gaussian Integrals

The n -dimensional generalisation of the Gaussian integral is given by

$$\mathcal{I}_n = \int_{-\infty}^{\infty} dx_1 \int_{-\infty}^{\infty} dx_2 \dots \int_{-\infty}^{\infty} dx_n \exp\left(-\frac{1}{2}x_i x_j A_{ij}\right) \quad (\text{C.13})$$

$$= \int_{-\infty}^{\infty} d^n x \exp\left(-\frac{1}{2}\mathbf{x}^T \cdot \mathbf{A} \cdot \mathbf{x}\right) \quad (\text{C.14})$$

$$= \sqrt{(2\pi)^n / |\mathbf{A}|}, \quad (\text{C.15})$$

where $\mathbf{A}$ is a real, symmetric, positive-definite $n \times n$ matrix.

Proof

Because $\mathbf{A}$ is symmetric its eigenvectors form an orthonormal basis in which $\mathbf{A}$ is diagonal. The proof proceeds by transforming the integral into this basis where it becomes a product of 1-dimensional Gaussian integrals.

As it involves little additional work, we will solve the more general integral

$$\int_{-\infty}^{\infty} d^n x \exp\left(-\frac{1}{2}\mathbf{x}^T \cdot \mathbf{A} \cdot \mathbf{x} + \mathbf{b}^T \cdot \mathbf{x} + c\right), \quad (\text{C.16})$$

where $\mathbf{b} \in \mathbb{R}^n$ and $c \in \mathbb{R}$ are constant. Completing the square gives

$$\text{int} = \int_{-\infty}^{\infty} d^n x \exp \left(-\frac{1}{2} [\mathbf{x} - \mathbf{A}^{-1} \cdot \mathbf{b}]^T \cdot \mathbf{A} \cdot [\mathbf{x} - \mathbf{A}^{-1} \cdot \mathbf{b}] + c' \right), \quad (\text{C.17})$$

where $c' = c + \mathbf{b}^T \cdot \mathbf{A}^{-1} \cdot \mathbf{b}/2$. Changing integration variables to $\mathbf{x}' = \mathbf{x} - \mathbf{A}^{-1} \cdot \mathbf{b}$ and then replacing $\mathbf{x}' \rightarrow \mathbf{x}$ in our notation gives

$$\text{int} = \exp(c') \int_{-\infty}^{\infty} d^n x \exp \left(-\frac{1}{2} \mathbf{x}^T \cdot \mathbf{A} \cdot \mathbf{x} \right). \quad (\text{C.18})$$

Because $\mathbf{A}$ is a real, symmetric matrix it may be written in diagonal form

$$\mathbf{A} = \mathbf{U}^T \cdot \mathbf{\Lambda} \cdot \mathbf{U}, \quad (\text{C.19})$$

where $\mathbf{U}$ is the matrix whose rows are the orthonormal eigenvectors of $\mathbf{A}$ (this implies that $\mathbf{U}$ is unitary, $\mathbf{U} \cdot \mathbf{U} = \mathbf{1}$, and hence $|\mathbf{U}| = 1$), and $\mathbf{\Lambda} = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_n)$ is the diagonal matrix of real eigenvalues (the fact that $\mathbf{A}$ is positive definite implies that all the eigenvalues are strictly positive). Note that the conditions placed on $\mathbf{A}$ are sufficient to ensure the existence of $\mathbf{A}^{-1}$.

Using equation (C.19) to eliminate $\mathbf{A}$ from (C.18), and changing integration variables from $\mathbf{x}$ to $\mathbf{y} = \mathbf{U} \cdot \mathbf{x}$, gives

$$\text{int} = \exp(c') \int_{-\infty}^{\infty} d^n y |\mathbf{U}^{-1}| \exp \left(-\frac{1}{2} \mathbf{y}^T \cdot \mathbf{\Lambda} \cdot \mathbf{y} \right). \quad (\text{C.20})$$

The factor $|\mathbf{U}^{-1}| = 1$ comes from the Jacobian of the transformation $\mathbf{x} \rightarrow \mathbf{y}$.

Using $\mathbf{\Lambda} = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_n)$, we can write the sum in the exponential as

$$\mathbf{y}^T \cdot \mathbf{\Lambda} \cdot \mathbf{y} = \sum_{i=1}^n y_i^2 \lambda_i. \quad (\text{C.21})$$

Therefore, the multivariate integral becomes a product of 1D integrals;

$$\text{int} = \exp(c') \prod_{i=1}^n \int_{-\infty}^{\infty} dy_i \exp \left(-\frac{1}{2} y_i^2 \lambda_i \right) \quad (\text{C.22})$$

$$= \exp(c') \prod_{i=1}^n \sqrt{2\pi/\lambda_i}, \quad (\text{C.23})$$

where we have used the result proven above. Finally, using the fact that the eigenvalue product is the determinant, i.e. $\prod_{i=1}^n \lambda_i = |\mathbf{A}|$, the result becomes

$$\text{int} = \exp \left(c + \frac{1}{2} \mathbf{b}^T \cdot \mathbf{A} \cdot \mathbf{b} \right) \sqrt{(2\pi)^n / |\mathbf{A}|}. \quad (\text{C.24})$$

Setting $c = 0$ and $\mathbf{b} = \mathbf{0}$ gives the desired result for $\mathcal{I}_n$.

Appendix D

χ^2 -squared Distribution

In section 6.4.4, it was stated that the sum of the squares of ν independent, standard Gaussian random variables follows a χ^2 distribution with ν degrees of freedom. This statement serves as a definition of the χ^2 distribution and needs no proof. However, here we will prove equation 6.18, which states that the PDF for the χ^2 distribution is

$$f(X; \nu) = \frac{X^{(\nu/2)-1} \exp(-X/2)}{2^{\nu/2} \Gamma(\nu/2)} \quad \text{for } X > 0. \quad (\text{D.1})$$

If we have ν independent, standard Gaussian random variables,

$$x_i \sim \mathcal{N}(0, 1), \quad \text{for } i = 1, 2, \dots, \nu, \quad (\text{D.2})$$

then we can form the new (positive) random variable

$$X = \sum_{i=1}^{\nu} x_i^2 > 0. \quad (\text{D.3})$$

The random variables x_i define a point in a ν -dimensional space; the random variable X is the square of the radius vector to this point;

$$r = \sqrt{X} \quad \Rightarrow \quad dr = \frac{dX}{2\sqrt{X}}. \quad (\text{D.4})$$

This setup is illustrated for $\nu = 3$ in figure D.1.

The probability that X lies in the range $(X, X + dX)$ can be expressed in two ways: in terms of X (left-hand side) and in terms of x_i (right-hand side). Equating these gives

$$f(X; \nu) dX = \int_{\text{shell}} p(x_1) p(x_2) \dots p(x_\nu) dx_1 dx_2 \dots dx_\nu, \quad (\text{D.5})$$

where the shell is the region between the two concentric spheres illustrated in figure [D.1](#). Each individual x_i follows a standard Gaussian distribution (see section [3.4.3](#));

$$f(X; \nu) dX = \int_{\text{shell}} \frac{\exp\left(-\frac{1}{2} \sum_{i=1}^{\nu} x_i^2\right)}{(2\pi)^{\nu/2}} dx_1 dx_2 \dots dx_\nu \quad (\text{D.6})$$

$$= \int_{\text{shell}} \frac{\exp\left(-\frac{1}{2} X\right)}{(2\pi)^{\nu/2}} dx_1 dx_2 \dots dx_\nu. \quad (\text{D.7})$$

The integrand depends only on r and is therefore constant over the shell; the exponential can be taken outside the integral. The integral is then simply the volume of the shell, which can in turn be written as the area, A , of the shell times the thickness dr . This gives

$$f(X; \nu) dX = \frac{\exp\left(-\frac{1}{2} X\right)}{(2\pi)^{\nu/2}} \int_{\text{shell}} dx_1 dx_2 \dots dx_\nu = \frac{\exp\left(-\frac{1}{2} X\right)}{(2\pi)^{\nu/2}} A dr. \quad (\text{D.8})$$

The area of a $(\nu - 1)$ -sphere with radius r is

$$A = \frac{2r^{\nu-1} \pi^{\nu/2}}{\Gamma(\nu/2)}. \quad (\text{D.9})$$

Combining equations [D.4](#), [D.8](#) and [D.9](#), we find

$$f(X; \nu) dX = \frac{X^{(\nu/2)-1} \exp(-X/2)}{2^{\nu/2} \Gamma(\nu/2)} dX, \quad (\text{D.10})$$

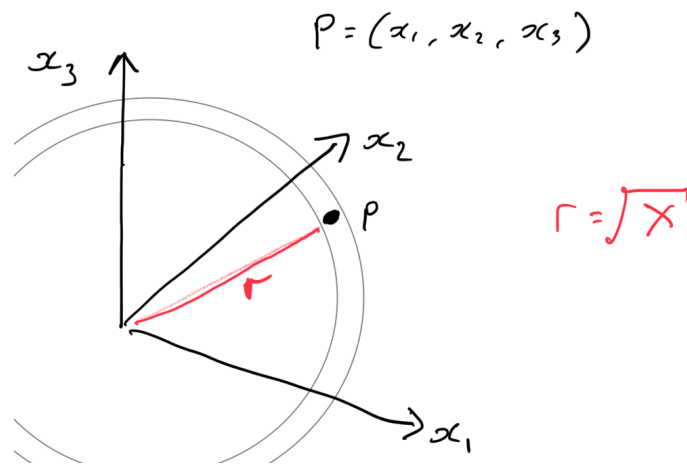


Figure D.1: A sketch of the geometric setup for the case $\nu = 3$.

Appendix E

The gamma function and Stirling's approximation

This appendix starts by giving a rough derivation of Stirling's approximation for the factorial of large numbers. The gamma function is then introduced. Due to its close relationship with the factorial function, the gamma function appears very often in problems of probability and statistics (see, for example, the χ^2 -distribution discussed in section 6.4.4 and appendix D). Finally, the gamma function is used to obtain an improved version of Stirling's formula.

A simple version of Stirling's formula can be found by approximating the following sum with an integral;

$$\log n! = \log \prod_{i=1}^n i = \sum_{i=1}^n \log i \sim \int_1^n dx \log x = [x \log x - x]_{x=1}^{x=n} = n \log n - n + 1. \quad (\text{E.1})$$

For large values of n the $+1$ on the right-hand side is negligible and can be discarded. Taking the exponential of both sides gives

$$n! \sim n^n \exp(-n). \quad (\text{E.2})$$

This is a very rough version of Stirling's formula.

The *gamma function* $\Gamma(z)$ is defined by the integral

$$\Gamma(z) = \int_0^\infty dt \, t^{z-1} \exp(-t), \quad (\text{E.3})$$

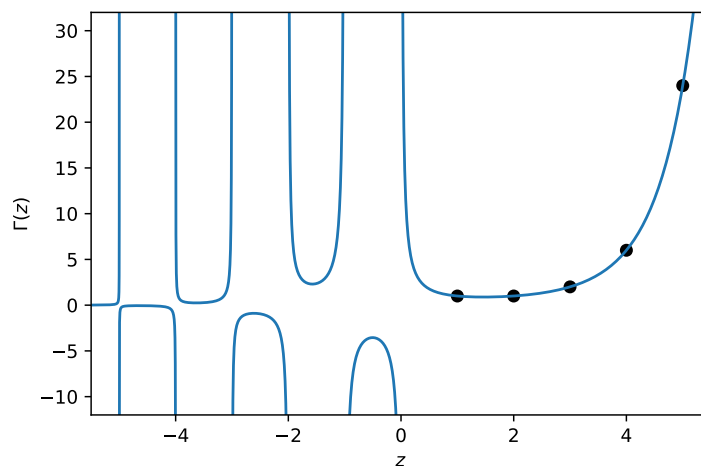


Figure E.1: The gamma function, $\Gamma(z)$. Black dots show the values $(n-1)!$ for the positive integers.

which converges for all arguments with $\text{Re}\{z\} > 0$. The *gamma function* is defined as the analytic continuation of this integral to the whole complex plane (except for a countable number of poles).

If $z = 1$, equation E.3 can be integrated directly to find $\Gamma(1) = 1$. Considering $\Gamma(z+1)$ and integrating by parts gives

$$\begin{aligned}\Gamma(z+1) &= \int_0^\infty dt \, t^z \exp(-t) \\ &= [-t^z \exp(-t)]_{t=0}^{t=\infty} + z \int_0^\infty dt \, t^{z-1} \exp(-t) \\ &= z\Gamma(z).\end{aligned}\tag{E.4}$$

Therefore, by induction, we must have that for any positive integer n

$$\Gamma(n+1) = n!\tag{E.5}$$

The gamma function is plotted as a function of a real-valued input in figure E.1. The gamma function has simple poles at the origin and at the negative integers.

Equation E.5 illustrates the gamma function as an extension of the factorial function to the real (or, more generally, the complex) numbers.

The factorial function is widely used in the fields of probability, statistics and combinatorics because it counts the number of ways of ordering n objects. For example, it appears in the PMF of the binomial distribution (see section 3.4.1). For this reason, the gamma function is also widely used in these fields.

The gamma function can be used to find an improved version of *Stirling's approximation*. For large integer n , we have

$$n! = \int_0^\infty dt \, t^n \exp(-t) \quad (\text{E.6})$$

$$= \int_0^\infty dt \, \exp(n \log t - t). \quad (\text{E.7})$$

The integrand is peaked around $t \approx n$. Letting $t = n + y$, so that $\log t = \log n + \log(1 + y/n)$, and expanding in y/n gives

$$\begin{aligned} n! &= \int_0^\infty dt \, \exp(n \log(n + y) - n - y) \\ &= \int_{-n}^\infty dy \, \exp\left(n \left[\log n + \frac{y}{n} - \frac{y^2}{2n^2} + \dots\right] - n + y\right) \\ &\approx \exp(n \log n - n) \int_{-n}^\infty dy \, \exp\left(\frac{-y^2}{2n}\right), \end{aligned} \quad (\text{E.8})$$

where, on the second line, we have used $\log(1 + x) = x - x^2/2 + x^3/3 - \dots$, for small $|x| < 1$. For large n this may be approximated by

$$n! \approx \exp(n \log n - n) \int_{-\infty}^\infty dy \, \exp\left(\frac{-y^2}{2n}\right). \quad (\text{E.9})$$

Using the standard result for the Gaussian integral (see appendix C) this becomes

$$n! \approx \sqrt{2\pi n} n^n \exp(-n). \quad (\text{E.10})$$

This is an improved version of Stirling's approximation. Further improvements can be obtained by keeping more terms in the expansion in equation E.8; however, these are rarely used in practice.